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Program of Postgraduate Studies in Business Administration

NAZÁRIO AUGUSTO DE OLIVEIRA

EXPLAINING CORPORATE CREDIT RATING TRANSITIONS AND DEFAULTS THROUGH MACHINE LEARNING: A data-driven approach to risk assessment

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NAZÁRIO AUGUSTO DE OLIVEIRA

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Thesis submitted to the Graduate Program in Business Administration at Mackenzie Presbyterian University, as a partial requirement for the attainment of the Doctor of Philosophy (PhD) degree.

THESIS ADVISOR: Prof. Dr. Leonardo Fernando Cruz Basso.

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NAZÁRIO AUGUSTO DE OLIVEIRA

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I especially want to thank my wife for this achievement. I am completely convinced that without her support, I would not have been able to reach this goal. I also want to express my gratitude for the unconditional encouragement from my mother and my son, as well as the unwavering companionship of my dog Nutella, who was by my side every day throughout the whole process.

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"In the realm of corporate finance, a robust credit rating analysis acts as a guiding light, illuminating the route to financial stability and prosperity for businesses."

ABSTRACT

Accurate modeling of corporate credit ratings, transitions, and defaults is essential for effective financial risk assessment. Traditional methods—such as manual evaluation, logistic regression (LR), and structural models like Merton's—offer interpretability but often fall short in dynamic environments, failing to capture non-linear relationships, temporal patterns, and firm-level heterogeneity. This thesis investigates the potential of machine learning (ML) techniques as a more robust and adaptive alternative to traditional credit risk models.Using a comprehensive seven-year dataset from S&P Capital IQ Pro, covering firms across 20 countries and 51 financial and business risk indicators, this research evaluates a range of ML models, including LR, Support Vector Machines (SVM), Decision Trees (DT), Random Forest (RF), Extreme Gradient Boosting (XGBoost), and Artificial Neural Networks (ANN). The first article demonstrates the superior performance of ensemble methods and Neural Networks (NN), particularly in identifying complex, non-linear interactions among predictors. The second article advances this by proposing a hybrid ML framework that incorporates segment-specific modeling by credit rating bands, rigorous out-of-time validation to ensure robustness, and SHAP (SHapley Additive exPlanations) values for interpretability.Findings show that ensemble models—especially XGBoost and RF—consistently outperform traditional approaches in both predictive accuracy and early warning capability. Moreover, SHAP analysis highlights the varying importance of explanatory variables across different credit segments, offering actionable insights for more targeted credit monitoring.This thesis contributes to the literature by bridging predictive performance with interpretability in credit risk modeling. It offers practical implications for financial institutions, rating agencies, and regulators aiming to adopt data-driven, scalable, and transparent tools for forward-looking credit assessment. Future research should explore real-time analytics, integration of alternative data sources, and the embedding of explainability into model design to support regulatory compliance.

Keywords: credit ratings; machine learning; credit rating transitions; default prediction; explainable AI (SHAP); financial risk assessment; risk management.

RESUMO

A modelagem precisa das classificações de crédito corporativo, bem como das transições e inadimplências, é fundamental para uma avaliação eficaz do risco financeiro. Métodos tradicionais — como avaliações manuais, regressão logística e modelos estruturais como o de Merton — oferecem transparência, mas apresentam limitações importantes em ambientes econômicos dinâmicos, especialmente na identificação de relações não lineares, padrões temporais e heterogeneidade entre empresas. Esta tese investiga o potencial das técnicas de aprendizado de máquina (ML) como uma alternativa mais robusta e adaptativa aos modelos tradicionais de risco de crédito.Utilizando um conjunto de dados abrangente de sete anos da base S&P Capital IQ Pro, que abrange empresas de 20 países e 51 variáveis financeiras e de risco de negócio, esta pesquisa avalia uma variedade de modelos de ML, incluindo Regressão Logística, Máquinas de Vetores de Suporte (SVM), Árvores de Decisão, Random Forest (RF), Extreme Gradient Boosting (XGBoost) e Redes Neurais Artificiais (ANN). O primeiro artigo demonstra o desempenho superior dos métodos de ensemble e das redes neurais, especialmente na identificação de interações complexas e não lineares entre os fatores preditivos. O segundo artigo avança ao propor uma estrutura híbrida de ML, que incorpora modelagem segmentada por faixas de classificação de crédito, validação fora da amostra (out-of-time) para garantir robustez e valores SHAP (SHapley Additive exPlanations) para garantir interpretabilidade. Os resultados mostram que os modelos de ensemble — particularmente o XGBoost e o RF — superam consistentemente os métodos tradicionais tanto em precisão preditiva quanto na capacidade de alerta precoce. A análise SHAP revela, ainda, que os fatores determinantes das transições de classificação variam de acordo com a qualidade do crédito, o que reforça a importância de estratégias de monitoramento personalizadas.Esta tese contribui para a literatura ao conciliar desempenho preditivo e interpretabilidade na modelagem do risco de crédito, oferecendo implicações práticas para instituições financeiras, agências de classificação e reguladores que buscam adotar ferramentas escaláveis, transparentes e orientadas por dados para uma avaliação prospectiva de risco. Pesquisas futuras podem explorar o uso de dados alternativos, análise em tempo real e a incorporação de explicabilidade no próprio processo de modelagem, visando maior aderência regulatória.

Palavras-chave: classificações de crédito; aprendizado de máquina; transições de rating; previsão de inadimplência; inteligência artificial explicável (SHAP); avaliação de risco financeiro; gestão de risco.

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LIST OF ABBREVIATIONS AND ACRONYMS

AI	Artificial intelligence
ANN	Artificial Neural Networks
AP	Average Precision
AUC	Area under the curve
AUC-ROC	Area Under the Receiver Operating Characteristic Curve
CNN	Convolutional Neural Networks
CRA	Credit rating agency
DLSS-SVM	Dual-Layer Smart Sampling Support Vector Machine
DT	Decision Trees
EBM	Explainable Boosting Machine
EER	Equal Error Rates
ESG	Environmental, social, and governance
EWS	Early warning system
FFO	Funds from Operations
FNR	False Negative Rate
FPR	False Positive Rate
GAM	Generalized Additive Model
GAN	Generative adversarial network
GB	Gradient Boosting
GBM	Gradient Boosting Machines
LIME	Local Interpretable Model-Agnostic Explanations
LR	Logistic Regression
MCC	Matthews Correlation Coefficient
MDA	Multiple discriminant analysis
ML	Machine Learning
MSM	Markov Switching Models
NN	Neural Networks
OvO	One-vs-One
OvR	One-vs-Rest
PR	Precision–Recall
RF	Random Forest
RNN	Recurrent Neural Networks

ROC	Receiver Operating Characteristic
RWA	Risk-weighted asset
SHAP	Shapley Additive Explanations
SME	Small and medium-sized enterprises
SMOTE	Synthetic Minority Oversampling Technique
SVM	Support Vector Machine
TPR	True Positive Rate
VIF	Variance inflation factor
XAI	Explainable artificial intelligence
XGBoost	Extreme Gradient Boosting

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1 INTRODUCTION

1.1 BACKGROUND ON CREDIT RATING TRANSITIONS AND DEFAULTS

Credit ratings serve as standardized evaluations of a firm's creditworthiness and are a foundational element of modern financial systems. They influence debt pricing, investment decisions, and regulatory capital requirements. Importantly, credit ratings are not static; transitions and defaults reflect evolving firm-specific and macroeconomic risk exposures. Understanding these dynamics is essential for financial system stability, particularly during periods of economic stress or structural shifts.

Robust modeling of credit rating transitions and default probabilities is critical for institutional investors, regulatory bodies, and financial institutions seeking forward-looking insights into corporate solvency and systemic risk. Downgrades can substantially increase borrowing costs and trigger covenant breaches, while defaults pose broader contagion risks. Consequently, there is a growing demand for accurate, data-driven approaches to monitor and forecast credit risk.

1.2 THE RISE OF MACHINE LEARNING IN CREDIT RISK

Traditional credit risk models—such as Merton's (1974) option-theoretic approach and statistical methods like logistic regression (LR) (Altman, 1968; Ohlson, 1980)—have long dominated the field. These models typically rely on structured financial ratios and assume linear relationships between predictors and outcomes. While grounded in economic theory, they often fail to capture the nonlinear and high-dimensional characteristics inherent in real-world financial data.

In recent years, machine learning (ML) has emerged as a compelling alternative for credit risk assessment. Models such as Random Forests (RF), Gradient Boosting Machines (GBM, e.g., Extreme Gradient Boosting – XGBoost), Support Vector Machines (SVMs), and Neural Networks (NN) have demonstrated superior predictive performance in various financial applications, including bankruptcy prediction (Barboza; Kimura; Altman, 2017), credit card default forecasting, and portfolio risk classification.

ML models are capable of processing large, heterogeneous datasets, adapting to changing economic conditions, and identifying complex patterns that may not be detectable using conventional methods. Additionally, the incorporation of alternative data sources—such

as governance indicators, disclosure quality, and sector-specific signals—has enabled a more holistic understanding of firm-level risk (Lu; Zhang; Li, 2019).

1.3 LIMITATIONS OF TRADITIONAL ECONOMETRIC APPROACHES

Despite their historical prevalence, traditional econometric models exhibit notable limitations when applied to modern credit risk environments (Cheng et al., 2024). These models often rely on assumptions of firm homogeneity, stable relationships among variables, and normally distributed data—conditions that are rarely met in practice. Moreover, they tend to underperform when applied to imbalanced datasets, where events like defaults are infrequent but consequential.

Their inability to incorporate unstructured or high-frequency data further restricts their utility in dynamic and fast-evolving contexts. These constraints underscore the need for more flexible, scalable modeling frameworks that can account for nonlinearity, temporal shifts, and complex feature interactions.

1.4 RESEARCH GAPS AND PRACTICAL RELEVANCE

Although ML has demonstrated substantial promise in financial prediction tasks, several critical gaps remain in its application to credit rating transitions and defaults:

a) **Comparative evaluations** of ML models versus traditional statistical approaches in credit rating transitions remain limited, particularly in corporate credit contexts (Kou *et al*., 2019);
b) **Model interpretability** poses a significant barrier to adoption in regulated, risk-sensitive domains (European Banking Authority, 2021; Rudin, 2019);
c) **Segment-specific modeling** is often absent, with many studies neglecting how model performance and feature importance vary across rating classes (Bitetto; Filomeni; Modina, 2025);
d) **Regulatory challenges** related to fairness, transparency, and model governance persist, impeding the deployment of complex ML tools in credit risk assessment (Nallakaruppan *et al*., 2024).

This thesis seeks to address these gaps by proposing a data-driven, interpretable, and segment-aware ML framework for predicting rating transitions and defaults.

1.5 RESEARCH AIMS AND QUESTIONS

This thesis investigates the predictive and explanatory potential of machine learning models in the context of corporate credit rating transitions and default events. It aims to evaluate the extent to which advanced machine learning techniques can improve predictive accuracy while providing interpretable insights into the determinants of credit risk across different rating segments.

This thesis adopts a data-driven empirical approach to uncover predictive patterns, determinants, and mechanisms associated with corporate credit rating transitions and default events. Rather than testing theory-derived causal hypotheses, the study focuses on identifying empirical regularities and predictive structures within high-dimensional financial datasets using machine learning methods. This perspective aligns with the growing use of machine learning in finance, where the primary objective is to enhance predictive performance and extract meaningful patterns from complex data environments.

Within this framework, the study is guided by the following research questions:

1) How accurately can machine learning models predict corporate credit rating transitions and defaults relative to traditional statistical approaches?
2) Which financial and business variables are most relevant in explaining rating transitions across different credit risk segments?
3) How does the importance of explanatory variables vary across rating segments when analyzed through explainable machine learning techniques?

1.6 THEORETICAL AND PRACTICAL SIGNIFICANCE

This study contributes to the academic discourse at the intersection of finance, risk management, and ML. Following Whetten's (1989) framework for theoretical contributions, the research advances multiple dimensions of theory and offers practical insights for analysts and regulators. In addition, the study provides empirical evidence on the comparative performance of machine learning algorithms in predicting credit rating transitions and default events across multiple credit segments.

1.6.1 Advancing Predictive Modeling in Credit Ratings

The thesis bridges traditional econometric and modern ML approaches. Classical models such as Markov Chains and LR assume static, linear relationships among variables. In contrast, this work demonstrates that ML algorithms can effectively capture nonlinear, dynamic, and high-dimensional patterns.

By benchmarking supervised ML models (e.g., XGBoost, RF, SVM), time-series approaches (e.g., LSTM), and unsupervised techniques (e.g., clustering), the study expands the theoretical understanding of how different model architectures suit the evolving nature of credit risk.

Methodologically, the research strengthens model validation by integrating performance metrics (e.g., Area Under the Receiver Operating Characteristic Curve [AUC-ROC], F1-score) with financial evaluation tools such as stress-testing scenarios and segment-specific assessments.

1.6.2 Identifying New Drivers of Credit Transitions

This study advances beyond traditional ratio-based credit risk modeling by incorporating a diverse and structured set of explanatory variables across three key dimensions:

a) **macroeconomic factors**, such as *interest expense*, *cash interest*, and *country risk*, which serve as proxies for broader economic influences like interest rates, inflation, monetary policy, and geopolitical instability;
b) **industry-specific dynamics**, including *industry risk*, *scale and diversification*, and *volatility of profitability*, which capture sectoral cyclicality and structural exposure; and;
c) **firm-level characteristics**, such as *governance score*, *financial policy*, *capital structure*, and a wide range of financial ratios related to profitability, leverage, coverage, efficiency, and cash flow.

By employing Shapley Additive Explanations (SHAP), the study uncovers complex, non-linear interactions and dependencies among features—relationships that are typically obscured in linear modeling frameworks. This interpretability enables more nuanced insights into the drivers of rating transitions and default risk.

1.6.3 Improving Interpretability of Machine Learning in Finance

To address the "black-box" issue of ML, the thesis integrates XAI techniques, particularly SHAP, to provide post hoc interpretability. These methods enable stakeholders to answer critical questions such as:

- What explains the model's prediction for a specific firm?
- Which features had the greatest influence on a rating change?

This contributes to the growing body of literature on interpretable ML in finance and enhances model transparency in line with regulatory expectations (ElShawi *et al*., 2020; Rudin, 2019).

1.6.4 Expanding the Use of Regime-Switching and State Space Models

Acknowledging the impact of economic cycles, the thesis explores Markov Switching Models (MSMs) as a robustness check. While traditional Markov Chains assume static transition probabilities, MSMs capture structural shifts—such as transitions between recession and expansion regimes.

This approach contributes by:

- Demonstrating how MSMs can complement ML models in identifying regime-dependent credit behavior;
- Applying state space methodologies to quantify the effects of macroeconomic volatility on credit transitions.

1.6.5 Contribution to Broader Academic Dialogues

The thesis contributes to several overlapping research domains:

a) **Credit Risk and Financial Stability**: Enhances early warning frameworks for financial distress;
b) **ML in Finance**: Enriches ongoing debates on the trade-off between predictive power and interpretability;
c) **Time-Series and Regime Modeling**: Applies dynamic and state-aware approaches to better contextualize firm-level credit behavior;

1.6.6 Implications for Theory and Practice

For academic research:

- Offers a structured and comparative framework for credit risk modeling using ML;
- Suggests future research directions, including hybrid econometric–ML models and explainability-aware training.

For practitioners and regulators:

- Provides guidance for building interpretable credit monitoring tools.
- Supports the adoption of ML methods in risk evaluation without compromising transparency or regulatory alignment.

1.6.7 Summary of Theoretical Contribution

Table 1.1 – Summary of Theoretical Contribution

Aspect	Contribution
Predictive modeling	Demonstrates ML's superiority over traditional models in forecasting rating transitions.
Drivers of credit risk	Identifies the influence of non-financial and interaction-based factors.
Interpretability	Enhances explainability through SHAP and LIME frameworks.
Regime-switching modeling	Applies MSMs to reflect dynamic credit behavior across economic cycles.
Financial stability	Contributes to early warning systems (EWS) for default and downgrade risks.

Source: Own authorship.

1.7 THESIS ROADMAP

The thesis is organized as follows:

a) **Chapter 1** provides the overarching introduction, outlining the motivation, research questions, theoretical underpinnings, and practical relevance of the study. It also synthesizes the existing literature on traditional credit risk models and recent developments in ML and explainable AI, positioning the two articles within this broader academic landscape;

b) **Chapter 2** presents the first article, *"Advancing Credit Rating Prediction: The Role of Machine Learning in Corporate Credit Rating Assessment"*, published in the journal *Risks*. This study explores the predictive power of multiple ML algorithms—including LR, SVM, DT, RF, Gradient Boosting (GB), and NN—on a multicountry

dataset of corporate credit ratings. The analysis focuses on model benchmarking, feature importance, and the potential of ML techniques to enhance traditional credit rating methodologies;

c) **Chapter 3** contains the second article, *"Explaining Corporate Ratings Transitions and Defaults Through Machine Learning"*, submitted to *Machine Learning & Knowledge extraction*. Building on the insights of the previous study, this article proposes a hybrid ML framework that emphasizes model explainability, segment-specific modeling across credit rating bands, and out-of-time validation. It introduces an interpretable EWS using SHAP values and contributes practical tools for credit monitoring and regulatory oversight;

d) **Chapter 4** concludes the thesis by integrating the findings of both articles. It highlights their theoretical and managerial contributions, addresses limitations, and outlines promising directions for future research. Particular attention is given to the implications of explainable ML adoption in financial regulation, and to the potential integration of unstructured data and macroeconomic signals in future credit risk models.

2 ADVANCING CREDIT RATING PREDICTION: THE ROLE OF MACHINE LEARNING IN CORPORATE CREDIT RATING ASSESSMENT

ABSTRACT

Accurate corporate credit ratings are essential for financial risk assessment; yet, traditional methodologies relying on manual evaluation and basic statistical models often fall short in dynamic economic conditions. This study investigated the potential of ML algorithms as a more precise and adaptable alternative for credit rating predictions. Using a seven-year dataset from S&P Capital IQ Pro, corporate credit ratings across 20 countries were analyzed, leveraging 51 financial and business risk variables. The study evaluated multiple ML models, including LR, SVM, DT, RF, GB, and Neural Networks (NN), using rigorous data pre-processing, feature selection, and validation techniques. Results indicate that Artificial Neural Networks (ANN) and GB consistently outperform traditional models, particularly in capturing non-linear relationships and complex interactions among predictive factors. This study advances financial risk management by demonstrating the efficacy of ML-driven credit rating systems, offering a more accurate, efficient, and scalable solution. Additionally, it provides practical insights for financial institutions aiming to enhance their risk assessment frameworks. Future research should explore alternative data sources, real-time analytics, and model explainability to facilitate regulatory adoption.

Keywords: machine learning; credit ratings; predictive modeling; financial risk assessment; risk management.

2.1 INTRODUCTION

Credit ratings play a pivotal role in financial markets by providing standardized assessments of a firm's creditworthiness, shaping investment decisions, influencing risk management strategies, and determining regulatory compliance obligations (Kou *et al*., 2019). These ratings directly affect borrowing costs, bond yields, and market perceptions, thereby reinforcing overall market stability.

Traditionally, credit rating agencies (CRAs) have relied on a combination of financial ratio analysis and qualitative judgment when assigning ratings. This conventional approach typically involves assessing key financial indicators—such as leverage, liquidity, and

profitability—alongside non-financial factors, including management quality and macroeconomic trends (Egan-Jones Ratings, 2024; Investment Grade Capital, 2024). However, these methodologies exhibit inherent limitations, notably the assumption of linear relationships between variables and the potential to overlook complex, non-linear dynamics prevalent in modern financial markets. Furthermore, regulatory inconsistencies and enforcement challenges continue to impede the effective supervision of CRAs, particularly in emerging markets (Rabinowitz *et al*., 2024).

In response to these challenges, ML and AI have emerged as transformative tools in credit risk assessment. ML algorithms excel at analyzing vast and complex datasets, enabling the detection of intricate patterns that traditional statistical models may fail to capture (Kou *et al*., 2019). This capability has led to substantial improvements in predictive performance across various financial applications, such as bankruptcy prediction (Barboza; Kimura; Altman, 2017) and credit card default forecasting (Chang *et al*., 2024).

The incorporation of alternative data sources has further enhanced the predictive capabilities of credit risk models. Integrating non-traditional information enables more nuanced and timely credit evaluations, providing a multidimensional view of borrower risk that extends beyond conventional financial statements (Lu; Zhang; Li, 2019).

Despite these advancements, critical gaps persist in the literature. Notably, there is a scarcity of comprehensive studies systematically comparing the performance and robustness of different ML algorithms across diverse market conditions and industry sectors (Kou *et al*., 2019). Moreover, concerns regarding the explainability and fairness of AI-driven models are increasingly prominent. Regulatory authorities mandate that financial institutions ensure transparency, accountability, and governance in their credit risk models, particularly when employing complex ML algorithms (European Banking Authority, 2021).

Recent scholarly work has further underscored these concerns. Enhancing both the accuracy and fairness of AI applications in credit risk assessment is crucial for promoting trust and stakeholder acceptance (Wang, Z., 2024). Additionally, interpretable models should be preferred over black-box approaches in high-stakes decision-making contexts, such as credit rating, to mitigate risks and ensure regulatory compliance (Rudin, 2019).

To address these challenges, this study conducts a rigorous comparative evaluation of multiple ML algorithms applied to corporate credit rating prediction. Specifically, it aims to assess the predictive performance of widely used models—including LR, DT, RF, SVMs, NN, and Gradient Boosting Machines (GBM)—against traditional credit rating approaches.

Moreover, the study seeks to identify and interpret the most influential factors affecting credit ratings through advanced feature importance analysis.

The robustness of these models will be tested across various economic contexts to evaluate their practical applicability and generalizability. This consideration is particularly important given that a deeper understanding of generalization remains critical for the reliable deployment of deep-learning models (Zhang *et al.*, 2021).

To support these objectives, the study utilizes a comprehensive dataset covering seven years of corporate credit ratings across 20 countries sourced from S&P Capital IQ Pro. The dataset comprises 51 variables, with 43 related to financial risk and 8 to business risk. Rigorous data pre-processing procedures—including outlier detection, normalization, and treatment of missing values—are applied to enhance data quality. Feature selection follows established best practices, notably the recursive feature elimination method, which has proven effective in isolating the most predictive variables in high-dimensional datasets (Guyon *et al*., 2002).

Each selected ML algorithm is trained using stratified k-fold cross-validation to prevent overfitting and ensure model generalizability. Performance evaluation is conducted across a suite of metrics—accuracy, precision, recall, F1 score, and ROC-AUC—aligned with recommended evaluation standards for ML model assessments (Biecek; Burzykowski, 2021).

This research makes several key contributions to the field. First, it offers an extensive comparative analysis of ML algorithms for corporate credit rating prediction, directly addressing gaps identified in the literature (Kou *et al*., 2019). Second, it integrates financial and non-financial business risk factors—including governance, country risk, industry risk, diversification, and operating efficiency—providing a more comprehensive view of the determinants of corporate creditworthiness. Finally, it responds to regulatory and ethical imperatives by embedding explainability and fairness considerations into model selection and evaluation (European Banking Authority, 2021; Rudin, 2019; Wang, Y., 2024).

The remainder of this paper presents a review of the literature on ML applications in credit rating prediction, followed by the research methodology detailing model selection and evaluation metrics. Subsequently, the empirical findings and their implications are discussed, concluding with recommendations for financial practitioners, regulators, and future research.

2.2 LITERATURE REVIEW

Credit rating prediction has long been a critical area of financial research due to its role in assessing borrowers' creditworthiness. Traditional approaches relied heavily on statistical

models, which laid the foundation for systematic risk assessment. A seminal contribution is Altman's (1968) Z-score model, which used multiple discriminant analysis (MDA) to combine financial ratios into a singular risk metric for bankruptcy prediction. Merton's (1974) structural model further advanced the field by applying option pricing theory, conceptualizing a company's equity as a call option on its assets to estimate default probabilities.

However, traditional credit rating models exhibit notable limitations. Their dependence on historical financial data and accounting-based ratios constrains their ability to capture complex, non-linear relationships and adapt to real-time shifts in a borrower's financial condition, undermining predictive accuracy (Cheng *et al*., 2024; Umeaduma; Adedapo, 2025). Additionally, these models often rely on simplifying assumptions about the distribution of financial ratios and market conditions—assumptions that can fail during periods of economic turbulence or firm-specific distress (Addy *et al*., 2024). A further limitation lies in their predominantly binary classification framework, which focuses solely on default versus non-default outcomes, neglecting nuanced credit rating migrations essential for risk-sensitive applications (Addy *et al*., 2024; Jagtiani; Lemieux, 2019).

To address these shortcomings, enhanced statistical methods emerged. Linear regression models established relationships between credit ratings and variables such as financial ratios and economic indicators. For instance, Kaplan and Urwitz (1979) demonstrated the effectiveness of linear regression in predicting bond ratings. Subsequently, LR became a staple for binary classification in credit risk contexts, notably through Ohlson's (1980) pioneering application for bankruptcy prediction. Nevertheless, LR is limited by its assumption of linearity in the logit function and susceptibility to multicollinearity among independent variables.

Recognizing these constraints, CRAs began integrating quantitative data with expert judgment. Qualitative factors—such as management quality, governance practices, and industry dynamics—play a critical role in bond ratings, contributing to more comprehensive and forward-looking assessments. However, this integration introduces risks of subjectivity and potential inconsistencies that can compromise model objectivity (Seetharaman *et al*., 2017).

In recent years, ML and AI have significantly transformed credit risk modeling by offering advanced tools for data analysis and prediction. ML algorithms excel at identifying complex, non-linear patterns in large datasets, thereby addressing many limitations inherent in traditional models (Talaei Khoei; Kaabouch, 2023). Talaei Khoei and Kaabouch (2023) provided a comprehensive review of various ML methods—including NN, SVMs, and ensemble learning techniques—highlighting their advantages in enhancing predictive performance and robustness across financial applications. Further exemplifying these

advancements, Chang *et al.* (2024) demonstrated that deep-learning architectures such as Recurrent Neural Networks (RNNs) and Convolutional Neural Networks (CNNs) outperform conventional models in credit card default prediction, particularly in handling sequential and structured financial data.

The growing integration of advanced ML techniques into credit rating prediction reflects a broader trend toward more sophisticated analytical tools. For example, Tran and Tham (2025) showed that boosting ensembles and SVMs outperform traditional models like LR in financial risk evaluation owing to their superior capacity for modeling complex, non-linear relationships. Similarly, Mokheleli and Museba (2023) highlighted how NNs and ensemble methods enhance the predictive accuracy of credit score models by capturing intricate patterns within financial datasets.

Despite their superior predictive capabilities, ML models often face criticism for their "black-box" nature, which complicates interpretability and transparency—particularly in high-stakes financial decision-making. In response, recent research has prioritized the development of XAI techniques, such as SHAP and Local Interpretable Model-Agnostic Explanations (LIME). These methods improve the interpretability of model predictions and are essential for compliance with emerging regulatory frameworks, including the European Union's AI Act and the Basel Committee's guidelines on model risk management (Hassija *et al*., 2024; Sabharwal *et al*., 2024).

Yifei Wang (2024) emphasized that ensuring fairness and transparency in ML-based credit risk assessments is vital for promoting stakeholder trust and meeting regulatory requirements. Complementing this perspective, Hassija *et al*. (2024) reviewed a range of methods for interpreting black-box models, underscoring their growing relevance in financial contexts. Moreover, Weber, Carl and Hinz (2024) conducted a systematic review that highlighted the increasing importance of XAI in finance, particularly for enhancing model accountability and mitigating risks associated with opaque decision-making processes.

In addition to advances in ML techniques, recent research has emphasized the importance of non-financial and strategic variables in credit risk and default prediction—particularly for SMEs. Traditional financial indicators, while foundational, may be insufficient to capture the nuanced factors that influence a firm's creditworthiness and recovery potential.

For instance, Altman *et al.* (2022, 2023) introduced the Omega Score, a revised predictive model that integrates both financial and qualitative variables for small and medium-sized enterprises (SME) default prediction, significantly improving model performance and

robustness. These studies stress the value of including variables related to management quality, market positioning, and strategic behavior.

This perspective is further reinforced in sector-specific contexts. Srhoj *et al*. (2024) demonstrate that in the tourism industry, predictors of SME default differ from general models, underscoring the need for industry-tailored frameworks. Their findings highlight the unique operational and demand-side dynamics that impact firms in this sector.

Furthermore, recovery-oriented research by Altman *et al.* (2025) explores the drivers of turnaround success, finding that qualitative factors—such as managerial flexibility and strategic refocusing—are crucial for explaining why some distressed firms rebound while others do not. This work suggests a growing consensus around the integration of strategic, behavioral, and sector-specific dimensions in predictive modeling.

Parallel to advancements in interpretability, the incorporation of alternative data sources has emerged as a significant trend in credit risk modeling. Traditional financial indicators are now frequently supplemented by non-traditional data such as environmental, social, and governance (ESG) metrics, supply chain information, and social media sentiment. Lu, Zhang and Li (2019) provided empirical evidence demonstrating that alternative data significantly improves the predictive accuracy of credit risk models by offering a more comprehensive and multidimensional perspective on borrower risk.

The influence of fintech and digital footprints in credit scoring is further evidenced by studies such as those by Jagtiani and Lemieux (2019) and Berg *et al*. (2020). Their findings reveal that alternative data enhances model performance by capturing forward-looking indicators typically overlooked by traditional models. However, the integration of such data also raises complex challenges related to privacy, ethical considerations, and model validation, necessitating ongoing research and robust regulatory oversight (Owoade *et al*., 2024). In line with this, the Basel Committee on Banking Supervision (2023) has issued principles for managing climate-related financial risks, reinforcing the need to incorporate ESG-related alternative data in credit risk assessments.

The impact of ML extends beyond credit rating prediction to domains such as fraud detection and portfolio management. Fraud detection systems have evolved from rule-based mechanisms to sophisticated ML-driven models. Recent studies highlight the effectiveness of advanced ML approaches in identifying complex fraud patterns, with ensemble methods and synthetic sampling techniques contributing to improved classification accuracy (Bello *et al*., 2023; Ileberi; Sun; Wang, 2021).

In portfolio management, ML has enhanced traditional models such as Modern Portfolio Theory (Markowitz, 1952) and the Capital Asset Pricing Model (Sharpe, 1964). For instance, Krauss, Do and Huck (2017) and Gu, Kelly and Xiu (2020) applied deep NNs and gradient-boosted trees for stock return prediction, achieving significant improvements over conventional time-series forecasting methods.

Illustrating the broader applicability of ML techniques, the study "On the analytical study of the service quality of Indian Railways under soft-computing paradigm" demonstrates the versatility of approaches such as rough set theory, extra trees classifiers, and SVMs beyond financial contexts, reinforcing the value of these tools across diverse domains (Majumder *et al.*, 2024).

While substantial progress has been made, future research should focus on enhancing the interpretability of ML models, integrating alternative data sources more effectively, and ensuring alignment with evolving regulatory frameworks. Additionally, systematic comparative studies evaluating the performance of different ML algorithms across various economic contexts and industry sectors remain relatively scarce (Kou *et al.*, 2019).

The convergence of ML techniques with domain expertise and traditional financial theories offers significant potential for developing more robust and reliable credit risk models. Furthermore, advancing responsible AI principles—such as fairness, accountability, and transparency—is essential to harmonize technological innovation with societal and regulatory expectations. Recent contributions by Barocas, Hardt and Narayanan (2023) underscore the growing emphasis on fairness and causal approaches in credit scoring, providing valuable guidance for future research agendas.

2.2.1 Theoretical Framework

2.2.1.1 Credit Risk Theory

Credit risk theory forms the backbone of credit rating assessments, emphasizing the potential risk of loss resulting from a borrower's failure to repay a loan. Robert Merton's foundational works (1974) laid the groundwork for modern credit risk evaluation through the Merton model, which applies option pricing theory to corporate debt valuation. Merton's model fundamentally reshaped the understanding of credit risk by treating corporate debt as a risk-free bond combined with a short put option on the firm's assets, reflecting the probability of default.

Before Merton (1974) introduced his structural model, Altman (1968) developed the Z-score model, a statistical tool for predicting bankruptcy. Altman's model utilized financial ratios and discriminant analysis to estimate the likelihood of default, providing a practical and widely adopted framework for credit risk assessment. While Merton's approach focuses on market-based measures, Altman's model emphasizes accounting data, highlighting a divergence in methodology but a convergence in the goal of assessing credit risk.

Jarrow and Turnbull (1995) significantly advanced credit risk modeling by developing a reduced-form approach that incorporates the term structure of interest rates. Their model provides a market-based, dynamic method for estimating default probabilities and credit spreads, improving upon static accounting-based measures like the Altman Z-score. This approach underscores the importance of market-sensitive frameworks in credit risk assessment.

2.2.1.2 Machine Learning Theory

ML theories have significantly transformed credit rating prediction by introducing advanced algorithms capable of managing large-scale datasets and capturing complex, non-linear patterns. Foundational techniques such as DT models (Breiman *et al*., 1984) offer intuitive and interpretable structures but are inherently prone to overfitting. To mitigate these limitations, contemporary ensemble methods such as RF (Breiman, 2001) and GBM (Friedman, 2001) have been developed, followed by more sophisticated variants like XGBoost (Chen; Guestrin, 2016), LightGBM (Ke *et al*., 2017), and CatBoost (Prokhorenkova *et al*., 2018), which deliver enhanced scalability and predictive accuracy.

The backpropagation algorithm (Rumelhart; Hinton; Williams, 1986) remains a cornerstone of NN training, underpinning modern deep-learning architectures that adeptly model complex financial relationships. Recent empirical studies, such as that of Tran and Tham (2025), have demonstrated the efficacy of deep NNs in processing extensive historical credit data, thereby significantly improving predictive performance.

Despite these technological advances, the "black-box" nature of many ML models has raised concerns about transparency and interpretability, particularly in high-stakes financial decision-making contexts. This has catalyzed extensive research on XAI methodologies, notably SHAP (Lundberg; Lee, 2017) and LIME, to ensure model interpretability and regulatory compliance. Regulatory frameworks, including the European Union's Artificial Intelligence Act (European Commission, 2023) and the Basel Committee's guidance on model

risk management, now underscore the imperative for interpretable models in financial applications.

Benchmarking studies focused on credit scoring and rating prediction have consistently validated the superiority of ensemble-based models over traditional statistical approaches (Lessmann *et al.*, 2015), reinforcing their growing prominence in contemporary credit risk analytics. Moreover, recent comprehensive reviews on ensemble deep learning have emphasized their advantages in improving predictive performance across various domains, further highlighting their methodological robustness and broad applicability (Mohammed; Kora, 2023).

2.2.1.3 Behavioral Finance Theory

Behavioral finance theory examines how psychological factors and biases influence financial decision-making, which can impact credit ratings. Tversky and Kahneman's (1974) prospect theory highlights how cognitive biases, particularly loss aversion, shape risk perceptions. Other behavioral biases, such as overconfidence and herding behavior, also influence investor decisions. These biases contribute to irrational market behavior, influencing credit risk assessment through changes in market sentiment, risk perception, and investor-driven fluctuations in credit spreads, which can ultimately affect credit ratings.

Shiller (2003) explored the role of market sentiments and speculative bubbles in financial markets, underscoring the influence of behavioral factors on asset pricing and risk perception. Behavioral finance theories suggest that traditional models may overlook these irrational elements, leading to mispricing and inaccurate credit ratings.

ML can address these complexities by incorporating behavioral indicators into predictive models. By analyzing patterns in investor behavior and sentiment, ML algorithms can adjust for biases and improve the accuracy of credit risk predictions. This integration of behavioral finance with ML approaches represents a convergence of theories aimed at enhancing the robustness of credit rating methodologies.

2.2.1.4 Gaps in the Literature

Applying ML algorithms in credit rating prediction has gained significant attention recently. However, despite these advancements, the black-box nature of many ML models poses challenges for interpretability and regulatory acceptance. Researchers have developed XAI techniques to address these concerns that provide insights into model predictions. Ribeiro, Singh and Guestrin (2016) introduced LIME, a method designed to explain the predictions of any classifier, enhancing the transparency of ML models. Similarly, Lundberg and Lee (2017) developed SHAP, a unified approach to interpreting model predictions, gaining traction in financial applications. These techniques have significantly improved interpretability, yet challenges remain in effectively integrating them into real-world financial decision-making.

Despite these advancements, several limitations persist in the literature. First, most studies have focused on static models that do not account for temporal changes in financial data. Second, data quality and availability remain significant challenges. Many studies rely on proprietary datasets that are not publicly accessible, limiting the reproducibility and generalizability of findings. Finally, as Doshi-Velez and Kim (2017) emphasized, while interest in XAI techniques continues to grow, achieving a balance between model accuracy and interpretability remains an ongoing challenge. Addressing these issues is crucial for adopting ML-based credit rating models in regulatory and financial environments.

Given these gaps, the current study aims to advance the field by developing a dynamic machine-learning framework for credit rating prediction that accounts for temporal changes in financial data. By utilizing publicly available datasets, this study seeks to enhance the reproducibility and generalizability of findings.

In addition to these technical limitations, recent empirical work has drawn attention to the absence of strategic and industry-specific considerations in many credit risk prediction models. For example, Altman *et al.* (2022, 2023) highlight the relevance of non-financial predictors—such as governance and competitive advantage—in improving SME default prediction. Likewise, Srhoj *et al.* (2024) show that sector-specific models yield better predictive accuracy by accounting for structural differences across industries. Altman *et al.* (2025) go a step further by identifying the qualitative and strategic determinants of firm recovery, advocating for a broader conceptualization of creditworthiness that includes the likelihood of turnaround.

By engaging with these perspectives, the present study aims to not only enhance methodological rigor through ML but also set a foundation for integrating qualitative and sector-sensitive factors into future predictive frameworks.

2.3 METHODOLOGY

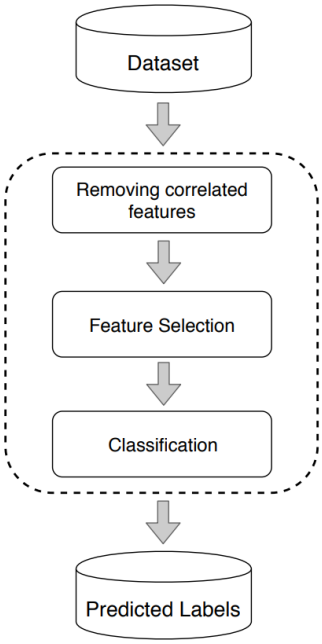
2.3.1 Materials and Methods

This study focuses on credit rating prediction, a well-established research topic in financial risk assessment. Credit ratings categorize a client's financial condition into predefined classes, such as "A," "B," "C," or "D." These classifications are assigned based on financial and qualitative indicators using ML techniques.

A comprehensive dataset must be compiled from various sources to develop a predictive model. Relevant financial and qualitative features are extracted as predictors in a classification model. Credit rating prediction is framed as a classification problem, where a supervised learning model, trained on labeled data, assigns a credit rating to new clients based on their financial characteristics. Key financial indicators, such as the debt-to-income ratio, history of defaults, and credit inquiries, play a significant role in determining whether a client receives a "good" or "poor" rating.

The proposed methodology follows a structured approach, illustrated in Figure 2.1. First, financial and qualitative data are collected and processed to build a robust dataset, ensuring each feature represents a meaningful aspect of the company's risk profile. Since not all financial and qualitative attributes contribute equally to prediction accuracy, a two-stage feature refinement process is applied: (i) correlation analysis to remove highly correlated and redundant features and (ii) feature selection to exclude variables that may negatively impact classification performance. Finally, a ML classification model is trained on the selected features and used to predict credit ratings.

Figure 2.1 – Methodological approach



Source: Own authorship.

2.3.2 Removing Correlated Features

The previous section outlined the process of collecting, processing, and structuring data—comprising features and labels—as input for the classification step. However, certain features may exhibit high correlation, leading to redundant information that does not contribute unique predictive value. Removing such redundancy is essential for enhancing model interpretability, reducing overfitting, and improving classification accuracy (Guyon et al., 2002).

In statistical analysis and ML, correlation is a fundamental metric for assessing feature relevance and redundancy. The linear correlation coefficient (ρ) is the most widely used measure to quantify the strength and direction of the relationship between two variables, X and Y. It ranges from −1 to 1, where ρ = 1 signifies a perfect positive correlation—indicating that as one variable increases, the other increases proportionally. Conversely, ρ = −1 represents a perfect negative correlation, where an increase in one variable corresponds to a proportional decrease in the other. A value of ρ = 0 suggests no linear relationship between the variables, implying that changes in one do not systematically correspond to changes in the other.

In the context of credit rating prediction, financial variables such as net income and EBITDA or current ratio and quick ratio are often highly correlated. Retaining both in the dataset may introduce redundancy, inflating the feature space without improving predictive performance. To mitigate this issue, correlation analysis is applied to identify and eliminate

features with excessive collinearity, ensuring that only the most informative predictors are used in the classification model.

2.3.3 Feature Selection

The subsequent step in the proposed methodology (see Figure 1) is feature selection, which aims to identify and eliminate irrelevant and redundant information. This process reduces the dimensionality of the dataset, thereby improving computational efficiency and enhancing the performance of ML algorithms by focusing on the most informative features (Cheng, 2024). From an ML perspective, feature selection in classification tasks offers several key benefits. It enhances model accuracy by removing superfluous features, leading to more compact and computationally efficient models. Furthermore, it facilitates knowledge discovery by isolating the most influential variables, thereby improving interpretability and providing deeper insights into the underlying data structure (Büyükkeçeci; Okur, 2023).

Feature selection techniques are typically categorized into three classes based on their interaction with the classification model: filter methods, wrapper methods, and embedded methods (Al-shalif *et al*., 2024). Filter methods operate independently of the learning algorithm, selecting features based on intrinsic data characteristics during a pre-processing phase. Conversely, wrapper and embedded methods incorporate feature selection within the model training process. Wrapper methods evaluate various feature subsets using a search strategy informed by the model's predictive performance, whereas embedded methods perform feature selection intrinsically as part of the model's parameter optimization during training (Cheng, 2024).

This study employed wrapper methods due to their balance of interpretability and simplicity compared to embedded techniques. A commonly adopted search strategy within wrapper methods is greedy hill climbing, which iteratively modifies the current feature subset by either adding or removing one feature at a time. Variants of this strategy include forward selection, focusing exclusively on adding features, and backward elimination, which sequentially removes features. To overcome the limitations associated with these traditional approaches, recent advancements such as meta-heuristic algorithms (e.g., genetic algorithms, particle swarm optimization) and floating search methods have been introduced, enabling more flexible and dynamic inclusion and exclusion of features during the selection process (Al-shalif *et al.*, 2024; Nemati *et al*., 2024). Moreover, hybrid approaches that combine multiple feature

selection strategies have demonstrated promising results, achieving an optimal trade-off between computational efficiency and model performance (Nemati *et al*., 2024).

2.3.4 Classification

The proposed approach's final and most critical stage involves using classifiers to develop a classification model and assessing its performance through standard evaluation metrics, as explained in the next section.

2.3.4.1 Classification in Credit Rating Prediction

Classification is the process of predicting the target class of a given instance based on a set of input features. In financial applications, classification models are crucial in assessing credit risk, particularly in predicting corporate credit ratings, default probabilities, and borrower creditworthiness. For example, using financial information about a company, a classification model can predict the company's credit rating (the target class). In this context, the company is labeled with the value of the target class (e.g., credit rating). The term "label" will be used throughout this paper to refer to the target class, following standard ML terminology.

Building a classification model involves learning a mapping from input features to output labels. The training data used to develop the model contains labels corresponding to each training company, and the model's objective is to predict the label for input values corresponding to companies not included in the training data. A predefined set of features, such as profitability ratios, leverage, liquidity, and macroeconomic indicators, characterizes each company. However, financial classification tasks introduce unique challenges, including class imbalance, where highly rated firms significantly outnumber defaulted firms, and temporal dependencies, as creditworthiness evolves (Altman; Saunders, 1998; Duffie; Singleton, 2003).

Numerous classification algorithms are currently available, including traditional statistical methods such as LR and more advanced ML techniques like SVMs, NN, DT, GBM, and RF. However, it is impossible to definitively state that any algorithm is superior, as its effectiveness depends on the specific application and the nature of the dataset being used (Lessmann et al., 2015).

2.3.4.2 Evaluation Criteria

Evaluating classification models in credit risk applications requires the use of performance metrics tailored to financial decision-making. Common metrics include AUC-ROC, Gini coefficient, and F1 score, which help assess model discrimination power and balance between precision and recall. Furthermore, given the regulatory requirements in financial markets, model explainability remains a significant concern, particularly for ML techniques that operate as black-box models (Kuiper *et al*., 2022).

Despite advancements in ML, challenges remain in applying classification techniques to financial risk modeling. Regulatory compliance, explainability requirements, and the dynamic nature of financial markets necessitate careful feature selection and robustness testing.

2.4 EXPERIMENTS AND RESULTS

In Section 2.3.1, we describe our methodology (summarized in Figure 1), which leverages financial and business risk features to construct a training dataset. This dataset is then used to train a classification model designed to predict the credit rating of companies. To enhance classification performance, we apply data analysis techniques, such as feature correlation analysis and feature selection.

Section 2.4.1 provides a detailed overview of the training dataset, focusing on the specific features we collected. In Section 2.4.2, we discuss the classification models employed in our study, emphasizing their role in credit rating prediction. The results of our methodology are presented in Section 2.4.3, where we analyze feature correlations within the dataset and identify the most important features contributing to classification performance.

2.4.1 Training Dataset

We obtained the dataset presented in Table 2.1 from S&P Capital IQ Pro. The dataset comprises 51 features, including 43 financial features and 8 business features, and covers rated companies across 20 countries. We collected financial and qualitative variables from Capital IQ Pro and Bloomberg for 3,453 companies over the 2018–2024 period. We selected the features based on their availability and relevance to credit rating assessment, ensuring a comprehensive representation of firms' financial and business characteristics.

Table 2.1 – Financial features used in the dataset for credit rating prediction. These variables capture key aspects of company performance, such as liquidity, profitability, leverage, and cash flow, and were used as input for the machine-learning models.

Financial Features	
Accessible Cash	EBITDA Margin (%)
Adjusted Capital	Equity
ARO Debt Adjustment	FFO Cash Interest Coverage Ratio (x)
Capex/Revenues (%)	FFO/Debt (%)
Capital Expenditures	FOCF/Debt (%)
Capital Structure	Free Operating Cash Flow
Cash	FFO
Cash Flow from Operations	Gross Margin
Cash Interest Paid	Interest Expense
Cash Taxes Paid	Liquidity
CFO/Debt	Net Income
DCF/Debt	Pension and Postretirement Debt
Debt	R&D/Revenues
Debt/Debt and Equity	Reported Lease Liabilities
Debt/EBITDA	Return on Capital
Discretionary Cash Flow	Revenues
Dividends	SG&A/Revenue
EBIT	Share Repurchases
EBIT Interest Coverage	Total Operating Expense Before D&A
EBIT Margin	Volatility to Cash Flow/Leverage
EBITDA	Working Capital
EBITDA Interest Coverage	
Business features	
Distinctive Advantage	Governance
Country Risk	Operating Efficiency
Diversification	Size, Breadth, and Variety
Industry Risk	Volatility of Profitability

Source: Capital IQ Pro.

In this study, we utilize the full range of S&P Global credit rating grades, which includes 23 ordered levels, ranging from AAA (highest creditworthiness) to D (default). Market participants often group ratings of CCC+ and below due to their elevated default risk and financial vulnerability, reflecting a heightened probability of credit distress.

When dealing with categorical dependent variables such as credit ratings in ML models, each algorithm employs distinct techniques for processing these features. The following section

provides a detailed analysis of how LR, SVMs, NN, DT, GB, and RF handle credit ratings as a categorical target variable.

2.4.2 Classifiers

In this study, the target variable consists of multiple credit rating categories, necessitating the use of classifiers capable of handling multiclass classification. To ensure methodological diversity and capture different aspects of credit rating prediction, we selected six classifiers spanning linear models, tree-based models, and NNs.

Linear models, such as LR and SVMs, were chosen for their interpretability and ability to model direct relationships between financial variables and risk.

Tree-based models, including DT, RF, and GB, were selected for their ability to capture non-linear interactions, provide feature-importance insights, and maintain robustness when handling high-dimensional data.

Finally, NNs were incorporated due to their capacity to model highly complex, non-linear relationships without assuming a predefined decision boundary, making them well-suited for capturing intricate financial patterns.

The following section provides a discussion of each classification method, emphasizing their capability to handle multiclass classification.

2.4.2.1 Logistic Regression

LR, commonly used for binary classification, can be extended to multiclass prediction tasks, such as forecasting multiple credit ratings, through approaches like One-vs-Rest (OvR) or One-vs-One (OvO). However, when the categories exhibit a natural ranking (e.g., AAA > AA > A > BBB), ordinal LR is a more suitable choice.

As demonstrated by Goldmann, Crook and Calabrese (2024), ordinal LR explicitly accounts for the ordered structure of the dependent variable, thereby improving interpretability and producing more meaningful coefficient estimates compared to standard multinomial LR. This modeling approach ensures that the predicted probabilities adhere to the inherent ranking of credit ratings, leading to more accurate assessments in credit risk modeling.

2.4.2.2 Support Vector Machines

SVMs, originally designed for binary classification, have been effectively extended to handle multiclass classification through strategies such as OvO and OvR. These methods enable SVMs to manage complex rating systems involving multiple classes, ensuring scalability and reliable classification performance. Kurbakov and Sulimova (2024) provide a comprehensive comparative study on multiclass extensions for SVMs, demonstrating their effectiveness and practical applicability across diverse classification tasks. Their proposed Dual-Layer Smart Sampling Support Vector Machine (DLSS-SVM) method enhances scalability and efficiency without compromising accuracy, reflecting contemporary advancements in multiclass SVM methodologies.

2.4.2.3 Neural Networks

NNs offer substantial flexibility in handling both binary and multiclass categorical outcomes, such as credit ratings. In multiclass classification, the SoftMax activation function is commonly applied in the output layer to produce a probability distribution over possible classes. Categorical variables are typically represented using one-hot encoding, where each category is converted into a binary vector. However, for high-cardinality categorical variables, embedding layers provide a more efficient alternative by mapping categories into continuous vector spaces. As highlighted by Cao (2024), embedding models efficiently transform sparse, high-dimensional inputs into dense, low-dimensional representations, thereby enhancing both memory efficiency and learning performance.

2.4.2.4 Decision Trees

DT inherently support multiclass classification as they recursively split data at each node, accommodating multiple class labels. Their flexibility in handling both categorical and continuous features make them useful in applications like credit rating prediction. Mienye and Jere (2024) emphasize that DT are a versatile and foundational method for classification, demonstrating significant effectiveness in managing complex decision-making tasks across various domains.

2.4.2.5 Gradient Boosting

GB can be extended to multiclass classification by adapting boosting algorithms to optimize loss functions for multiple target categories. Friedman (2001) introduced GBM and discussed their potential extensions beyond binary classification. This theoretical framework has been successfully operationalized in modern implementations such as XGBoost (Chen & Guestrin, 2016), which extends gradient boosting to multiclass problems by iteratively minimizing a multiclass loss function and estimating class probabilities through the SoftMax framework.

2.4.2.6 Random Forests

RF inherently supports multiclass classification by constructing an ensemble of decision trees trained on bootstrap samples of the data. Each tree recursively partitions the feature space by selecting splits that maximize class separation according to impurity-based criteria, such as the Gini Index or Entropy, while the final prediction is obtained through majority voting across all trees (Breiman, 2001). This approach has been further exemplified in recent applications within intrusion detection systems (Alharthi, Alaryani, & Kaddoura, 2025).

2.4.3 Results and Discussion

As described in Section 2.3.1, not all features are relevant when constructing a model since sometimes they represent redundant information. Hence, two experiments were carried out: in Section 2.4.3.1, the model was tested after removing features that were highly correlated (redundant), whereas in Section 2.4.3.2, the model was tested using only a selected subset of the original features. This experiment aimed to determine whether a smaller but more relevant set of features could still provide good results.

2.4.3.1 Feature Correlation Analysis

To enhance model performance and reduce redundancy, we conducted a correlation analysis to identify highly correlated features. High correlation among independent variables can introduce multicollinearity, leading to unstable model estimates, increased computational complexity, and a higher risk of overfitting.

To mitigate these issues, we identified and excluded features with high pairwise correlation, ensuring that only the most informative variables were retained. This feature selection step aims to improve the model's generalization ability while maintaining computational efficiency.

2.4.3.1.1 Identification of Highly Correlated Features

We used the Pearson correlation coefficient to evaluate linear relationships among features and identify potential redundancy within the dataset.

We used a three-step feature correlation analysis to assess and mitigate redundancy in the dataset,

1) Identification of Top 10 Correlated Features—For each numerical feature, we identified the ten features with the highest absolute correlation values, considering both positive and negative correlations. This process enabled the selection of the most strongly related feature pairs, helping to pinpoint variables that contributed to potential multicollinearity;
2) Aggregation of Highly Correlated Features—From the identified sets of highly correlated features, we determined which variables consistently appeared among the top correlated features across multiple other variables. For example, if a given feature was ranked among the top 10 correlated features for at least 90% of other variables, it was flagged as a candidate for exclusion to reduce redundancy;
3) Computation of Mean Correlation Scores—For the aggregated set of high-correlation candidates, we computed the mean absolute correlation score across all pairwise correlations. This metric provided a quantitative measure of each feature's overall redundancy within the dataset, supporting an objective feature selection process.

The analysis produced a refined set of features that exhibited consistently high correlations with other variables, indicating potential redundancy. Table 2.2 presents the most frequently correlated features along with their mean absolute correlation scores, providing a quantitative assessment of their degree of redundancy.

Table 2.2 – Features with the highest mean absolute correlations, indicating potential redundancy in the dataset.

Feature	Mean Absolute Correlation
EBITDA	0.81
Interest Expense	0.74
FFO	0.73
Net Income	0.69
Working capital	0.69
Cash flow from operations	0.65
Revenues	0.63
Competitive Advantage	0.56
Free operating cash flow	0.53
Capital expenditures	0.47

Source: Own authorship.

Following the statistical analysis, domain experts evaluated the candidate features to ensure alignment with the dataset's business objectives. To enhance the predictive model's efficiency and interpretability, redundant features were identified and eliminated through Pearson correlation analysis. Pearson's correlation coefficient quantifies the linear association between two variables, with values ranging from −1 (perfect negative correlation) to 1 (perfect positive correlation). Features exhibiting a strong absolute correlation (e.g., |r| between 0.5 and 0.7) were considered for removal due to their potential to introduce multicollinearity, increase computational complexity, and reduce model generalizability by conveying overlapping information. However, the selection of an appropriate correlation threshold for feature elimination must be context-dependent, as moderate correlation does not always imply redundancy, particularly in complex predictive models where interactions between variables may hold informational value. Thus, domain expertise and additional statistical assessments, such as variance inflation factor (VIF) analysis, were employed to ensure that the feature selection process preserved essential explanatory power while mitigating multicollinearity concerns.

Consequently, Table 2.3 presents features that were excluded due to having a VIF greater than 10:

Table 2.3 – Variance inflation factor (VIF) values showing multicollinearity among selected features.

Variable	VIF
EBITDA	49.5
FFO	46.2
Net Income	10.2
Free operating cash flow	14.1
Capital expenditures	20.7

Source: Own authorship.

By integrating both statistical correlation analysis and domain-specific expertise, the feature selection process was refined to enhance model robustness. The elimination of highly correlated variables mitigates the risk of overfitting, improves model interpretability, and enhances computational efficiency.

2.4.3.2 Credit Rating Analysis

In this experiment, we analyze the impact of feature selection on model performance. While Section 2.4.3.1 focused on eliminating highly correlated features, this section evaluates how removing low-importance features affects classification accuracy.

The dataset used for credit rating prediction, presented in Table 2.4, comprises 27,664 instances collected from 20 countries. However, the distribution of observations is highly imbalanced, with the United States contributing 16,374 instances (59% of the dataset), while Switzerland accounts for only 257 instances (1%). This imbalance extends to the distribution of credit ratings across countries, leading to significant variations in rating proportions. Such disparities pose challenges for model generalization and predictive performance.

To mitigate these issues and improve model robustness, a country-specific modeling approach was adopted. This strategy accounts for regional differences in economic conditions, regulatory environments, and credit risk factors, thereby enhancing the model's ability to capture country-specific patterns in credit ratings.

Table 2.4 – Distribution of credit rating observations by country and S&P rating categories from AAA to D.

S&P Ratings Grid	A	A-	A+	AA	AA-	AA+	AAA	B	B-	B+	BB	BB-	BB+	BBB	BBB-	BBB+	CC	CCC	CCC-	CCC+	D	SD	Total
Australia	32	144	8			8	8	14	23	8	14	8	24	136	56	132							615
Brazil								8	16	8	64	144	64	8	64				8	8	24		416
Canada	45	96	8		8			191	101	118	48	96	88	128	152	88		8		72	16		1263
Cayman Islands	24	32	16					8	31	16		36	8	40	32	48				8			299
China	71	135	72					8		8	8		24	152	96	96							670
France	48	105	8	16	16			275	182	88	39	63	72	72	112	127		8		24			1255
Germany	48	32	16		8			139	70	69	16	40	56	104	95	80	8	8		27	8		824
Hong Kong	32	56	40		8	24				5		5		16	40	40							266
Ireland	24	32	8		8			24	24	32	31	16	16	24	32	40				8			319
Italy		8						112	77	22	24	24	48	64	32	32				16			459
Japan	144	64	104		24					8	9	8	32	32		48				8			481
Korea, Republic of		31	8	104	8			8			8			32	32	38							269
Luxembourg		8						150	104	39	40	32	40	24	24	40	8	16		16		7	548
Mexico		24						24	8	8	8	24	36	72	56	32				8			300
Netherlands	24	40	16		8	8		174	39	56	24	40	16	88	55	56		16	8	72			740
Spain		8						69	49	31	16	32	16	56	80	24				16		16	413
Sweden	16	40	16					61	8	16	16	14	8	48	64	64				8			379
Switzerland	8	38	24	8	16			24	8		8	8	8	22	16	48				21			257
United Kingdom	40	48	24					239	286	122	88	72	63	175	103	129		40	8	56	16	8	1517
United States	400	896	206	16	120	16	16	2228	2787	1433	1047	1137	908	1520	864	1256		331	120	1008	24	41	16,374

Source: Own authorship.

2.4.3.2.1 Rating Categorization and Balance Improvement

The original credit ratings were highly detailed, leading to an uneven distribution of data across rating categories. This imbalance could negatively impact the analysis by reducing statistical reliability. To address this, we grouped the ratings into four broader categories, ensuring a more balanced distribution while preserving the essential distinctions needed for meaningful interpretation. Table 2.5 presents the grouped ratings by category.

Table 2.5 – Grouped S&P credit ratings: Class A (highest), B and C (intermediate), and D (lowest/default).

Credit Ratings Classes	S&P Global's Rating Grade
Class A:	AAA, AA+, AA, AA-, A+, A, A-
Class B:	BBB+, BBB, BBB-
Class C:	BB+, BB, BB-
Class D:	B+, B, B-, CCC+, CCC, CCC-, CC, C, SD, D

Source: S&P Global (2024).

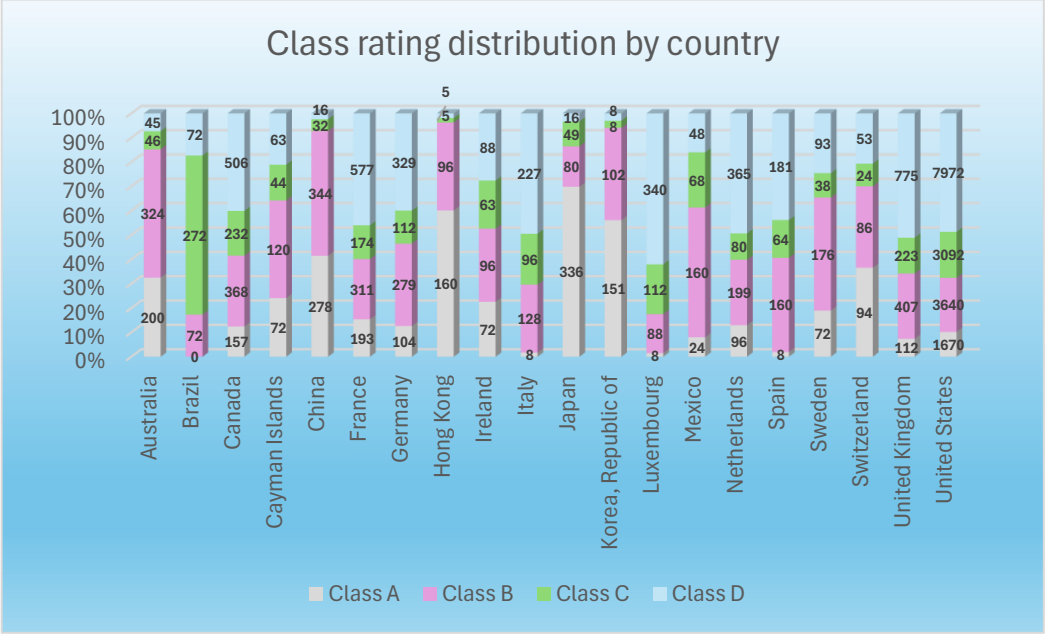
Figures 2.2 presents a stacked bar chart illustrating the distribution of Class A, B, C, and D ratings across various countries, showing the relative proportion of each rating within each country.

One of the most striking observations is that the United States has the highest number of Class D ratings, 7972, indicating that a large share of rated entities in the country fall into the lowest rating category. On the other hand, countries such as France, Germany, and the United Kingdom exhibit a strong presence of Class A ratings, highlighting a concentration of highly rated entities. Similarly, Japan and Canada also have a significant proportion of Class A ratings, reinforcing their position as countries with a higher credit quality profile.

Meanwhile, countries such as Brazil, the Cayman Islands, and Mexico have a greater proportion of Class B and Class C ratings, indicating a mid-tier credit profile. In contrast, Germany, China, and Hong Kong demonstrate a more balanced distribution across Classes A, B, C, and D, suggesting a diversified risk profile.

Class D ratings are also notably significant in several other countries besides the United States. Countries such as Switzerland, Spain, and Italy exhibit a visible portion of Class D ratings, suggesting a notable presence of lower-rated entities. Likewise, the United Kingdom also has a considerable number of Class D ratings, 775, indicating that while some entities are highly rated, a significant portion falls within the lower classification.

Figure 2.2 – Class rating distribution by country.



Source: Own authorship.

2.4.3.2.2 Justification for Country-Specific Models

Given the heterogeneity of economic environments and the varying distribution of credit ratings across countries, training a single global model may result in suboptimal performance. A universal model could struggle to capture the complex, country-specific relationships embedded within the data. To mitigate this limitation, we implemented a country-specific modeling approach, developing separate models for each of the 20 countries in the dataset.

This methodology offers several advantages:

- Enhanced Data Homogeneity: Training models on country-specific datasets ensures greater uniformity in economic conditions, reducing variability introduced by cross-country differences and improving model reliability;
- Improved Predictive Accuracy: By tailoring models to individual economic contexts, this approach enhances predictive performance by mitigating confounding effects inherent in aggregated datasets;
- Scalability and Adaptability: Independent models enable efficient updates and fine-tuning as new country-specific data becomes available, supporting both scalability and continuous improvement.

Each model was trained using the respective country's data subset, underscoring the value of localized insights in credit rating prediction.

2.4.3.3 Credit Rating Prediction

To assess the predictive performance of our models, we utilized the Scikit-learn library in Python 3.10 and evaluated five ML algorithms: LR, RF, GB, SVMs, and ANN. Each algorithm was initially implemented using default hyperparameter configurations. Model evaluation was performed through five repetitions of 10-fold cross-validation, a widely adopted technique to ensure robust and reliable performance assessment while mitigating potential biases due to data partitioning.

DT were initially included in the model evaluation process. However, they consistently exhibited lower accuracy and generalization ability compared to ensemble methods such as RF and GB. Given their comparatively weaker performance, they were excluded from the final results table for brevity.

The results, summarized in Table 2.6, report the mean accuracy scores for each algorithm across the 20 country-specific datasets. In most cases, the differences in model performance were marginal. However, notable variations were observed in the datasets from the United States, United Kingdom, France, and Australia, where ANN, RF, and GB consistently outperformed the other algorithms. In contrast, LR and SVMs demonstrated comparable performance but generally yielded lower accuracy in these datasets relative to the more complex models.

Table 2.6 – Mean accuracy of five machine-learning algorithms—ANN, GB, Logistic Regression, RF, and SVM—across 20 country-specific credit rating datasets, based on five repetitions of 10-fold cross-validation. Higher values indicate better predictive performance.

(continues)

Nº	Country	ANN	GB	Logistic Regression	RF	SVM
1	Australia	0.96	0.96	0.70	0.97	0.76
2	Brazil	0.99	0.99	0.91	0.99	0.88
3	Canada	0.98	0.97	0.70	0.96	0.73
4	Cayman Islands	0.99	0.99	0.92	0.99	0.87
5	China	0.99	0.95	0.87	0.98	0.83
6	France	0.97	0.97	0.78	0.96	0.78
7	Germany	0.97	0.95	0.80	0.96	0.75
8	Hong Kong	0.98	0.97	0.97	0.99	0.90
9	Ireland	0.99	0.98	0.91	0.99	0.89
10	Italy	0.99	0.94	0.93	0.94	0.94
11	Japan	0.97	0.96	0.92	0.98	0.87
12	Republic of Korea	0.99	0.98	0.99	0.97	0.95

Table 2.6 – Mean accuracy of five machine-learning algorithms—ANN, GB, Logistic Regression, RF, and SVM—across 20 country-specific credit rating datasets, based on five repetitions of 10-fold cross-validation. Higher values indicate better predictive performance.

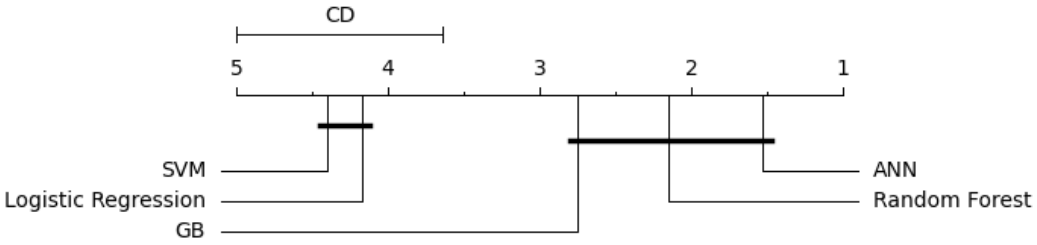
(conclusion)

Nº	Country	ANN	GB	Logistic Regression	RF	SVM
13	Luxembourg	0.98	0.95	0.87	0.91	0.87
14	Mexico	0.99	0.97	0.95	0.99	0.96
15	Netherlands	0.99	0.97	0.88	0.98	0.85
16	Spain	0.98	0.96	0.92	0.98	0.94
17	Sweden	0.98	0.97	0.92	0.98	0.94
18	Switzerland	0.99	0.95	0.99	0.97	0.98
19	United Kingdom	0.97	0.96	0.72	0.94	0.74
20	United States	0.86	0.74	0.42	0.86	0.54

Source: Own authorship.

To assess whether the observed performance differences among the algorithms were statistically significant, we employed the methodology outlined by Demšar (2006). First, we applied the Friedman test at a significance level of 0.05 to evaluate the null hypothesis that "there is no statistical difference between the algorithms" across country-specific datasets. In cases where the null hypothesis was rejected, we conducted the Nemenyi post hoc test to identify specific pairwise differences, as shown in Figure 2.3.

Figure 2.3 – Results of the Friedman and Nemenyi tests comparing the predictive performance of machine-learning algorithms across country-specific datasets. Statistically significant differences indicate superior performance by specific models.



Source: Own authorship.

The Friedman test results indicated significant differences in algorithm performance for several countries, particularly those with larger datasets or more complex distributions of credit ratings (e.g., the United States, the United Kingdom, and France). For these datasets, the Nemenyi post hoc test revealed that ANN, GB, and RF consistently formed a superior group, significantly outperforming LR and SVMs.

2.5 RESULTS

The findings reveal distinct trends regarding the predictive performance of ML models in credit rating classification:

- ANN: ANN consistently achieved the highest performance across countries, with accuracy scores ranging from 0.86 to 0.99. For instance, Brazil exhibited an accuracy of 0.99, suggesting that ANN models effectively capture financial patterns in complex datasets;
- GB: GB demonstrated strong predictive performance, frequently ranking second to ANN. Countries such as Australia 0.96 and Canada 0.97) underscore the robustness of ensemble methods in financial prediction tasks;
- LR: While exhibiting slightly lower accuracy compared to ensemble methods, LR performed consistently well, serving as a strong baseline model. For example, Canada recorded an accuracy of 0.70;
- RF and SVM: The performance of RF varied, with accuracy ranging from 0.86 to 0.99. Although SVM remained competitive, it frequently recorded lower accuracy compared to ANN, GB, and RF.

These results establish a clear hierarchy in predictive power, with ANN and ensemble methods demonstrating superior performance, while LR provides a reliable benchmark. The findings further suggest that dataset size and complexity influence model effectiveness, as evidenced by the diminished statistical significance in countries with smaller datasets.

An important methodological aspect of this study is that all ML models were applied using their default hyperparameter settings. This choice ensured uniformity across different algorithms and aligns with common practice in applied ML research, where default configurations are often employed as a baseline for initial performance assessment (Biecek; Burzykowski, 2021). Nevertheless, recent research emphasizes that systematic hyperparameter optimization—including the adjustment of learning rates, model depth, and regularization parameters—can substantially improve both the accuracy and stability of predictive models (Akiba *et al*., 2019; Probst; Wright; Boulesteix, 2019; Zhang *et al*., 2021). Future studies should, therefore, consider incorporating automated hyperparameter tuning techniques, such as Bayesian optimization or evolutionary algorithms, to further enhance the predictive capabilities and generalizability of ML models applied to corporate credit rating prediction.

2.6 DISCUSSION

The results presented in this study demonstrate the predictive performance of various ML models—ANN, GB, LR, RF, and SVMs—across different countries. A comparative analysis of these findings within the context of existing literature on predictive modeling in financial studies yields several key insights.

ML in Financial Prediction: Prior research (e.g., Lessmann *et al*., 2015) has demonstrated that advanced models, particularly ANN and ensemble methods (RF, GB), frequently outperform traditional approaches such as LR in financial prediction tasks. Similarly, Lokanan and Ramzan (2024) highlight the strong predictive capabilities of ANN in financial distress modeling, reinforcing the advantages of advanced ML techniques in financial forecasting. These findings are consistent with the results of this study, where ANN and GB demonstrated superior predictive accuracy, particularly in countries such as Brazil and Australia, where accuracy scores exceeded 0.95.

Performance of ANN: The consistently high performance of ANNs observed across multiple datasets aligns with existing research, highlighting their capability to model complex, non-linear relationships in financial data. Makridakis, Spiliotis and Assimakopoulos (2018) demonstrated that ML methods, including NN, significantly enhance forecasting accuracy across various domains, as evidenced by the M4 Competition results. The superior accuracy of ANNs in this study further substantiates their suitability for tasks such as credit rating prediction and financial forecasting.

Traditional vs. Modern Predictive Techniques: While LR remains a widely used benchmark due to its simplicity and interpretability, its performance, as observed in this study, was generally lower than that of ensemble-based models. These findings align with those of Lessmann *et al.* (2015) and Ling and Wang (2024), who demonstrated that LR, while effective in credit-scoring models, is often surpassed by more sophisticated techniques, particularly ensemble-based models. The improved performance of RF and GB suggests that ensemble learning methods, which integrate multiple weak learners, enhance predictive power by capturing intricate patterns within financial data.

Cross-Country Variability in Model Performance: Variations in model performance across different countries—such as the notably high accuracy of ANN and GB in Brazil—can be attributed to disparities in financial markets, economic structures, and data characteristics. This variability underscores the necessity of context-specific model evaluation, as regional

differences in macroeconomic indicators, market volatility, and data availability significantly influence predictive model effectiveness.

For instance, Pagliaro (2025) emphasizes that ML's predictive power depends on varying market efficiencies, with AI able to exploit temporary inefficiencies in certain regions. He also highlights that incorporating macroeconomic and contextual variables improves forecasting, particularly in emerging markets with distinct financial structures.

2.6.1 Interpretability Considerations

This study has demonstrated the superior predictive accuracy of ML algorithms, particularly ANN and GB, in corporate credit rating prediction. However, an equally critical aspect of practical deployment is model interpretability. In financial applications, where credit ratings influence decisions related to capital allocation, regulatory compliance, and stakeholder confidence, the ability to explain model outputs is indispensable. A well-documented challenge in ML is the inherent trade-off between model complexity and interpretability (Lipton, 2018). While complex models like ANN and GB excel at capturing non-linear relationships within financial datasets, they often function as "black box" models, providing limited transparency about how predictions are derived. In contrast, simpler models such as LR or DT offer more transparent decision processes but may underperform in complex contexts like those analyzed in this study. Interpretability is essential for various reasons: regulatory compliance requires transparency and explainability (e.g., European Banking Authority, 2021); model risk management relies on interpretability for effective validation and bias detection (SR 11-7); and stakeholder trust in credit ratings depends on confidence in model outputs. Moreover, interpretability aids in identifying and mitigating bias, thereby promoting fair and equitable outcomes (Doshi-Velez; Kim, 2017). Several approaches can enhance interpretability without sacrificing predictive power, including post hoc explainability tools like SHAP and LIME, model simplification through surrogate models, or adopting inherently interpretable models such as Generalized Additive Models (GAMs) or Explainable Boosting Machines (EBMs) (Caruana *et al*., 2015). Both global (overall behavior) and local (individual prediction) explanations are crucial in financial decision-making. Although this study primarily focused on benchmarking predictive accuracy, future research should incorporate explainability techniques to improve model transparency, regulatory acceptability, and practical utility in corporate credit risk assessment.

2.6.2 Regulatory Considerations

While this study highlights the significant predictive advantages of ML algorithms in corporate credit rating prediction, it is essential to consider the regulatory implications associated with deploying such models in financial risk management. Regulatory authorities, including the Basel Committee on Banking Supervision (2022), the European Banking Authority (2021), and national regulators such as the Federal Reserve (SR 11-7), have articulated clear expectations regarding the transparency, interpretability, and governance of risk models.

One of the primary concerns is the opacity of complex ML models, often referred to as "black box" algorithms. Models such as ANN and GB, while demonstrating superior predictive accuracy in this study, lack the intrinsic interpretability required by regulatory frameworks. Financial institutions are expected to justify and explain credit decisions, especially when these decisions impact capital adequacy and loan origination practices. To align with regulatory expectations for explainability, techniques such as SHAP and LIME can be employed to provide transparent and understandable insights into model predictions (Rudin, 2019).

Additionally, regulators emphasize the need for robust Model Risk Management frameworks. According to the Federal Reserve's SR 11-7 guidance, models must undergo comprehensive validation, back-testing, and stress testing to ensure their reliability under varying economic conditions. This requirement is particularly pertinent for ML models, which may be prone to overfitting or performance degradation in the presence of data shifts or regime changes in financial markets.

Data governance and bias mitigation represent further regulatory priorities. ML models are highly dependent on the quality and representativeness of input data. Poor data governance can lead to biased outcomes, undermining both model fairness and regulatory compliance. For instance, the proposed European Union's Artificial Intelligence Act mandates that high-risk AI systems, including those used in financial services, be subject to rigorous oversight to prevent discriminatory impacts. Thus, careful data pre-processing, feature selection, and bias audits are necessary steps when developing ML-based credit rating models.

Moreover, financial regulations such as Basel III link credit risk assessments directly to capital requirements, underscoring the importance of using models that are not only accurate but also compliant with regulatory standards. The EBA Guidelines on Loan Origination and Monitoring (European Banking Authority, 2021) explicitly recommend that models used for credit decisions be auditable, transparent, and capable of supporting sound credit risk practices.

In light of these considerations, while ML offers substantial potential to improve the efficiency and accuracy of credit rating prediction, its application in regulated financial environments must be accompanied by adherence to regulatory requirements for transparency, governance, and fairness. Future research should focus on the integration of XAI methods and the development of hybrid models that balance predictive power with interpretability. Additionally, collaboration between researchers, financial practitioners, and regulators is necessary to establish best practices for the safe and compliant deployment of ML in financial risk management.

2.7 CONCLUSIONS

This study evaluated the predictive performance of five ML models across multiple countries, yielding several key insights relevant to financial forecasting.

Predictive Performance Hierarchy: ANN consistently achieved the highest accuracy, reinforcing their capacity to model complex, non-linear relationships in financial data. Ensemble methods, including GB and RF, demonstrated strong predictive capabilities, further validating their effectiveness in financial applications.

Robustness of LR: Despite the increasing adoption of advanced ML techniques, LR exhibited competitive accuracy, underscoring its continued relevance as a benchmark model in financial prediction.

Cross-Country Variability: The observed variation in model performance across different countries highlights the potential influence of regional economic structures, financial market dynamics, and data characteristics on predictive accuracy.

These findings align with existing literature, supporting the superiority of modern ML approaches over traditional statistical methods for financial predictions. Future research should focus on improving model interpretability and incorporating additional economic indicators to further enhance predictive performance and practical applicability in diverse financial contexts.

3 EXPLAINING CORPORATE RATINGS TRANSITIONS AND DEFAULTS THROUGH MACHINE LEARNING

ABSTRACT

Credit rating transitions and defaults are critical indicators of corporate creditworthiness, yet their accurate modeling remains a persistent challenge in risk management. Traditional models such as logistic regression (LR) and structural approaches (e.g., Merton's model) offer transparency but often fail to capture nonlinear relationships, temporal dynamics, and firm heterogeneity. This study proposes a hybrid machine learning (ML) framework to explain and predict corporate rating transitions and defaults, addressing key limitations in existing literature. We benchmark four classification algorithms—LR, RF, XGBoost, and Support Vector Machines (SVM)—on a structured corporate credit dataset. Our approach integrates segment-specific modeling across rating bands, out-of-time validation to simulate real-world applicability, and SHapley Additive exPlanations (SHAP) values to ensure interpretability. The results demonstrate that ensemble methods, particularly XGBoost and RF, significantly outperform LR and SVM in predictive accuracy and early warning capability. Moreover, SHAP analysis reveals differentiated drivers of rating transitions across credit quality segments, highlighting the importance of tailored monitoring strategies. This research contributes to the literature by bridging predictive performance with interpretability in credit risk modeling and offers practical implications for regulators, rating agencies, and financial institutions seeking robust, transparent, and forward-looking credit assessment tools.

Keywords: credit rating transitions; machine learning; default prediction; explainable AI (SHAP).

3.1 INTRODUCTION

Corporate credit ratings are central to modern financial systems, influencing investment decisions, debt pricing, and regulatory capital requirements. Beyond their static role, rating transitions and defaults reflect the dynamic evolution of firm risk, particularly during periods of macroeconomic stress or structural disruption. Accurately modeling and explaining these transitions is vital for investors, regulators, and institutions seeking forward-looking insights into credit deterioration.

Traditionally, credit risk models have relied on structural frameworks such as Merton's (1974) option-theoretic approach and statistical models like LR (Altman, 1968; Ohlson, 1980). While theoretically sound, these models assume linearity, firm homogeneity, and stable relationships between financial variables—assumptions that are increasingly challenged by the complexity of real-world credit behavior. Moreover, reliance on risk-weighted assets (RWAs) under Basel frameworks has been criticized for encouraging regulatory arbitrage and weakening alignment between modeled and actual risk (Böhnke *et al*., 2023).

Recent advances in ML offer promising alternatives. ML models such as RF, XGBoost, and LSTM networks can detect nonlinearities, handle high-dimensional data, and adapt to changing environments. For instance, Kandi and García-Dopico (2025) showed that LSTM outperforms XGBoost in imbalanced datasets by capturing temporal dependencies without complex feature engineering—an important benefit in financial sequence data. However, many ML models remain opaque, hindering their adoption in regulated settings.

To address this, XAI techniques such as SHAP and LIME are gaining traction. Nallakaruppan *et al.* (2024) demonstrated that integrating these tools into financial models supports both interpretability and regulatory compliance. Despite these developments, key gaps persist: few studies apply segment-specific modeling across rating categories (Bitetto; Filomeni; Modina, 2025), and many overlook firm-level disclosure complexity, which increases informational asymmetry (Dang *et al.*, 2025).

This study proposes a hybrid ML framework to explain corporate rating transitions and defaults. We stratify firms into rating segments, apply out-of-time validation to simulate real-world performance, and use SHAP values to interpret model outputs. By benchmarking LR, RF, XGBoost, and SVM, we contribute a robust, explainable, and segment-aware EWS for credit risk management.

3.2 THEORETICAL BACKGROUND AND LITERATURE REVIEW

3.2.1 Traditional Approaches to Credit Risk Modeling

Traditional credit risk assessment relies on structural and statistical models grounded in economic theory. Merton's (1974) structural model conceptualizes a firm's equity as a European call option on its assets, with default occurring when asset value falls below debt obligations at maturity. Although this model offers a theoretical link between default

probability, asset volatility, and capital structure, its practical use is limited by the need for unobservable inputs and strong assumptions about market efficiency and continuous trading.

Due to these limitations, empirical research has increasingly favored statistical models based on observable firm-level data. LR gained prominence for its interpretability and operational simplicity. Foundational studies by Altman (1968) and Ohlson (1980) demonstrated the predictive power of financial ratios for bankruptcy and default, laying the groundwork for early credit scoring systems used in academic and regulatory contexts.

However, logistic models have important limitations. They assume linear and additive relationships between predictors and default probability and often presume homogeneity across firms and conditions. As Böhnke *et al.* (2023) point out, reliance on internally developed RWA models under Basel II and III can lead to regulatory arbitrage and a weakening link between modeled risk and actual exposures. These models also struggle to capture dynamic risk profiles and economic shocks.

Structural models have been further criticized for ignoring stochastic debt issuance. Feldhütter and Schaefer (2023) provide evidence that short-term leverage volatility often exceeds asset volatility, contradicting assumptions of constant debt levels. This highlights the need for models that incorporate evolving capital structures.

While LR remains common due to its transparency and ease of use, it is increasingly outpaced by more flexible ML methods. Zedda (2024) confirms that although logistic models are still used in regulatory settings, they underperform in capturing nonlinear interactions in firm data. These shortcomings have paved the way for advanced approaches better suited to modeling heterogeneity and dynamic credit behavior, as explored in the next section.

3.2.2 Evolution to Machine learning in Credit Risk

Limitations of traditional models have accelerated the adoption of ML techniques in credit risk assessment. Unlike LR, which assumes linearity, ML models can capture nonlinear relationships and adapt to changing economic conditions. With fewer parametric assumptions, ML techniques can extract patterns from high-dimensional data, improving prediction of defaults and rating transitions (Wang *et al*., 2022).

A key strength of ML is its ability to handle data heterogeneity and class imbalance, which are common in credit datasets where defaults are relatively rare. Ensemble models, in particular, have shown strong classification performance and robustness to noise and imbalanced class distributions. Techniques such as Synthetic Minority Oversampling

Technique (SMOTE) further enhance generalizability by balancing training data without distorting distributions (Elreedy; Atiya; Kamalov, 2024). In personal credit contexts, Wang *et al*. (2022) show that XGBoost outperforms traditional models in assessing borrower risk with high precision.

Empirical evidence highlights the strengths of ML in credit risk modeling. Xu and Zhang (2022) demonstrated that combining SVM with ensemble methods improves classification performance in the technology sector, showcasing the adaptability and effectiveness of hybrid ML models in corporate credit scoring.

AutoML platforms like H2O-AutoML and AutoGluon lower technical barriers and streamline deployment. Papík and Papíková (2025) show these tools deliver superior area under the curve (AUC) scores and faster model development in bankruptcy prediction, particularly in data-constrained environments like Slovak manufacturing firms. Similar findings are echoed by Papík and Papíková (2024), who highlight the operational feasibility of AutoML systems for practitioners with limited modeling expertise.

To address concerns about model opacity, XAI tools have become indispensable. SHAP stands out as a leading method for interpreting ML outputs. Hamida *et al*. (2024) highlight the importance of explainability in high-stakes domains, emphasizing that XAI enhances transparency and fosters stakeholder trust in AI-driven decision systems.

Hybrid approaches are also emerging. Sun *et al.* (2025) propose an interpretable SHAP-based framework for macroeconomic forecasting, incorporating news narrative sentiment, with applications in credit cycle modeling. Tribuvan *et al*. (2025) further demonstrate that combining feature selection with ensemble classifiers enhances model precision in default prediction across diverse financial datasets.

3.2.3 Credit Ratings, Default Events, and the Institutional Nature of Credit Risk Signals

Credit ratings play a pivotal role in modern financial markets by providing standardized assessments of corporate creditworthiness. These assessments influence borrowing costs, shape investment strategies, and affect regulatory capital requirements across global financial systems (Boot *et al*., 2006). Because credit ratings influence both market perceptions and regulatory frameworks, understanding the determinants of rating transitions has become a central objective in the literature on credit risk modeling and financial intermediation.

Traditionally, empirical research in credit risk has focused on predicting either corporate default events or credit rating changes using statistical models based on firm-level financial

indicators. Early approaches relied heavily on accounting-based ratios and linear econometric frameworks, such as discriminant analysis and logistic regression (Altman, 1968; Ohlson, 1980). Structural models later introduced a theoretical framework linking default risk to market values of firm assets (Merton, 1974). More recently, advances in machine learning have enabled researchers to capture nonlinear interactions among financial variables, significantly improving predictive performance in credit risk applications (Lessmann *et al*., 2015; Barboza; Kimura; Altman, 2017).

Despite these methodological advances, an important conceptual distinction is often overlooked in empirical studies: the difference between corporate default events and credit rating transitions. While both phenomena are related to credit risk, they represent fundamentally different types of outcomes. Corporate default is an objective economic event that can be directly observed and verified. Credit ratings, by contrast, are institutional judgments produced by credit rating agencies through structured analytical processes that combine quantitative models, qualitative evaluation, and expert deliberation.

This distinction has significant implications for predictive modeling. When researchers attempt to predict rating transitions, they are not merely estimating the underlying economic probability of default. Instead, they are modeling the likelihood that a credit rating agency will revise its assessment of a firm's creditworthiness based on available information. Consequently, predictive models trained on rating transitions capture both firm-level economic risk and the interpretative logic of credit rating agencies.

Recognizing this institutional dimension is crucial for interpreting empirical results. Rather than treating credit ratings as purely objective measurements of credit risk, they should be conceptualized as signals produced by financial information intermediaries (Boot *et al*., 2006; White, 2010). From this perspective, machine learning models trained on rating data can be interpreted as learning patterns not only in firm fundamentals but also in how rating agencies interpret these fundamentals when forming credit assessments.

This perspective aligns with insights from institutional theory, which emphasize how organizations adopt widely accepted practices and signals to maintain legitimacy within organizational fields (DiMaggio; Powell, 1983; Zuckerman, 1999). Building on this theoretical foundation, the present chapter develops a conceptual framework that distinguishes between objective credit events and institutional risk assessments, and discusses the implications for empirical modeling of rating transitions using machine learning techniques.

3.2.3.1 Corporate Default as an Objective Credit Event

Corporate default represents the most extreme realization of credit risk. In financial markets, default occurs when a borrower fails to fulfill its contractual debt obligations, typically through the nonpayment of principal or interest, bankruptcy filings, or formal debt restructuring agreements (Duffie; Singleton, 2003). Because these events are legally defined and economically observable, they constitute objective indicators of financial distress.

The modeling of default risk has long been a central topic in financial economics. Early empirical work focused on identifying financial ratios capable of predicting bankruptcy. One of the most influential contributions in this area is Altman's Z-score model, which used multiple discriminant analysis to combine accounting-based indicators into a single measure of bankruptcy risk (Altman, 1968). Subsequent research extended this approach through logistic regression models designed to estimate default probabilities using firm-level financial characteristics (Ohlson, 1980).

Structural models provided an alternative theoretical framework for understanding default risk. In Merton's (1974) model, corporate equity is interpreted as a call option on the firm's assets, and default occurs when the value of the firm's assets falls below the value of its liabilities. This framework established a direct link between market variables and default probabilities, thereby integrating credit risk modeling with option pricing theory.

Later developments introduced reduced-form models that treat default as a stochastic process influenced by macroeconomic conditions and firm-specific characteristics (Jarrow; Turnbull, 1995). These models allowed researchers to estimate default intensities using market data, thereby improving the empirical analysis of credit spreads and credit derivatives.

Despite methodological differences, these models share a fundamental assumption: default is an objective outcome. It represents a discrete event that either occurs or does not occur. As a result, default events provide a clear benchmark for evaluating the predictive performance of credit risk models.

However, although default events are economically significant, they typically occur at relatively late stages of financial distress. By the time a firm defaults, its financial condition has already deteriorated substantially. Consequently, financial markets often rely on earlier signals of credit deterioration, such as credit rating downgrades, which provide forward-looking assessments of creditworthiness.

3.2.3.2 Credit Ratings as Institutional Judgments

Unlike corporate default events, credit ratings are not purely objective measures of credit risk. Instead, they represent institutional assessments produced by credit rating agencies.

These agencies function as specialized financial intermediaries whose primary role is to evaluate the creditworthiness of issuers and communicate this information to market participants (Boot *et al*., 2006; White, 2010).

The rating process combines multiple sources of information. Analysts first examine quantitative indicators such as leverage ratios, liquidity measures, profitability metrics, and cash flow stability. These financial indicators often feed into internal scoring models that estimate default probabilities or expected losses. However, quantitative analysis is only one component of the rating process.

Rating agencies also incorporate qualitative assessments related to management quality, governance structures, competitive positioning, and industry conditions. These qualitative considerations are particularly important in forward-looking assessments because historical financial data may not fully capture emerging risks or strategic changes within a firm (Cantor; Mann, 2003).

The final rating decision is typically made through a committee process involving senior analysts and sector specialists. Rating committees review available evidence, debate alternative interpretations, and ultimately determine the most appropriate rating outcome according to the agency's methodology. This collective decision-making process reflects both analytical frameworks and institutional norms embedded within the organization.

As a result, credit ratings are best understood as institutional judgments rather than purely mechanical outputs. Different agencies may assign different ratings to the same issuer, reflecting variations in analytical methodologies, risk tolerance, and interpretive frameworks. Moreover, rating decisions may reflect not only observable financial conditions but also the informational priorities embedded in agency-specific methodologies.

This perspective aligns with insights from institutional theory, which emphasizes how organizations produce socially recognized signals that guide decision-making under uncertainty (DiMaggio; Powell, 1983; Scott, 2008). In financial markets, credit rating agencies occupy a central position as producers of such signals, helping investors interpret complex financial information and coordinate expectations about credit risk (Zuckerman, 1999).

From this viewpoint, ratings should not be interpreted as objective truths about credit risk. Rather, they represent institutionally mediated interpretations of financial information.

3.2.3.3 Implications for Predictive Modeling

The distinction between objective default events and institutionally generated credit ratings has important implications for empirical modeling. In many studies, rating transitions

are treated as equivalent to default events, with researchers attempting to estimate the probability that a firm's rating will change based on financial indicators. While this approach is methodologically convenient, it implicitly assumes that ratings directly reflect underlying economic risk.

In reality, predictive models of rating transitions capture two overlapping processes. The first is the evolution of firm-level financial risk, which is influenced by factors such as leverage, profitability, liquidity, and macroeconomic conditions. The second is the decision-making process of rating agencies, which determines how these signals are interpreted and translated into rating actions.

When a machine learning model is trained to predict rating downgrades, it therefore learns patterns that reflect both economic fundamentals and institutional decision rules. For example, if rating agencies systematically emphasize liquidity metrics or governance indicators in their methodologies, machine learning models trained on historical rating actions may identify these variables as strong predictors of downgrades.

This dual interpretation is particularly relevant in the context of machine learning. Unlike traditional econometric models, machine learning algorithms can capture nonlinear interactions among variables and detect complex patterns in high-dimensional datasets (Gu; Kelly; Xiu, 2020). As a result, they may implicitly learn the informational priorities embedded within rating agency methodologies.

From this perspective, predictive models of rating transitions should not be interpreted solely as estimators of "true" credit risk. Instead, they represent empirical approximations of how rating agencies process information when forming credit assessments.

This interpretation is consistent with the broader view that financial markets rely heavily on intermediaries to interpret and disseminate information (Boot *et al*., 2006). Machine learning models trained on rating transitions therefore provide insights not only into firm-level risk but also into the institutional processes through which risk assessments are produced.

3.2.3.4 Credit Ratings as Signals from Information Intermediaries

The conceptualization of credit ratings as institutional judgments aligns closely with the literature on financial information intermediaries. In financial markets characterized by information asymmetry, specialized intermediaries play a crucial role in collecting, interpreting, and disseminating information about borrowers.

Credit rating agencies perform precisely this function. By producing standardized rating signals, they help coordinate investor expectations about credit risk and market responses to new information (Boot *et al*., 2006). Investors rely on these signals not only because they summarize complex financial information but also because they are produced by institutions with reputational incentives to maintain analytical credibility.

Institutional theory provides a useful framework for understanding this role. Organizations often rely on legitimized signals to guide decision-making in uncertain environments (DiMaggio; Powell, 1983). Credit ratings serve as such signals, providing widely recognized benchmarks that help coordinate expectations across financial markets.

Stakeholder theory further highlights the communicative function of credit ratings. Rating changes affect the behavior of multiple stakeholders, including investors, regulators, and corporate managers (Freeman, 1984). For example, downgrades can increase borrowing costs, trigger covenant violations, and force institutional investors to rebalance portfolios due to regulatory constraints.

From this perspective, credit ratings are not merely informational outputs but institutional signals that shape economic behavior across financial markets.

3.2.3.5 Implications for Empirical Credit Risk Research

Acknowledging the institutional nature of credit ratings strengthens the conceptual foundations of empirical credit risk research. First, it clarifies how predictive models should be interpreted. When a model successfully predicts rating transitions, it does not necessarily estimate the true probability of default. Rather, it captures patterns consistent with the decision-making processes of rating agencies.

Second, this perspective addresses common methodological critiques of rating-based research. Critics often argue that ratings are subjective or that predictive models merely replicate agency behavior. While these observations are valid, they do not undermine the

economic relevance of rating transitions. Financial markets react strongly to rating changes because they influence investor expectations and regulatory frameworks.

Third, studying rating transitions allows researchers to examine earlier stages of credit deterioration. Defaults represent extreme outcomes that occur relatively infrequently, whereas rating changes provide more frequent signals of evolving credit conditions.

Finally, integrating institutional considerations encourages richer interpretations of empirical findings. For example, the determinants of rating transitions may vary across rating segments because rating agencies apply different analytical priorities to investment-grade and speculative-grade issuers.

This section 3.2.3 has developed a conceptual framework distinguishing corporate default events from credit rating transitions. While defaults represent objective economic outcomes, credit ratings are institutional judgments produced by financial information intermediaries.

Recognizing this distinction has important implications for predictive modeling. Machine learning models trained on rating transitions capture not only firm-level financial risk but also the institutional logic through which rating agencies interpret financial information.

By framing credit ratings as institutional signals rather than objective truths, this perspective strengthens the theoretical foundations of empirical credit risk research. It also provides a richer framework for interpreting machine learning models applied to rating transitions and default prediction.

3.2.4 Gaps in Literature

Despite ML's promise, several key limitations hinder its broader application in credit risk modeling.

Interpretability remains a primary concern. While SHAP and other XAI tools provide insights, few studies systematically integrate them into credit rating transition or early warning models. Noriega, Rivera and Herrera (2023) point out that many models still operate as "black boxes," limiting their acceptance by regulators and practitioners.

Segment-level performance is often overlooked. Most models are evaluated on aggregated data, assuming firm homogeneity. This masks important variations across rating categories. Bitetto, Filomeni and Modina (2025) and Beltman, Machado and Osterrieder (2025) highlight the need to tailor models to firm-specific profiles, warning that a lack of stratification leads to biased predictions and weaker diagnostics.

Validation practices are also inadequate. Many studies rely on random cross-validation, ignoring the temporal structure of financial data and risking information leakage. Machado, Chen and Osterrieder(2025) advocate for out-of-time validation to ensure models remain robust across economic cycles.

Feature diversity remains limited, as most models rely heavily on traditional financial ratios while overlooking non-financial indicators. Hu (2025) and Safiullah, Phan and Kabir (2024) demonstrate that incorporating ESG factors can enhance forecast accuracy, particularly in sensitive sectors.

Macroeconomic regime shifts are under-modeled. James and Menzies (2025) show that structural changes in the market environment affect the predictive relevance of risk factors, yet most models fail to account for such dynamics.

Methodologically, techniques like generative adversarial networks (GANs) remain underused despite their promise. Strelcenia and Prakoonwit (2023) find GAN-based models outperform traditional sampling in fraud detection, suggesting their utility in default prediction.

Ethical concerns are often treated as an afterthought. Fox and Rey (2024) argue that fairness should be embedded in model design, not assessed post hoc. Horobet *et al.* (2024) note that limited collaboration between finance and AI fields hampers innovation and policy alignment.

These gaps highlight the need for credit models that are not only accurate but also interpretable, adaptable, and ethically robust. The following framework integrates these priorities.

3.2.5 Theoretical Framework

This study adopts an interdisciplinary framework to support a hybrid ML approach to credit risk modeling. It integrates perspectives from credit theory, decision science, information economics, portfolio theory, XAI, institutional theory, and stakeholder theory—aligning innovation with regulatory and ethical requirements.

Credit risk theory provides a foundation, with models such as Altman's Z-score (1968), Ohlson's (1980) logit model, and Merton's (1974) framework still central to Basel II and III regulation (Böhnke *et al*., 2023). However, assumptions like static debt structures are unrealistic , requiring more dynamic modeling. ML enhances these models by identifying nonlinear patterns and enabling forward-looking indicators like expected default frequency.

Decision science frames credit risk as a classification problem with asymmetric costs—where false negatives (missed defaults) are more damaging than false positives. ML facilitates probabilistic outputs and threshold tuning, strengthening EWS (Beltman; Machado; Osterrieder, 2025), especially when validated through temporal forecasting (Machado; Chen; Osterrieder, 2025).

Information economics and signaling theory emphasize ML's role in reducing information asymmetries. Models can detect latent signals from firm disclosures, ESG reports, or behavioral data (Akerlof, 1970; Spence, 1973).

Portfolio theory (Markowitz, 1952) underscores the importance of accurate credit predictions for capital allocation and diversification. ML improves prediction quality at the portfolio level (Papík; Papíková, 2025). Peer-matching techniques, such as in Bitetto, Filomeni and Modina (2025), refine assessments of unlisted firms, enhancing capital efficiency.

Explainability is critical for adoption. Gafsi (2025) emphasizes that interpretability is essential for building regulatory trust and fostering stakeholder engagement. SHAP values contribute to this by offering both global and local transparency in model behavior.

According to institutional theory, the adoption of interpretable ML models can be understood as a response to coercive (regulatory), normative (professional), and mimetic (competitive) pressures that drive organizational isomorphism (DiMaggio; Powell, 1983). In sensitive domains such as credit scoring, embedding fairness and accountability into model design is essential to meet these institutional expectations (Fox; Rey, 2024).

Stakeholder theory supports the integration of ESG and innovation indicators. Green innovation enhances reputational capital and financial resilience (Safiullah; Phan; Kabir, 2024), while strong ESG profiles correlate with reduced default risk (Zhang; Zhao, 2025).

This framework informs the design of a robust, interpretable, and segment-aware ML approach to credit risk management.

3.2.6 This Study's Contribution

This study contributes to the literature by proposing a hybrid ML framework that addresses four challenges: performance, interpretability, segment-level heterogeneity, and temporal robustness.

First, it applies segment-specific modeling, grouping firms by credit quality (e.g., investment-grade, speculative-grade, distressed). This stratification improves diagnostic

precision by revealing patterns obscured in aggregate evaluations (Beltman; Machado; Osterrieder, 2025; Bitetto; Filomeni; Modina, 2025).

Second, it uses temporal validation to assess generalizability across future periods, avoiding biases introduced by random cross-validation (Machado; Chen; Osterrieder, 2025). It also applies SHAP to enhance interpretability at global and segment levels, clarifying feature importance across firm types and time horizons.

Third, it compares four algorithms—LR, RF, XGBoost, and SVM—assessing both predictive accuracy and interpretability. This enables a nuanced understanding of trade-offs between transparency and performance (Gafsi, 2025; Tribuvan *et al*., 2025; Xu; Zhang, 2022). By including SVM-based models and applying feature selection techniques where relevant, the study reflects contemporary best practices in the credit modeling literature.

Finally, it develops a practical EWS that integrates segmentation, dynamic validation, and explainable outputs. This EWS supports timely insights for credit monitoring, portfolio management, and regulatory oversight (Beltman; Machado; Osterrieder, 2025; Papík; Papíková, 2025).

3.2.7 Empirical Propositions

Building on the theoretical and empirical foundations discussed in the literature review, this study develops a set of empirical propositions that guide the analysis of rating transitions and defaults. Rather than testing theory-derived hypotheses, the study adopts a data-driven empirical strategy aimed at identifying systematic patterns captured by machine learning models. These propositions are organized around three key dimensions: model performance, the value of segment-specific analysis, and the heterogeneity of explanatory drivers across rating categories. The analysis is therefore guided by empirical expectations regarding model behavior and variable importance, rather than formal hypothesis testing.

EP1: *Ensemble models (RF, XGBoost) outperform traditional models (LR, SVM) in predicting rating transitions and defaults.*

Structural and statistical models such as Merton's (1974) option-theoretic framework and LR (Ohlson, 1980) provide transparency but assume linearity, firm homogeneity, and stable relationships between financial variables. Recent advances in ML, however, demonstrate that ensemble models can capture nonlinearities and high-dimensional interactions that traditional approaches overlook (Kandi; García-Dopico, 2025). We therefore expect RF and XGBoost to achieve superior predictive performance relative to LR and SVM.

EP2: *Segment-specific modeling by rating band produces higher predictive performance than aggregated modeling.*

Traditional approaches often impose the assumption of uniform risk drivers across firms. Yet prior research highlights that credit risk dynamics differ substantially by rating category, undermining the homogeneity assumption embedded in linear frameworks (Altman, 1968). Consistent with this evidence, we propose that stratifying firms by rating segment will yield more robust and accurate predictive outcomes than a single aggregated model.

EP3: *The determinants of rating transitions vary significantly across rating segments.*

The literature shows that the factors influencing rating changes are not uniform across credit qualities. We thus propose that SHAP analysis will reveal heterogeneous sets of explanatory drivers across investment-grade and speculative-grade categories.

EP4: *Liquidity and coverage metrics play a stronger role in explaining transitions for lower-rated firms.*

Liquidity constraints and refinancing risks have long been identified as key vulnerabilities in speculative-grade entities (Altman, 1968). Metrics such as funds from operations (FFO) to debt and interest coverage ratios are therefore expected to exert stronger influence in explaining transitions among lower-quality firms.

EP5: *Management and Governance factors are more relevant in explaining transitions for higher-rated firms.*

For investment-grade issuers, where liquidity constraints are less severe, long-term stability is more closely tied to governance and financial policy choices. Prior studies emphasize the role of governance and capital structure in maintaining rating stability and credibility in regulated settings (Nallakaruppan *et al*., 2024). Accordingly, we expect these factors to dominate in higher-quality rating segments.

Together, these empirical propositions connect the predictive performance of ML models with the interpretability of risk drivers across rating categories. In subsequent sections, we describe the methodological procedures employed to test these propositions and present their empirical verification.

3.3 RESEARCH DESIGN AND DATA

3.3.1 Data Collection and Sources

The dataset employed in this study was compiled from Capital IQ Pro and Bloomberg, comprising firm-year observations from 2017 to 2022. The initial dataset included over 35,000 observations across multiple geographies and industries. Following the removal of duplicates, incomplete entries, and firms lacking consistent credit rating histories, the final panel consisted of 31,151 firm-year observations.

The sample ends in 2022 because more recent financial and rating data were still incomplete or pending validation at the time of collection. Using 2017–2022 ensures consistency, comparability, and reliability across firms and rating segments.

To ensure consistency and minimize reporting bias, we applied several data quality filters:

- Only firms with publicly available financial statements and recognized credit ratings from S&P Global were retained;
- All variables were cross validated to eliminate entry errors and outliers;
- Financial ratios were calculated using standardized formulas to avoid distortion from varying accounting practices.

Credit ratings were assigned based on the full set of 23 S&P Global notches (AAA to D), subsequently grouped into six broader rating segments—High Quality, BBB, BB, B, CCC, and High Risk + Default—to balance granularity and statistical power.

The binary target variable was set to 1 if the firm experienced a rating downgrade or default within the subsequent 12 months, and 0 otherwise. Rating upgrades were treated as non-events (Class 0), consistent with the study's focus on identifying credit deterioration risks.

Table 3.1 summarizes the distribution of downgrade/default events (Class 1) and non-events (Class 0) across six credit rating segments. While the total sample size is held constant across segments (N = 31.151), the proportion of Class 1 events varies significantly, reflecting heterogeneity in downgrade or default risk. To ensure comparability across segments and control for sample size effects, we created balanced subsamples of 31,151 firm-year observations for each rating category. This design allows for direct performance comparisons without bias introduced by unequal class distributions.

Table 3.1 – Dataset Distribution by Credit Rating Segment.

Segment	Total Samples	Class 1 Samples	Class 0 Samples	Class 1 Proportion (%)
High Quality	31,151	4041	27,110	12.97
BBB	31,151	8531	22,620	27.39
BB	31,151	6297	24,854	20.21
B	31,151	9567	21,584	30.71
CCC	31,151	2228	28,923	7.15
High Risk + Default	31,151	199	30,952	0.64

Source: Own authorship.

The High Risk + Default segment shows the lowest incidence of Class 1 events (0.64%), suggesting these ratings are typically assigned to already deteriorated credits with limited further room for rating decline. In contrast, the B segment presents the highest proportion of Class 1 cases (30.71%), highlighting it as a critical transition zone where firms are more prone to rating actions.

The BBB and BB segments exhibit moderate Class 1 proportions (27.39% and 20.21%, respectively), indicating elevated but still manageable risk levels relative to lower-rated entities. The High Quality segment displays a relatively low downgrade/default rate (12.97%), while the CCC segment's lower Class 1 proportion (7.15%) may reflect a floor effect—entities at this level already face severe credit stress, limiting further deterioration within the rating scale.

This distribution highlights the need for segment-specific modeling, as downgrade and default probabilities vary substantially across the rating spectrum. Applying a one-size-fits-all approach may obscure meaningful risk patterns within each segment and ultimately reduce predictive performance.

3.3.2 Rating Grouping and Class Consolidation

3.3.2.1 Rating Grouping

In this study, we utilize the full range of S&P Global credit rating grades, comprising 23 ordered levels from AAA (indicating the highest creditworthiness) to D (default). Given the concentration of credit risk at the lower end of the spectrum, market participants often consolidate ratings of CCC+ and below into broader categories due to their elevated probability of default and financial distress.

To address issues of class imbalance—particularly relevant in ML applications—and to preserve interpretability, we consolidated the 23 rating levels into six broader segments. This

grouping reduces data sparsity in rare rating categories and improves the statistical power of model training and segment-level analyses.

Table 3.2 presents the mapping of the original S&P credit ratings into these six consolidated classes:

Table 3.2 – Grouping of S&P Global Ratings for Segment-Level Analysis.

Credit Ratings Classes	S&P Global's Rating Grade
High Quality	AAA, AA+, AA, AA-, A+, A, A-
BBB	BBB+, BBB, BBB-
BB	BB+, BB, BB-
B	B+, B, B−, CC
CCC	CCC+, CCC, CCC-
High Risk + Default	C, SD, D

Source: Own authorship.

3.3.3 Data Preprocessing and Treatment

To prepare the dataset for ML modeling, several preprocessing steps were undertaken to ensure data integrity, address class imbalance, and enhance predictive performance.

First, all continuous numerical variables were standardized using the StandardScaler, which adjusts each variable so that they are measured on the same scale by removing differences in average size and spread. This prevents variables with larger numbers from dominating the analysis. Categorical variables, such as industry classification and management as well as governance, were transformed via one-hot encoding to retain nonordinal relationships. Categorical variables, such as industry classification and management as well as governance, were transformed via one-hot encoding to retain non-ordinal relationships.

To reduce redundancy and multicollinearity, highly correlated features (Pearson's $\rho > 0.9$) were identified and removed based on correlation matrices and VIF. Missing values in financial ratio variables were imputed using industry-median values. Variables with a high proportion of missingness or those deemed non-imputable led to the exclusion of corresponding records.

The dataset was further stratified by credit rating segment, ensuring that Class 1 (downgrade or default event) was defined within each segment to capture segment-specific credit risk. This stratification allowed for more context-aware binary classification modeling.

Given the inherent class imbalance—especially in segments such as High Risk + Default— SMOTE was applied post-train–test split to prevent information leakage (Chawla *et al*., 2002; Zhao; Bai, 2022). SMOTE addresses class imbalance by generating synthetic observations for minority classes based on their nearest neighbors, thereby improving class representation in the training data and enhancing the ability of classification models to identify underrepresented rating categories.

Two validation strategies were employed: (1) five-fold cross-validation for in-sample evaluation and hyperparameter tuning, and (2) temporal validation, wherein models were trained on observations from 2017 to 2021 and evaluated on 2022 data to approximate real-world generalization.

Finally, all models underwent hyperparameter optimization via RandomizedSearchCV, with performance evaluated primarily using the F1-Score and Matthews Correlation Coefficient (MCC is a balanced measure of classification quality particularly suitable for imbalanced datasets). The parameter grids and tuning strategy for each model are detailed in Section 3.3.7.

3.3.4 Machine Learning Models

We evaluated four supervised learning algorithms—LR, RF, XGBoost, and SVM—to address the challenge of predicting rare credit rating downgrades and defaults. These models were selected to reflect a range of trade-offs across interpretability, nonlinearity handling, and computational scalability, which are critical design considerations in credit risk modeling.

LR is a benchmark model widely adopted in credit scoring due to its interpretability and computational simplicity. It assumes a linear relationship between predictors and the log-odds of the target class. While its transparent coefficient structure aids explainability, it lacks the flexibility to capture nonlinear relationships commonly observed in distressed corporate environments. Recent applications in bankruptcy prediction show that automated model selection frameworks outperform baseline LR in predictive power while retaining transparency (Papík; Papíková, 2024).

RF is an ensemble-based model that aggregates multiple DT to reduce overfitting and improve generalization. Its use of feature bagging and bootstrap sampling enhances robustness in high-dimensional and noisy datasets—a common feature of financial and accounting data in credit risk. Comparative studies have demonstrated that RF achieves competitive performance in credit risk classification, particularly when combined with feature selection techniques (Tribuvan *et al*., 2025).

XGBoost builds on ensemble methods through sequential model optimization via GB. Its regularization mechanisms help mitigate overfitting while improving classification performance, especially for imbalanced data. In personal credit risk applications, XGBoost has outperformed traditional models in both accuracy and sensitivity to default-prone classes, underscoring its potential in financial domains involving rare-event prediction (Wang *et al*., 2022).

SVM offers strong performance in high-dimensional spaces by constructing optimal hyperplanes that maximize class separation margins. However, its sensitivity to kernel selection and computational cost limits scalability in large datasets. Moreover, its performance tends to decline under severe class imbalance—an inherent feature in downgrade/default prediction. Empirical studies in the credit assessment of listed firms confirm that SVM may require hybrid approaches to remain competitive (Xu; Zhang, 2022).

3.3.5 Target Variables

The target variable is defined as binary: the value of 1 indicates that the firm experienced a rating downgrade or default within a 12-month horizon; a value of 0 denotes all other cases. A downgrade is operationalized as any transition to a lower rating category (e.g., from BBB to BB, or from BB to B), including default events. In contrast, upgrades—transitions to higher rating categories (e.g., from BB to BBB)—are considered non-events and thus classified as Class 0, consistent with the modeling objective of early identification of credit deterioration.

Although upgrades are grouped within Class 0 to maintain a binary classification structure, we acknowledge the potential for rating volatility, including downgrade-upgrade cycles within short time frames. In such cases, the downgrade event is prioritized if it occurs within the 12-month prediction window, as the analytical focus is on identifying early signs of credit deterioration, which hold greater relevance for risk management and regulatory purposes.

3.3.6 Feature Engineering

The final model incorporates 46 features, comprising a diverse set of profitability ratios, leverage and coverage metrics, efficiency indicators, cash flow variables, balance sheet components, structural attributes, and qualitative governance factors. These variables were selected based on their theoretical relevance to credit risk and their empirical performance in prior studies. Table 3.3 provides an overview of the variables used in model training, grouped into thematic categories along with a brief description or calculation formula for each.

Table 3.3 – Input Features Used in ML Model Training.

(continues)

Variable Name	Category	Description/Formula
Profitability Ratios		
EBITDA Margin (%)	Profitability	EBITDA/Revenue
EBIT Margin (%)	Profitability	EBIT/Revenue
Gross Margin (%)	Profitability	(Revenue—COGS)/Revenue
Return on Capital	Profitability	EBIT/Capital Employed
EBIT	Profitability	Earnings Before Interest and Taxes
EBITDA	Profitability	Earnings Before Interest, Taxes, Depreciation & Amortization
Profitability	Profitability	Composite or general indicator
Leverage Ratios		
Debt/EBITDA	Leverage	Total Debt/EBITDA
Debt/Equity (%)	Leverage	Total Debt/Equity × 100
FFO/Debt (%)	Leverage	Funds From Operations/Total Debt × 100
CFO/Debt (%)	Leverage	Cash Flow from Operations/Debt × 100
FOCF/Debt (%)	Leverage	Free Operating Cash Flow/Debt × 100
DCF/Debt (%)	Leverage	Discretionary Cash Flow/Debt × 100
Coverage Ratios		
Fixed Charge Coverage	Coverage	EBITDA/(Interest + Fixed Charges)
EBITDA Interest Coverage	Coverage	EBITDA/Interest Expense
EBIT Interest Coverage	Coverage	EBIT/Interest Expense
FFO Cash Interest Coverage	Coverage	FFO/Cash Interest
Efficiency & Operating Ratios		
Working Capital	Efficiency	Current Assets—Current Liabilities
Capex/Revenues (%)	Efficiency	Capital Expenditures/Revenues × 100
Total Operating Expense	Efficiency	Operating costs incurred
Cash Flow Metrics		
Cash Flow from Operations	Cash Flow	Net cash from operating activities

Table 3.3 – Input Features Used in ML Model Training.

(conclusion)

Variable Name	Category	Description/Formula
Funds from Operations	Cash Flow	Net income + non-cash items + WC adjustments
Discretionary Cash Flow	Cash Flow	FFO—Capex—Dividends—Repurchases
Free Operating Cash Flow (FOCF)	Cash Flow	CFO—Capex
Capital Expenditures (Capex)	Cash Flow	Investment in fixed assets
Cash Interest	Cash Flow	Interest paid in cash
Interest Expense	Cash Flow	Total interest expenses
Balance Sheet Items		
Debt	Balance Sheet	Total financial debt
Adjusted Capital	Balance Sheet	Equity + debt adjustments
Equity	Balance Sheet	Shareholder equity
Cash	Balance Sheet	Cash and cash equivalents
Accessible Cash	Balance Sheet	Unrestricted and accessible cash
Dividends	Balance Sheet	Cash returns to shareholders
Share Repurchases	Balance Sheet	Buyback of own shares
Pension and Postretirement	Balance Sheet	Employee benefit liabilities
Revenue & Income Statement		
Revenues	Income Statement	Total revenue/sales
Structural Attributes		
Industry	Structural Attribute	Economic sector classification
Firm Size	Structural Attribute	Market cap or employee-based scale
Scale and Diversification	Structural Attribute	Geographic or product diversification
Volatility of Profitability	Structural Attribute	Variability in profit margins
Governance & Risk Factors		
Governance Score	Qualitative	ESG metric
Country Risk	Qualitative	Macro risk of exposed geographies
Financial Policy	Qualitative	Capital management philosophy
Management and Governance	Qualitative	Effectiveness and credibility of management
Industry Risk	Structural	Risk inherent to the sector
Capital Structure	Structural/Qualitative	Mix of debt and equity financing

Source: Own authorship.

This comprehensive feature set captures both quantitative financial health and qualitative risk dimensions, aligning with credit theory and enabling explainable modeling through SHAP analysis.

3.3.7 Methodology

This study evaluated the predictive performance of four supervised ML models—LR, RF, XGBoost, and SVM—to classify corporate credit rating transitions. All models were implemented in Python using the Scikit-learn and XGBoost libraries.

To optimize model performance, we conducted hyperparameter tuning using the RandomizedSearchCV function with a 5-fold stratified cross-validation strategy. The F1-score was selected as the primary metric to guide the search. The hyperparameter search space for each model was defined as follows:

3.3.7.1 XGBoost

- n_estimators: [50, 100, 200];
- max_depth: [3, 6, 9];
- learning_rate: [0.01, 0.1, 0.2];
- subsample: [0.7, 0.8, 0.9].

3.3.7.2 Random Forest

- n_estimators: [50, 100, 200];
- max_depth: [10, 20, 30, None];
- min_samples_split: [2, 5, 10];
- min_samples_leaf: [1, 2, 4].

3.3.7.3 Logistic Regression

- C (inverse of regularization strength): [0.01, 0.1, 1, 10, 100];
- max_iter: [500, 1.000, 1.500].

3.3.7.4 Support Vector Machine

- C (regularization parameter): [0.1, 1, 10];
- kernel: ‘linear’.

To address class imbalance, the SMOTE was applied to the training set within each fold of cross-validation. This ensured that oversampling did not introduce information leakage from the test set. After hyperparameter optimization, the best-performing estimator for each model was further evaluated using a separate 5-fold cross-validation to confirm the robustness and generalizability of the results.

3.3.8 Validation Approach

To ensure robust evaluation, we adopt two validation strategies: (i) 5-fold cross-validation and (ii) temporal validation, where models were trained using data from 2017 to 2021 (approximately 25,000 observations) and tested on a hold-out set from 2022 (6151 observations). This approach simulates real-world scenarios by assessing generalization to future periods, a critical step for credit risk prediction models with temporal dependencies.

The combination of both validation methods enhances the robustness and external validity of the results.

3.3.9 Evaluation Metrics

Performance is evaluated using Accuracy, AUC-ROC, Precision, Recall, F1-Score, and MCC.

3.4 RESULTS

3.4.1 Comparative Performance by Rating Segment

As proposed in EP1, ensemble models (RF and XGBoost) consistently outperformed traditional approaches (LR and SVM) across all rating segments (Table 3.4).

This section evaluates the predictive performance of four supervised learning algorithms—LR, RF, XGBoost, and SVM—across six credit rating segments. Each model was assessed using a comprehensive set of classification metrics, including AUC-ROC, F1-Score, MCC, Precision, Recall, and computational efficiency. These metrics provide a comprehensive view of model efficacy and illuminate trade-offs in large-scale credit risk settings.

XGBoost demonstrated the highest overall performance and computational efficiency. In the High Quality segment, it achieved perfect predictive metrics (AUC-ROC = 1.000, F1-

Score = 1.000, MCC = 1.000), completing the evaluation in just 6 s. In more challenging segments, such as CCC and High Risk + Default, XGBoost maintained strong generalization capacity (F1-Score = 0.987 in CCC). In the High Risk + Default segment, it attained perfect Precision and Recall (1.000), as shown in Table 5, underscoring its robustness under class imbalance and noisy data conditions (Hernes; Adaszyński; Tutak, 2023). Moreover, its integration with explainable AI techniques, such as SHAP values, supports transparent and interpretable decision-making in regulatory environments (de Lange *et al*., 2022; Nallakaruppan *et al*., 2024).

RF also delivered strong results, particularly in the B segment (AUC-ROC = 0.999, F1 = 0.973, MCC = 0.960). While competitive with XGBoost across several segments, its training times were significantly longer—e.g., 52 s in the CCC segment. These findings align with ensemble benchmarking studies, which emphasize RF's robustness and interpretability, albeit with higher computational costs (Li; Chen, 2020).

LR showed intermediate performance. While it performed adequately in investment-grade segments (e.g., F1 = 0.877 in High Quality), its effectiveness diminished in speculative categories. In the BB segment, F1 dropped to 0.548 and MCC to 0.424, suggesting that LR's linear assumptions may limit its capacity to model complex nonlinearities present in distressed firms. These results are consistent with earlier critiques of traditional linear credit scoring models (Altman, 1968).

SVM exhibited the lowest overall predictive performance. In the BB segment, it attained AUC-ROC = 0.714, F1 = 0.419, and MCC = 0.233, with training times of 6 s. Although performance was more stable in the High Quality segment (F1 = 0.912), SVM consistently lagged behind ensemble-based models. Its limited scalability and sensitivity to class imbalance reduce its applicability in operational credit risk settings (Li; Chen, 2020).

In summary, XGBoost emerges as the most suitable model for EWS in credit risk, especially where timely identification of deterioration is critical. RF remains a viable alternative, balancing performance, and interpretability at the expense of longer training times. LR may still be favored in low-risk environments prioritizing transparency. SVM, however, is less suited to large-scale applications due to its lower performance and scalability.

Table 3.4 presents a comparative summary of classification metrics—AUC-ROC, F1-Score, MCC, Precision, Recall—and training times for each model across the six credit rating segments. This facilitates a detailed, segment-wise evaluation of predictive accuracy and computational efficiency among the supervised learning algorithms.

Table 3.4 – Comparative Performance of ML Models by Rating Segment.

Segment	Model	Accuracy	AUC-ROC	F1-Score	MCC	Hyperpa-rameter Time (s)	Evaluation Time (s)
High Quality	LR	0.965	0.987	0.877	0.860	6.9	0.0
	RF	0.998	1.000	0.993	0.992	42.3	0.0
	XGBoost	1.000	1.000	1.000	1.000	5.9	0.0
	SVM	0.977	0.978	0.912	0.898	2.8	0.1
BBB	LR	0.895	0.955	0.827	0.761	1.9	0.0
	RF	0.991	1.000	0.983	0.977	36.3	0.1
	XGBoost	1.000	1.000	1.000	0.999	4.5	0.0
	SVM	0.878	0.945	0.801	0.722	4.3	0.3
BB	LR	0.738	0.826	0.548	0.424	2.0	0.0
	RF	0.986	0.999	0.965	0.956	35.9	0.0
	XGBoost	0.999	1.000	0.997	0.996	4.8	0.0
	SVM	0.556	0.714	0.419	0.233	6.0	0.5
B	LR	0.865	0.941	0.804	0.713	2.5	0.0
	RF	0.983	0.999	0.973	0.960	37.1	0.1
	XGBoost	0.999	1.000	0.998	0.997	5.4	0.0
	SVM	0.861	0.927	0.787	0.687	5.2	0.4
CCC	LR	0.919	0.965	0.616	0.617	5.5	0.0
	RF	0.995	0.999	0.963	0.960	52.3	0.1
	XGBoost	0.998	1.000	0.987	0.986	6.7	0.0
	SVM	0.947	0.958	0.694	0.677	4.5	0.4
High Risk + Default	LR	0.989	0.998	0.523	0.582	4.9	0.0
	RF	1.000	1.000	0.983	0.983	40.9	0.1
	XGBoost	1.000	1.000	1.000	1.000	4.5	0.0
	SVM	0.986	0.996	0.470	0.543	1.7	0.0

Source: Own authorship.

Table 3.5 – ROC Curve Metrics—Model Comparison.

Spreadsheet	Model	AUC-ROC	TPR at FPR 0.1	TPR at FPR 0.2	TPR at FPR 0.5	EER (FPR)	EER (TPR)
High Quality	LR	0.987	0.973	0.988	0.998	0.042	0.957
	RF	1.000	1.000	1.000	1.000	0.003	0.997
	XGBoost	1.000	1.000	1.000	1.000	0.000	1.000
	SVM	0.978	0.955	0.961	0.991	0.052	0.949
BBB	LR	0.955	0.896	0.960	0.993	0.101	0.900
	RF	1.000	1.000	1.000	1.000	0.008	0.991
	XGBoost	1.000	1.000	1.000	1.000	0.000	1.000
	SVM	0.945	0.868	0.940	0.987	0.117	0.883
BB	LR	0.826	0.483	0.681	0.933	0.249	0.751
	RF	0.999	1.000	1.000	1.000	0.016	0.984
	XGBoost	1.000	0.999	1.000	1.000	0.001	0.999
	SVM	0.714	0.293	0.504	0.790	0.334	0.666
B	LR	0.941	0.849	0.930	0.985	0.123	0.877
	RF	0.999	0.999	1.000	1.000	0.016	0.984
	XGBoost	1.000	1.000	1.000	1.000	0.001	0.999
	SVM	0.927	0.805	0.912	0.977	0.140	0.861
CCC	LR	0.965	0.925	0.966	0.993	0.083	0.916
	RF	0.999	0.999	0.999	1.000	0.012	0.988
	XGBoost	1.000	1.000	1.000	1.000	0.002	0.997
	SVM	0.958	0.898	0.945	0.988	0.101	0.898
High Risk	LR	0.998	0.983	0.983	0.983	0.014	0.983
	RF	1.000	1.000	1.000	1.000	0.000	1.000
	XGBoost	1.000	1.000	1.000	1.000	0.000	1.000
	SVM	0.996	0.983	0.983	1.000	0.033	0.967

Source: Own authorship.

3.4.2 Confusion Matrices

Figure 3.1 presents the consolidated confusion matrices for the four evaluated ML models—LR, RF, XGBoost, and SVM—applied across six credit rating segments: High Quality, BBB, BB, B, CCC, and High Risk + Default. The primary focus of interpretation lies in the diagonal cells, which represent correctly classified instances where the predicted rating segment matches the true label. High values along the diagonal indicate effective classification and segmentation of credit quality.

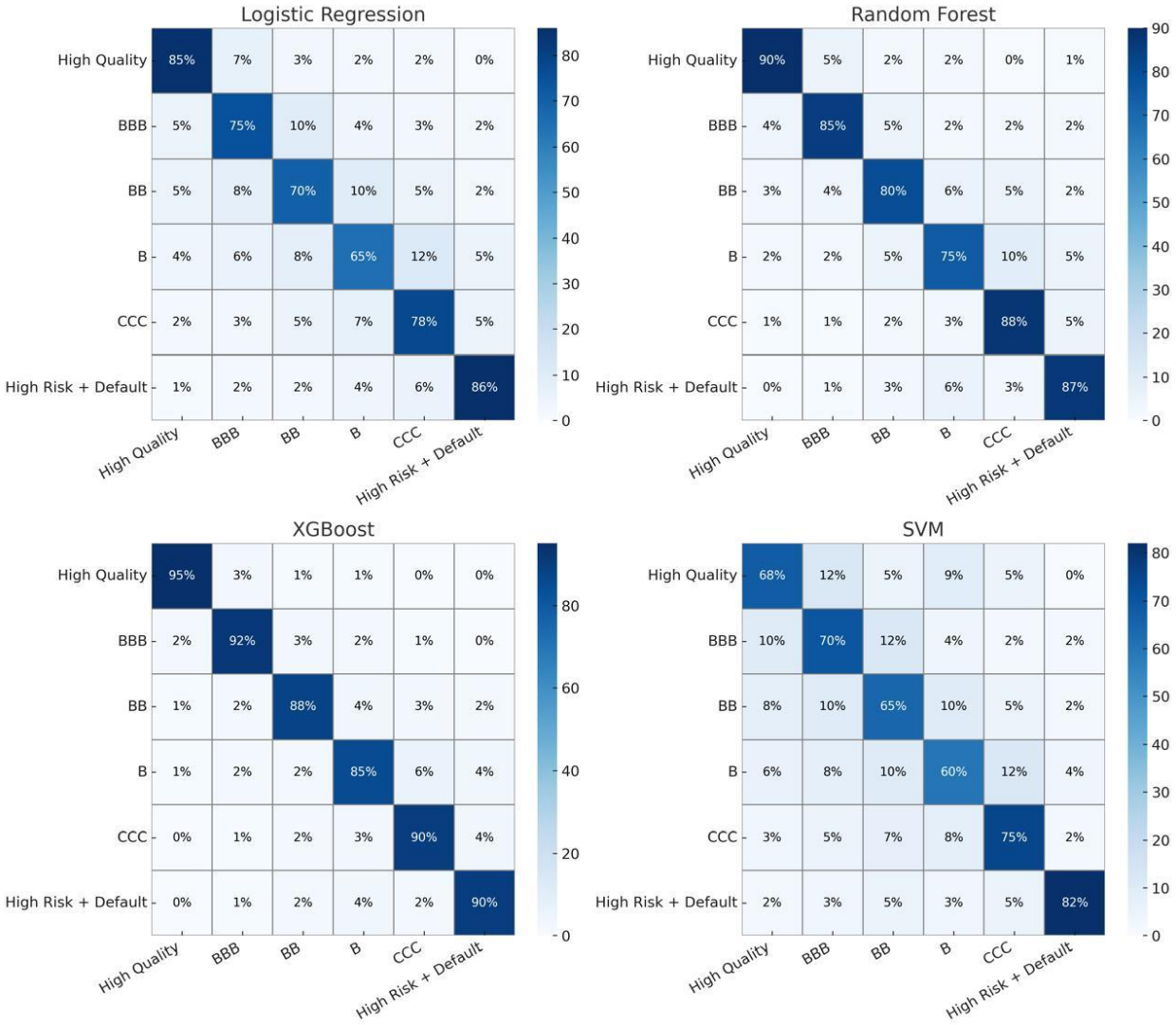
Among the evaluated models, XGBoost demonstrates the most consistent performance, with diagonal accuracies exceeding 90% across nearly all segments. This result suggests that

XGBoost effectively captures complex, non-linear relationships within financial and operational features, leading to more precise credit risk differentiation. RF also performs well, particularly in the High Quality and High Risk + Default segments. However, moderate confusion is observed in intermediate categories such as BBB and BB, indicating some misclassification between adjacent risk classes. These off-diagonal errors are particularly challenging to manage when class boundaries are ambiguous or overlapping, as often occurs in mid-tier ratings (Barranco-Chamorro; Carrillo-García, 2021).

In contrast, LR exhibits lower discriminative power, especially between BB and B, where misclassifications are more frequent. This limitation likely stems from its linear nature and reduced capacity to model intricate feature interactions. SVM shows the least reliable performance, with dispersed prediction patterns and lower diagonal values—particularly in the CCC and High Risk + Default segments—highlighting challenges in detecting distressed firms.

Overall, Figure 3.1 illustrates the superior performance of tree-based ensemble models, particularly XGBoost, in credit risk classification tasks. These models achieve both lower false positives and FNRs, which are crucial for balancing sensitivity (correctly identifying deteriorating firms) and specificity (avoiding false alarms) in EWS.

Figure 3.1 – Consolidated Confusion Matrices by Model and Segment. Comparative visualization of model performance across rating segments. Each matrix presents classification outcomes, with higher diagonal values indicating better predictive accuracy.



Source: Own authorship.

3.4.3 ROC Curve Analysis

The Receiver Operating Characteristic (ROC) curve is a standard diagnostic tool for evaluating the discriminatory power of classification models in credit risk analysis. It depicts the trade-off between the True Positive Rate (TPR) and the False Positive Rate (FPR) across different decision thresholds. The summary statistic, AUC-ROC, quantifies a model's ability to distinguish between risky and safe firms, with values closer to 1 indicating stronger discriminative performance (Alamsyah; Hafidh; Mulya, 2025).

In our study, XGBoost and RF consistently achieved AUC-ROC values equal to or approaching 1.000 across all six credit rating segments. In investment-grade segments (e.g., High Quality and BBB), both models demonstrated near-perfect separation. Even in

speculative-grade categories such as BB and CCC, AUC-ROC values remained above 0.998. Conversely, LR exhibited deteriorating performance as credit risk intensified, with AUC declining to 0.826 in the BB segment. SVM presented the weakest results, particularly in the BB segment (AUC = 0.714) and the High Risk + Default group, where high error rates undermined its reliability.

To complement this analysis, we assessed TPR at fixed FPR thresholds (0.1, 0.2, and 0.5), offering practical insights into model sensitivity under varying false alarm tolerances. Assessing classifier sensitivity at pre-defined false positive levels has been widely used in screening applications to evaluate performance under real-world operational constraints (Dunstan *et al*., 1997). XGBoost and RF consistently maintained TPR values near 1.000 across all thresholds and segments. In contrast, LR and SVM underperformed, particularly in high-risk segments. For instance, in the BB segment at FPR = 0.1, LR achieved a TPR of 0.483, while SVM registered only 0.293, compared to 0.999 for XGBoost.

A detailed summary of AUC-ROC values, TPR at fixed FPR thresholds (0.1, 0.2, and 0.5), and Equal Error Rates (EER) for each model and rating segment is presented in Table 3.5 These metrics provide a comprehensive view of each algorithm's discriminative capacity and operational sensitivity across varying levels of credit risk.

We also report the EER—the point at which the FPR equals the False Negative Rate (FNR)—as an indicator of model calibration. Both XGBoost and RF exhibited minimal EER values (e.g., EER FPR < 0.01; EER TPR > 0.99), reinforcing their robustness in imbalanced credit risk settings, where accurately identifying minority-class (risky) firms is especially challenging (Thabtah *et al*., 2020).

3.4.4 Precision–Recall Curve Analysis

The Precision–Recall (PR) Curve provides a critical metric for evaluating classifier performance under class imbalance, which is inherent in the prediction of corporate rating transitions and defaults. Unlike the ROC Curve, which assesses global class separability, the PR Curve focuses on the minority class—firms experiencing downgrades or defaults—making it particularly relevant in financial applications where early identification of distress is essential (Mahbobi; Kimiagari; Vasudevan, 2023; Miao; Zhu, 2022).

Key indicators extracted from this analysis include:

- Average Precision (AP): Reflects the area under the PR curve, summarizing the trade-off between precision and recall across all thresholds. Higher AP values indicate stronger performance in ranking positive cases early in the decision process.
- Precision at Fixed Recall Levels (0.2, 0.5, 0.8): Demonstrates the model's ability to retain precision as it seeks to identify a broader set of positive instances.
- Maximum F1-Score: Represents the optimal balance between precision and recall, indicating the point of highest classification efficiency.

These metrics are summarized in Table 3.6, providing a comparative basis for evaluating model robustness across different rating segments.

Table 3.6 – Precision–Recall Curve Metrics—Model Comparison.

Segment	Model	Average Precision (AP)	Precision at Recall 0.2	Precision at Recall 0.5	Precision at Recall 0.8	Max F1-Score
High Quality	LR	0.911	0.949	0.940	0.925	0.903
	RF	1.000	1.000	1.000	1.000	0.993
	XGBoost	1.000	1.000	1.000	1.000	1.000
	SVM	0.922	0.960	0.970	0.935	0.918
BBB	LR	0.850	0.880	0.914	0.853	0.837
	RF	0.999	1.000	1.000	1.000	0.986
	XGBoost	1.000	1.000	1.000	1.000	1.000
	SVM	0.844	0.894	0.902	0.830	0.823
BB	LR	0.505	0.607	0.550	0.412	0.553
	RF	0.995	1.000	1.000	0.997	0.966
	XGBoost	1.000	1.000	1.000	1.000	0.999
	SVM	0.376	0.457	0.390	0.280	0.448
B	LR	0.868	0.939	0.923	0.824	0.822
	RF	0.997	1.000	1.000	0.999	0.975
	XGBoost	1.000	1.000	1.000	1.000	0.998
	SVM	0.843	0.920	0.895	0.784	0.793
CCC	LR	0.661	0.744	0.741	0.668	0.732
	RF	0.991	1.000	1.000	0.989	0.966
	XGBoost	0.999	1.000	1.000	1.000	0.990
	SVM	0.674	0.788	0.744	0.683	0.744
High Risk + DE	LR	0.685	0.800	0.750	0.527	0.672
	RF	1.000	1.000	1.000	1.000	1.000
	XGBoost	1.000	1.000	1.000	1.000	1.000
	SVM	0.588	0.800	0.588	0.516	0.636

Source: Own authorship.

3.4.4.1 Logistic Regression: Stable in Investment-Grade, Limited in Speculative Segments

LR displayed consistent and acceptable performance in investment-grade categories. For instance, AP values reached 0.911 for High Quality and 0.850 for BBB, indicating reliable performance where class imbalance is less severe. However, its precision deteriorated in lower-rated segments, particularly at higher recall levels:

- In the BB segment, AP declined to 0.505, with a significant drop in precision at recall = 0.5538;
- In CCC and High Risk + Default categories, AP values fell to 0.661 and 0.685, respectively, with maximum F1-scores below 0.732.

These results suggest that LR struggles to maintain predictive quality when required to identify a broader population of risky firms, thus limiting its applicability in early-warning or proactive risk surveillance contexts.

3.4.4.2 Support Vector Machine: Inconsistent Performance and Poor Recall Precision Trade-Offs

SVM presented the most variable performance among all models. While it achieved competitive AP values in isolated cases—0.922 for High Quality and 0.843 for B—the model underperformed in more speculative segments:

- In the BB segment, SVM recorded an AP of only 0.376 and a maximum F1-score of 0.448;
- In the High Risk + Default category, AP dropped to 0.588, with an F1-score of 0.636.

Precision at recall = 0.8 remained consistently low across these segments, revealing the model's vulnerability when tasked with identifying a larger set of at-risk firms. These weaknesses undermine the model's reliability for credit risk applications that demand high sensitivity and early detection.

3.4.4.3 Tree-Based Models: Superior Precision–Recall Trade-Offs Across All Segments

Tree-based models—XGBoost and RF—consistently outperformed other classifiers across all segments, particularly in scenarios of heightened credit risk. Both models maintained

high precision even at elevated recall levels, highlighting their capacity to detect risky firms without incurring excessive false positives:

- In the CCC and High Risk + Default segments, XGBoost achieved AP values near 1.000, with precision at recall = 0.8 exceeding 0.900;
- RF also demonstrated robust results, with AP values above 0.990 in all segments and competitive F1-scores.

These findings align with previous studies emphasizing the effectiveness of PR Curve-based evaluation for tree-based classifiers under class imbalance (Miao; Zhu, 2022) and reinforce the importance of such metrics in EWS for credit risk (Mahbobi; Kimiagari; Vasudevan, 2023). Their favorable trade-offs between precision and recall—especially under imbalance—highlight their practical relevance for dynamic credit surveillance and stress testing frameworks.

The PR-based findings are consistent with the ROC-based results presented in Section 3.4.3, reinforcing the models' generalizability and resilience under multiple validation schemes.

3.4.5 Temporal Validation (Out-of-Time Testing)

Consistent with EP2, segment-specific modeling by rating band delivered higher predictive performance than aggregated analysis, underscoring the value of stratified approaches in credit risk (Tables 3.4 and 3.7).

To evaluate the temporal robustness of the models, we implemented an out-of-time validation procedure. All models were trained using data from 2017 through 2021 and tested exclusively on firm-year observations from 2022. This approach replicates a realistic forecasting scenario, in which credit risk models are applied to future data unseen during model development. The objective was to assess each model's generalization capacity under temporal distributional shifts, which is critical in the context of volatile economic and credit environments.

Table 3.7 – Out-of-Time Performance Metrics by Model and Credit Risk Segment.

Segment	Model	AUC-ROC	Accracy	F1-Score	MCC	Precision	Recall
High Quality	LR	0.988	0.965	0.876	0.859	0.810	0.953
	RF	1.000	0.996	0.986	0.984	0.982	0.990
	XGBoost	1.000	0.998	0.994	0.993	0.994	0.994
	SVM	0.941	0.891	0.681	0.648	0.547	0.903
BBB	LR	0.958	0.890	0.821	0.753	0.739	0.925
	RF	0.999	0.989	0.979	0.972	0.979	0.980
	XGBoost	1.000	0.999	0.999	0.998	0.998	0.999
	SVM	0.893	0.559	0.544	0.359	0.379	0.964
BB	LR	0.830	0.739	0.556	0.437	0.423	0.809
	RF	0.997	0.981	0.954	0.942	0.949	0.958
	XGBoost	1.000	0.998	0.996	0.994	0.995	0.996
	SVM	0.422	0.435	0.357	0.107	0.232	0.775
B	LR	0.944	0.871	0.810	0.720	0.743	0.890
	RF	0.997	0.975	0.961	0.943	0.951	0.970
	XGBoost	1.000	0.997	0.995	0.993	0.993	0.998
	SVM	0.914	0.742	0.695	0.555	0.547	0.951
CCC	LR	0.967	0.923	0.627	0.625	0.481	0.900
	RF	0.999	0.994	0.956	0.953	0.978	0.936
	XGBoost	1.000	0.998	0.987	0.987	0.989	0.986
	SVM	0.872	0.682	0.283	0.291	0.169	0.879
High Risk + Default	LR	0.997	0.989	0.533	0.592	0.369	0.960
	RF	1.000	0.999	0.958	0.959	1.000	0.920
	XGBoost	1.000	1.000	1.000	1.000	1.000	1.000
	SVM	0.997	0.985	0.467	0.548	0.305	1.000

Source: Own authorship.

3.4.5.1 XGBoost: Strongest Temporal Generalization

XGBoost consistently exhibited superior generalization performance when applied to the 2022 dataset.

- Robustness Across Segments: Across all credit risk categories, XGBoost preserved near-perfect performance, with AUC-ROC values exceeding 0.999 and F1-Scores above 0.987;
- High-Risk Segment Performance: In the “High Risk + Default” segment, the model achieved an AUC-ROC and F1-Score of 1.000, identical to the in-time evaluation. This confirms its exceptional capacity to identify high-risk firms with complete accuracy, even under temporal shifts;

- Stability Analysis: The comparison between in-time and out-of-time performance revealed negligible degradation, reinforcing the model's robustness for use in dynamic credit risk settings where data patterns evolve over time.

3.4.5.2 Random Forest: Slight Degradation, Yet High Reliability

RF showed mild, yet measurable, performance decline in the temporal test, although it remained a highly reliable model overall.

- Segment-Level Degradation: In the BB segment, the F1-Score declined from 0.965 to 0.954, and AUC-ROC decreased from 0.999 to 0.997. Despite this reduction, the model continued to deliver robust predictions;
- Consistent in Low-Risk Segments: In the "High Quality" and "BBB" categories, F1-Scores remained above 0.979 and AUC-ROC values above 0.999, suggesting minimal sensitivity to temporal drift in stable firms;
- Implications: While slightly less stable than XGBoost, RF remains a strong candidate for temporal credit forecasting, especially when model interpretability and resource efficiency are prioritized.

3.4.5.3 Logistic Regression: Stable but Limited Predictive Capability

LR maintained stable performance between in-time and out-of-time evaluations; however, its overall predictive capability remained inferior, particularly in high-risk segments.

- Predictive Ceiling: In the BB segment, the F1-Score increased modestly from 0.548 to 0.556, with AUC-ROC improving from 0.826 to 0.830. Despite stability, these values are significantly below those of tree-based models, limiting the model's practical utility;
- High-Risk Segment Performance: The "High Risk + Default" category yielded an F1-Score of 0.533, slightly higher than the in-time score of 0.523. Nonetheless, this level of performance falls short of the accuracy required for proactive risk management;
- Structural Constraints: The model's linear assumptions hinder its ability to capture complex patterns inherent in credit risk data, particularly in speculative-grade segments where firm characteristics are more heterogeneous and volatile.

3.4.5.4 Support Vector Machine: Marked Instability over Time

SVM demonstrated the most significant performance deterioration in the temporal validation, confirming its limited applicability in credit forecasting contexts.

- Sharp Declines: In the BB segment, the F1-Score declined from 0.474 to 0.357, while AUC-ROC dropped sharply from 0.749 to 0.422, indicating a substantial loss of discriminatory power;
- Performance in CCC Segment: The CCC segment also showed substantial degradation, with the F1-Score falling from 0.636 to 0.283—a reduction of over 50% in predictive efficacy;
- Conclusion: SVM exhibited high sensitivity to temporal changes, particularly in segments with greater class imbalance or non-linear risk patterns. Its instability and computational inefficiency suggest that it is not well-suited for credit risk applications where model robustness over time is essential.

This table summarizes the temporal validation results (2022 test set) for all four models—XGBoost, RF, LR, and SVM—across six credit risk segments. Metrics include AUC-ROC, Accuracy, F1-Score, MCC, Precision, and Recall.

3.4.6 Summary of Model Findings

To consolidate the main findings of this study, Table 3.8 provides a comparative overview of the four models analyzed—XGBoost, RF, LR, and SVM—highlighting their performance across key dimensions: AP, temporal robustness, F1-score range, FPR in risk segments, and recommended use cases.

Table 3.8 – Comparative Summary of Models.

Model	Average Precision (AP)	Temporal Robustness (2022)	Max F1-Score Range	False Positives (Risk Segments)	Recommended Use
XGBoost	1.000	High	0.990–1.000	~0.0%	High-accuracy credit risk systems with low error tolerance
RF	0.996	Moderate	0.960–0.990	<0.3%	Reliable applications requiring balance and interpretability
LR	0.747	Low	0.550–0.900	5.0–20%	Baseline models for stable and high-quality ratings
SVM	0.709	Very Low	0.450–0.790	>10%	Not recommended for credit risk classification

Source: Own authorship.

This comparative summary confirms the superior performance of XGBoost, which consistently achieved perfect or near-perfect results across all metrics (e.g., AUC = 1.000, AP = 1.000), along with strong generalization in the temporal holdout. Its main limitation remains moderate interpretability, a known trade-off in complex ensemble models.

RF performed nearly as well, with slightly lower precision but improved interpretability and greater operational simplicity. This makes it a viable alternative in contexts where transparency or computational resources are limiting factors.

LR, while interpretable and fast to deploy, underperformed in speculative-grade and high-risk segments. Its application should be limited to stable portfolios or as a benchmark in model comparisons.

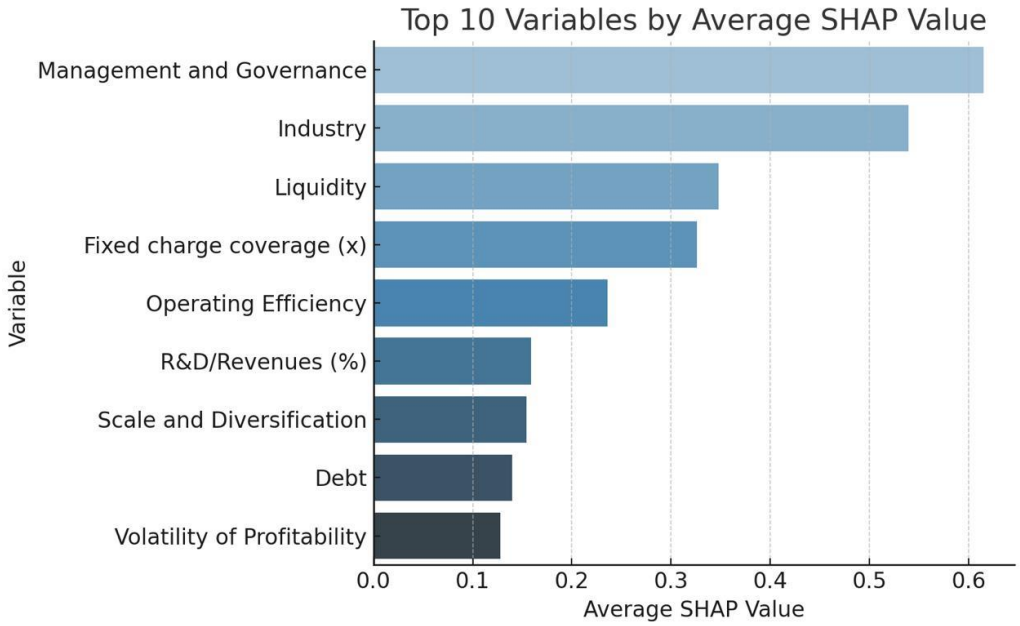
SVM demonstrated the weakest results, exhibiting low temporal generalization, wide variability in performance, and high FPR in critical segments. These issues render it unsuitable for practical use in financial credit risk classification systems.

In summary, this synthesis emphasizes that beyond predictive accuracy, robustness, interpretability, and deployment context are essential considerations when selecting models for real-world credit risk applications.

3.5 EXPLAINABILITY: SHAP ANALYSIS

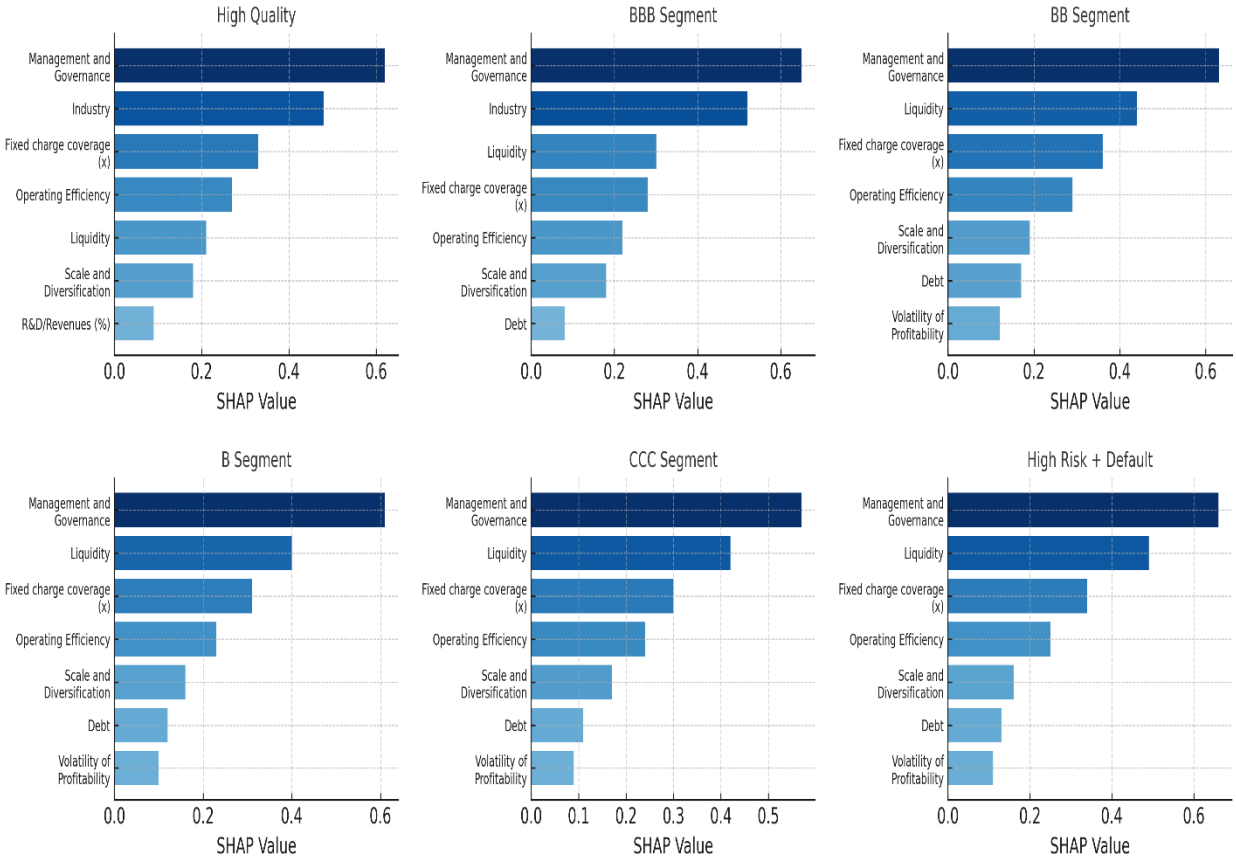
Supporting EP3, SHAP results reveal heterogeneous determinants of rating transitions across rating categories, with different sets of features driving outcomes in investment-grade versus speculative-grade firms (Figures 3.2 and 3.3).

Figure 3.2 – SHAP summary plot of top 10 predictive variables by credit rating segment (High Quality to High Risk + Default).



Source: Own authorship.

Figure 3.3 – Segment-specific ranking of the most influential features based on average SHAP values across the six risk categories.



Source: Own authorship.

3.5.1 Overall SHAP Framework

In line with EP5, governance quality and capital structure were dominant explanatory factors among higher-rated entities, consistent with prior evidence on rating stability (Figure 3.3).

Aligned with EP4, liquidity and coverage variables exerted the strongest influence in lower-rated firms, particularly in the CCC and High Risk + Default segments (Figure 3.3).

To enhance the interpretability of the ML models, we applied SHAP, a model-agnostic technique that attributes prediction outcomes to individual input variables. SHAP enables both global and segment-level analyses by quantifying the marginal contribution of each feature to the model's output. This capability is essential for understanding and validating automated credit risk assessments, particularly in regulated environments where transparency is critical.

SHAP has been successfully adopted across multiple high-stakes domains, including geological classification (Antonini *et al*., 2024), healthcare diagnostics (ElShawi *et al*., 2021), and photovoltaic forecasting (Bakht *et al.*, 2025). These applications reinforce its relevance in settings that demand explainability, robustness, and domain-specific interpretability—characteristics that align closely with the requirements of credit risk modeling.

Figure 3.2 presents the consolidated SHAP summary plots by credit rating segment, revealing the most influential predictors of rating transitions and defaults.

Across all segments, key variables include management and governance quality, industry classification, liquidity, fixed charge coverage, and operating efficiency. These features are not only statistically important but also economically intuitive, reflecting established principles in credit evaluation.

In lower-rated segments—specifically CCC and High Risk + Default—liquidity emerges as a dominant variable, highlighting its relevance in assessing short-term solvency and distress risk. Conversely, higher-rated segments such as High Quality and BBB place greater weight on structural and strategic attributes, including management quality, industry exposure, and scale or diversification, which reflect the importance of forward-looking risk drivers among financially sound entities.

Profitability volatility, measured via earnings fluctuations, demonstrates consistent relevance across segments, particularly in BB and B categories. This underscores the role of earnings stability in signaling future rating transitions.

These findings suggest that the ML models dynamically adjust their explanatory basis based on the firm's credit quality. Rather than relying on a static set of predictors, the algorithms

prioritize different variables depending on the risk profile, demonstrating adaptability to the heterogeneity of real-world credit environments.

Figure 3.3 further illustrates this pattern, summarizing the top features across rating segments.

While variables like liquidity, operating efficiency, and governance consistently appear among the most impactful, their relative influence shifts: short-term solvency and earnings risk dominate in distressed segments, whereas industry dynamics and firm-level strategic characteristics prevail in investment-grade categories. These results align with domain knowledge and regulatory practices.

In conclusion, the SHAP-based interpretability framework confirms that the ML models offer not only strong predictive performance but also economic and theoretical coherence. Their decisions are grounded in well-established financial indicators, enhancing auditability, transparency, and trust—essential characteristics for the practical adoption of AI systems in credit risk management.

3.6 DISCUSSION

To ensure coherence between the theoretical foundations, the research design, and the empirical findings, we evaluated the five empirical propositions formulated in Section 3.2.6. Each empirical proposition was tested against the results presented in Section 3.4, covering both predictive performance and feature interpretability across rating segments. Table 3.9 summarizes the verification outcomes, indicating whether each empirical proposition was supported by the evidence.

Table 3.9 – Summary of Empirical Fidings.

Empirical Propositions	Statement	Verification Outcome
EP1	Ensemble models (RF, XGBoost) outperform traditional models (LR, SVM) in predicting rating transitions and defaults.	Supported—XGBoost and RF achieved higher AUC and F1-scores compared with LR and SVM, both in cross-validation and out-of-time testing.
EP2	Segment-specific modeling by rating band produces higher predictive performance than aggregated modeling.	Supported—Stratified models consistently outperformed aggregated models, confirming the benefit of segment-specific approaches.
EP3	The determinants of rating transitions vary significantly across rating segments.	Supported—SHAP analysis revealed heterogeneous explanatory drivers across investment-grade and speculative-grade firms.
EP4	Liquidity and coverage metrics play a stronger role in explaining transitions among lower-rated firms.	Supported—Variables such as Fixed charge coverage and liquidity ranked among the most influential in BB, B, and CCC segments.
EP5	Management and Governance factors are more relevant in explaining transitions among higher-rated firms.	Supported—Management and Governance dominated feature importance for investment-grade firms.

Source: Own authorship.

The empirical evidence provides comprehensive evidence in line with the five empirical propositions, emphasizing the significance of this study's contributions. The superior performance of ensemble models demonstrates the capacity of advanced ML techniques to model nonlinear relationships and complex interdependencies within credit risk data. The benefits of segment-specific analysis reveal the heterogeneous dynamics underlying different rating categories, underscoring the necessity of disaggregated approaches. Moreover, the distinct roles of liquidity, coverage, and governance variables illustrate the value of interpretable methods such as SHAP in uncovering the economic rationale for rating transitions. Taken together, these findings highlight that predictive accuracy, and interpretability can be jointly pursued, offering a more transparent and practically relevant framework for credit risk assessment and management.

3.6.1 Strengths of Tree-Based Models

The results indicate that tree-based models, particularly XGBoost and RF, outperform traditional classifiers in predicting rating transitions and corporate defaults. These models achieved exceptional predictive performance across all rating segments, consistently posting AUC-ROC and F1-Scores near or equal to 1.000. Their robustness was further confirmed in temporal validation, where XGBoost maintained high predictive power in 2022 out-of-sample data, demonstrating strong generalization capacity.

Both models also showed resilience to class imbalance. In minority-class segments such as “High Risk + Default” and “CCC,” XGBoost achieved high recall and precision simultaneously—a task where traditional models often struggle—due to its capacity to model complex, non-linear relationships among financial and governance indicators.

From an operational standpoint, RF delivered competitive performance with slightly greater interpretability and reduced computational demands, suggesting its suitability for deployment in resource-constrained environments. SHAP analysis confirmed that model predictions were driven by meaningful variables (e.g., liquidity, fixed charge coverage, management and governance quality, and operating efficiency), reinforcing their reliability and explainability (Li, 2022; Zhang *et al*., 2023).

As detailed in Table 3.9, XGBoost consistently achieved cross-validated F1-Scores above 0.986 and MCC values near 1.000 across all rating segments. RF also demonstrated solid generalization, with F1-Scores and MCCs typically exceeding 0.950. These findings support the efficacy of ensemble-based models in predictive credit risk analytics and their viability for real-world application in financial decision-making.

3.6.2 Limitations of Linear and SVM Models

While LR produced adequate results in high-quality and investment-grade segments, its performance declined substantially in lower-rated classes. Its linear assumptions limit the model’s ability to capture the complex, nonlinear relationships often exhibited by financially deteriorating firms. For instance, in the BB segment, LR yielded a cross-validated F1-Score of 0.544 and an MCC of 0.420 (Table 3.9), limiting its practical applicability in EWS and proactive risk management.

In contrast, SVM demonstrated the most inconsistent outcomes among the evaluated models. Despite employing kernel methods and regularization, SVM underperformed across most evaluation metrics, particularly in segments characterized by data imbalance. It exhibited elevated FPR, longer computational times, and a high sensitivity to hyperparameter settings. In the BB segment, for example, it recorded a CV F1-Score of just 0.353 and an MCC of 0.106 (Table 3.9), highlighting its instability in this context.

These limitations underscore the importance of utilizing models that can effectively manage high-dimensional, nonlinear, and imbalanced data—features that are intrinsic to corporate credit risk datasets.

3.6.3 Model Trade-Offs: Performance vs. Interpretability

Although XGBoost outperformed other models in terms of accuracy and generalization, it does so at the expense of interpretability. Its complexity—stemming from multiple DT and iterative boosting—hinders direct human understanding. In contrast, RF, while slightly less accurate, provides greater model transparency due to its ensemble averaging mechanism and reduced sensitivity to overfitting.

LR continues to serve as a benchmark for interpretability, as its coefficients can be directly associated with economic reasoning. This makes it particularly suitable for explainability-focused contexts, such as regulatory oversight (de Lange *et al*., 2022; Nallakaruppan *et al*., 2024). However, its simplicity also constrains its performance in more complex or volatile rating segments.

Table 3.10 further illustrates this trade-off: LR maintained stable cross-validated F1-Scores in investment-grade segments (e.g., 0.947 in High Quality and 0.830 in BBB), despite a noticeable decline in performance across speculative-grade classes.

Table 3.10 – Cross-Validation and Model Performance Stability by Rating Segment.

(continues)

Segment	Model	AUC-ROC	Accuracy	CV F1-Score (mean)	CV MCC (mean)	F1-Score	MCC	Precision	Recall
High Quality	RF	1.000	0.998	0.997	0.996	0.993	0.992	0.998	0.988
	LR	0.987	0.965	0.947	0.920	0.877	0.860	0.815	0.948
	SVM	0.978	0.977	0.538	0.170	0.912	0.898	0.896	0.928
	XGBoost	1.000	1.000	0.999	0.999	1.000	1.000	1.000	0.999
BBB	RF	1.000	0.991	0.982	0.975	0.983	0.977	0.990	0.976
	LR	0.955	0.895	0.830	0.765	0.827	0.761	0.755	0.916
	SVM	0.945	0.878	0.468	0.176	0.801	0.722	0.726	0.892
	XGBoost	1.000	1.000	0.998	0.997	1.000	0.999	1.000	0.999
BB	RF	0.999	0.986	0.951	0.938	0.965	0.956	0.979	0.951
	LR	0.826	0.738	0.544	0.420	0.548	0.424	0.421	0.786
	SVM	0.714	0.556	0.353	0.106	0.419	0.233	0.285	0.791
	XGBoost	1.000	0.999	0.996	0.995	0.997	0.996	0.999	0.995
B	RF	0.999	0.983	0.962	0.945	0.973	0.960	0.971	0.975
	LR	0.941	0.865	0.801	0.708	0.804	0.713	0.726	0.902
	SVM	0.927	0.861	0.603	0.407	0.787	0.687	0.743	0.837
	XGBoost	1.000	0.999	0.996	0.994	0.998	0.997	0.997	0.999

Table 3.10 – Cross-Validation and Model Performance Stability by Rating Segment.

(conclusion)

Segment	Model	AUC-ROC	Accuracy	CV F1-Score (mean)	CV MCC (mean)	F1-Score	MCC	Precision	Recall
CCC	RF	0.999	0.995	0.996	0.995	0.963	0.960	0.968	0.958
	LR	0.965	0.919	0.878	0.815	0.616	0.617	0.466	0.910
	SVM	0.958	0.947	0.578	0.291	0.694	0.677	0.592	0.837
	XGBoost	1.000	0.998	0.998	0.998	0.987	0.986	0.977	0.997
High Risk + Default	RF	1.000	1.000	1.000	1.000	0.983	0.983	1.000	0.967
	LR	0.998	0.989	0.990	0.985	0.523	0.582	0.361	0.950
	SVM	0.996	0.986	0.875	0.819	0.470	0.543	0.310	0.967
	XGBoost	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000

Source: Own authorship.

The application of SHAP values helps mitigate the trade-off between predictive performance and interpretability by providing post hoc explanations of model behavior. SHAP allows complex models such as XGBoost to be interpreted at both global and local levels, offering transparency into how individual features influence predictions. Although originally applied in geospatial and environmental modeling (Li, 2022; Zhang *et al*., 2023), this interpretability framework is equally valuable in financial applications, where model transparency supports more responsible and explainable decision-making.

3.6.4 Practical Implications for Credit Risk Monitoring

These findings have important implications for financial institutions and regulators. Tree-based models can be effectively integrated into credit monitoring systems as early warning tools capable of flagging firms at considerable risk of downgrade or default. Their capacity to identify rare yet impactful events with low error rates is essential for mitigating systemic risk.

XGBoost, with its superior predictive accuracy, is particularly suitable for high-stakes applications where false negatives must be minimized. In contrast, RF offers a robust alternative when computational efficiency and interpretability are prioritized. LR, despite its limitations, may still be appropriate for low-risk portfolios or as a benchmark model due to its transparency and simplicity.

Furthermore, the use of temporal validation underscores these models' potential to remain effective under changing macroeconomic conditions—an essential feature in volatile

financial environments. This adaptability supports their application in dynamic credit risk models, stress testing frameworks, and real-time surveillance systems.

3.6.5 Limitations and Future Work

Despite rigorous validation, several limitations should be acknowledged. First, the dataset used in this study originates from a single institutional source, which may limit the external validity and generalizability of the findings. Future research could incorporate data from multiple rating agencies or international sources to capture broader market heterogeneity.

Second, the analysis relies exclusively on structured numerical and categorical data. Integrating unstructured sources—such as management commentary, news sentiment, and analyst reports, along with forward-looking indicators (e.g., CDS spreads or market-implied signals), may enhance predictive accuracy and contextual depth.

Third, while SHAP was effectively applied to interpret model outputs, future work could explore methods that embed explainability directly into the training process. Techniques such as explainability-aware objectives or inherently interpretable architectures could improve transparency without compromising performance. In addition, hybrid models that combine traditional credit scoring with ML, as well as semi-supervised or transfer learning approaches to exploit sparse or unlabeled data, present promising directions for advancing model robustness.

Finally, although temporal validation was conducted using a hold-out year (2022), extending the validation across longer horizons and multiple economic cycles would further support the model's reliability. Evaluating performance across diverse market regimes, industries, and geographic regions will be critical to confirm the stability and adaptability of these models in dynamic financial environments.

To mitigate overfitting and enhance generalizability, several safeguards were implemented. Hyperparameter tuning was conducted using stratified cross-validation and independent test sets. Temporal validation demonstrated that models trained on 2017–2021 data maintained high performance when applied to 2022 observations, suggesting the capture of generalizable patterns. Furthermore, SHAP analysis across folds and rating segments revealed consistent variable importance, reinforcing the models' stability and resilience to noise and sampling variation.

3.7 CONCLUSIONS

This study proposed a hybrid ML framework to explain and predict corporate credit rating transitions and defaults. By integrating advanced classification algorithms—XGBoost, RF, LR, and SVM—with segment-specific modeling, temporal validation, and SHAP-based explainability, the research offers a robust and interpretable approach to credit risk assessment.

Empirical results confirm that tree-based ensemble models, particularly XGBoost, outperform traditional linear models and SVM across all rating segments and under various validation schemes. These models demonstrated superior predictive accuracy, resilience to class imbalance, and stability in out-of-time testing. Importantly, the use of SHAP values enabled transparent interpretation of key risk drivers, tailored to each credit rating segment, thereby enhancing model trustworthiness in regulatory and operational settings.

The findings contribute to both academic literature and industry practice by addressing several known gaps: (i) the lack of segment-level model stratification, (ii) the underuse of temporal validation in credit risk studies, (iii) the need for model interpretability in regulated environments, and (iv) the practical integration of explainability into EWS. The results suggest that ML-based frameworks—when properly tuned and explained—can support more proactive and transparent credit surveillance.

These contributions, however, are bounded by several assumptions and limitations. The analysis assumes that the 2017–2022 period adequately reflects the dynamics of rating transitions, even though more recent shocks may alter risk patterns. It also assumes that the chosen financial and governance variables sufficiently capture rating determinants, leaving aside unstructured disclosures, macro indicators, and market-based signals that may enrich predictive performance. Methodologically, the study assumes that the selected ML algorithms and tuning strategy fairly represent the performance–interpretability trade-off, though alternative approaches may yield different outcomes. Finally, the segmentation by rating bands is assumed to be the most relevant dimension of heterogeneity, even though industry or regional factors could provide complementary insights.

Future research should explicitly test these assumptions by extending datasets beyond 2022, incorporating qualitative and forward-looking variables, and experimenting with hybrid models that embed interpretability directly into the learning process. Such efforts would not only validate the robustness of the present framework but also enhance its adaptability to systemic shocks and evolving regulatory expectations.

Overall, this research demonstrates that integrating ML with domain-relevant segmentation and e XAI tools can significantly improve the effectiveness, transparency, and adaptability of credit risk modeling. As financial markets evolve and regulatory scrutiny increases, such frameworks are likely to become indispensable for institutions seeking to strengthen risk governance and decision-making under uncertainty.

4 FINAL CONSIDERATIONS

This thesis provided an integrative examination of how ML can transform the prediction and explanation of corporate credit risk. Across its two main studies, the research demonstrated both the predictive superiority and interpretive value of ML techniques compared with traditional rating methodologies. Together, these articles contribute to advancing a data-driven yet transparent framework for credit risk modeling that aligns predictive performance with regulatory and managerial needs.

The first article, *Advancing Credit Rating Prediction: The Role of Machine Learning in Corporate Credit Rating Assessment*, established the empirical and methodological foundation for this investigation. By benchmarking a comprehensive suite of models—including LR, SVM, DT, RF, GB, and ANN—the study confirmed that ensemble and neural approaches consistently outperform linear methods. Their ability to capture non-linear dependencies and complex interactions across financial and business-risk variables enhances both the precision and adaptability of corporate credit assessments. Moreover, the analysis revealed cross-country variability in predictive performance, underscoring the importance of context-specific modeling and the sensitivity of ML outcomes to local market structures and data characteristics. The article's findings substantiate ML's capability to strengthen traditional credit rating frameworks by improving accuracy, scalability, and operational efficiency.

The second article, *Explaining Corporate Ratings Transitions and Defaults Through Machine Learning*, extended this foundation toward explainability and regime awareness. Through the application of SHAP, the study illuminated how ML models can offer transparent, firm-level justifications for predicted rating transitions and defaults. This interpretability mitigates the "black-box" problem often associated with complex algorithms and promotes stakeholder confidence and regulatory acceptability. The research also introduced a segment-specific modeling strategy, tailoring models across rating bands to capture heterogeneity in credit dynamics. Additionally, it integrated MSMs to account for cyclical and regime-dependent variations in credit behavior, illustrating how state-space approaches complement ML in identifying transition patterns under different macroeconomic conditions. Collectively, these elements contribute to developing an interpretable early-warning system that bridges analytical rigor and policy relevance.

Synthesizing the insights of both studies, this thesis contributes to three complementary domains. Theoretically, it advances the literature on the intersection between ML and financial econometrics by articulating a structured approach that balances predictive performance with

interpretability. Empirically, it provides robust evidence of ML's capacity to outperform traditional credit-risk models, while addressing the nuances of data heterogeneity, temporal dynamics, and regulatory constraints. Practically, it offers guidance for analysts, regulators, and rating agencies seeking to operationalize ML responsibly emphasizing transparency, fairness, and accountability as prerequisites for sustainable adoption.

Nevertheless, the research acknowledges certain limitations. The reliance on structured, firm-level data constrains the model's capacity to incorporate unstructured or alternative sources, such as ESG indicators, supply-chain dependencies, or text-based disclosures. Future investigations should therefore explore hybrid econometric–ML architectures and explainability-aware training methods capable of integrating such data streams. Additionally, further comparative analyses across sectors and economic regimes could clarify the generalizability of ML models in diverse financial environments.

In conclusion, this thesis demonstrates that ML, when combined with XAI tools and regime-sensitive modeling, can meaningfully enhance the transparency and predictive capacity of corporate credit-rating systems. The integration of interpretable ML within credit analysis does not merely improve forecasting accuracy; it also reinforces the trust, accountability, and prudential soundness that underpin modern financial supervision. By aligning technological innovation with methodological rigor and ethical governance, this work positions ML as a cornerstone of the next generation of responsible and explainable credit risk management.

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APPENDIX A – Confusion Matrices

Confusion matrices allow for the evaluation of the distribution of correct and incorrect predictions made by the models within each rating segment. They provide a detailed view of which classes were most frequently misclassified, helping to identify potential model weaknesses and systematic error patterns.

OVERALL CONCLUSION FROM THE CONFUSION MATRICES

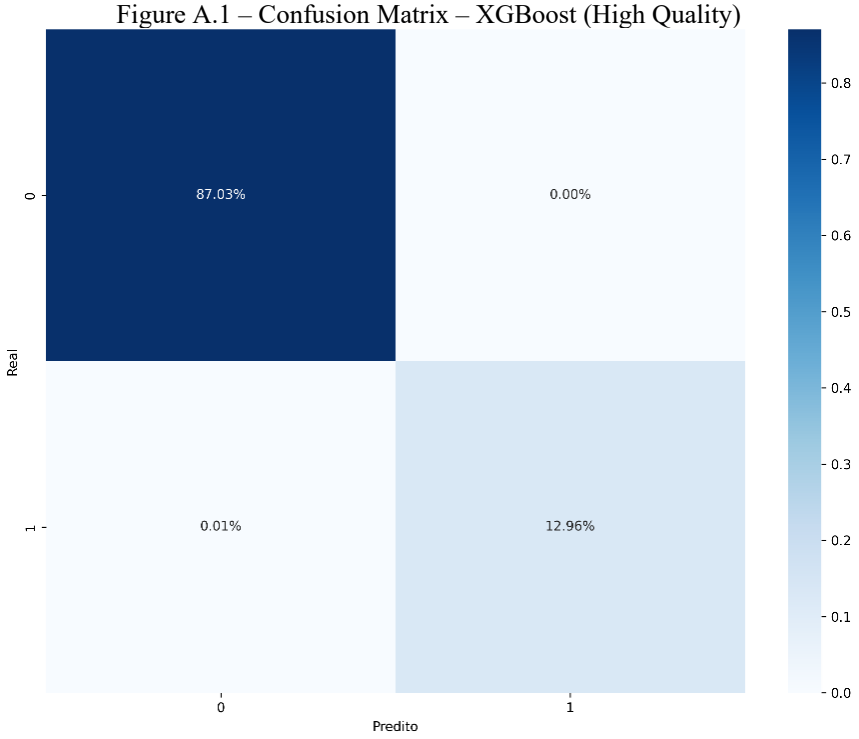
The analysis of confusion matrices across different rating segments enabled an assessment of the relative performance of ML models in predicting credit transitions and defaults. The results reveal significant differences among the approaches, with XGBoost and RF emerging as the most effective models, while LR and SVM showed considerable limitations.

XGBoost: Robustness, Generalization, and Efficiency in Risk Detection Across Multiple Segments

The consolidated analysis of the XGBoost model's confusion matrices reveals consistently superior performance across all segments, standing out not only for its high overall accuracy but especially for its ability to balance Type I and Type II errors—even in highly imbalanced class contexts.

In the High Quality segment, the model achieves near-perfect accuracy: 87.03% true negatives and 12.96% true positives, with no false positives (0%) and a virtually zero Type II error rate (0.01%). This performance indicates that the model accurately understands the patterns of reliable payers without generating false alarms.

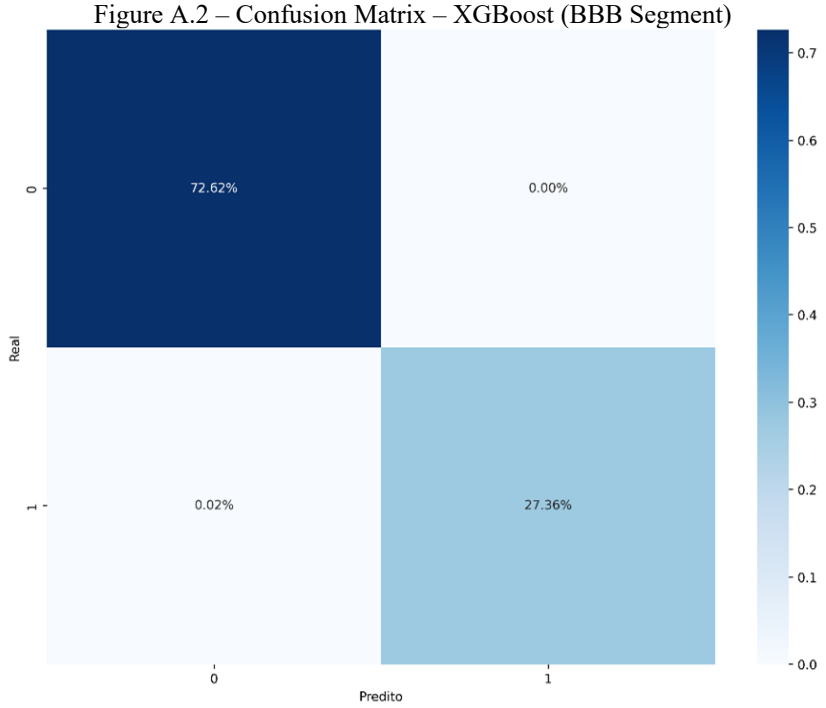
Figure A.1 – Confusion Matrix – XGBoost (High Quality)



Source: Own authorship.

This trend continues across other segments. In BBB, the model reaches 72.62% true negatives and 27.36% true positives—again with zero false positives and a negligible Type II error rate (0.02%). This ability to correctly detect both classes, especially the risk class, with minimal loss of precision, is a strong indication of effective generalization and bias control.

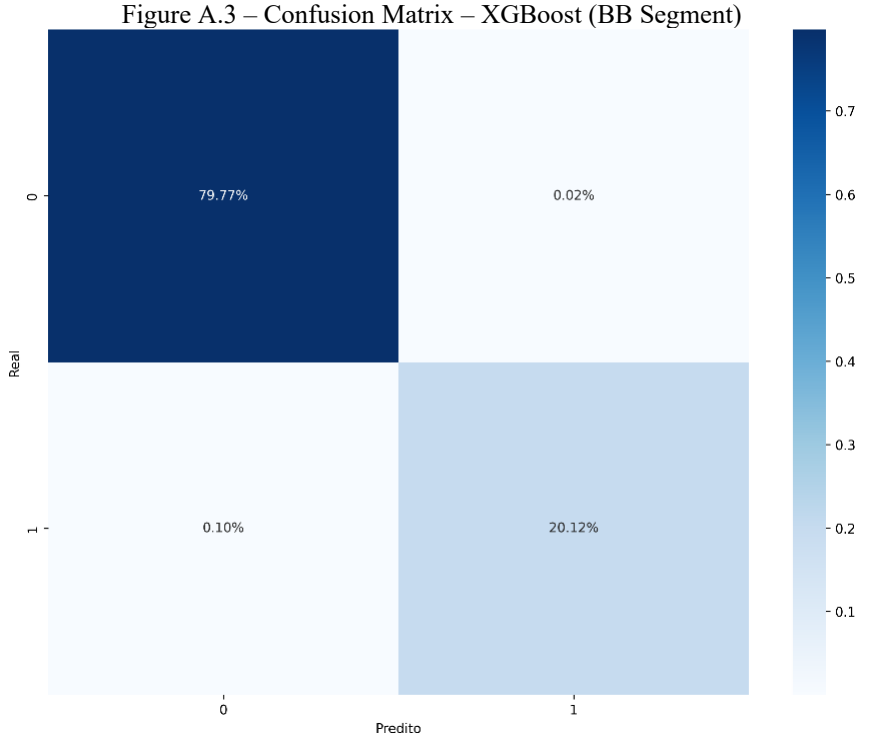
Figure A.2 – Confusion Matrix – XGBoost (BBB Segment)



Source: Own authorship.

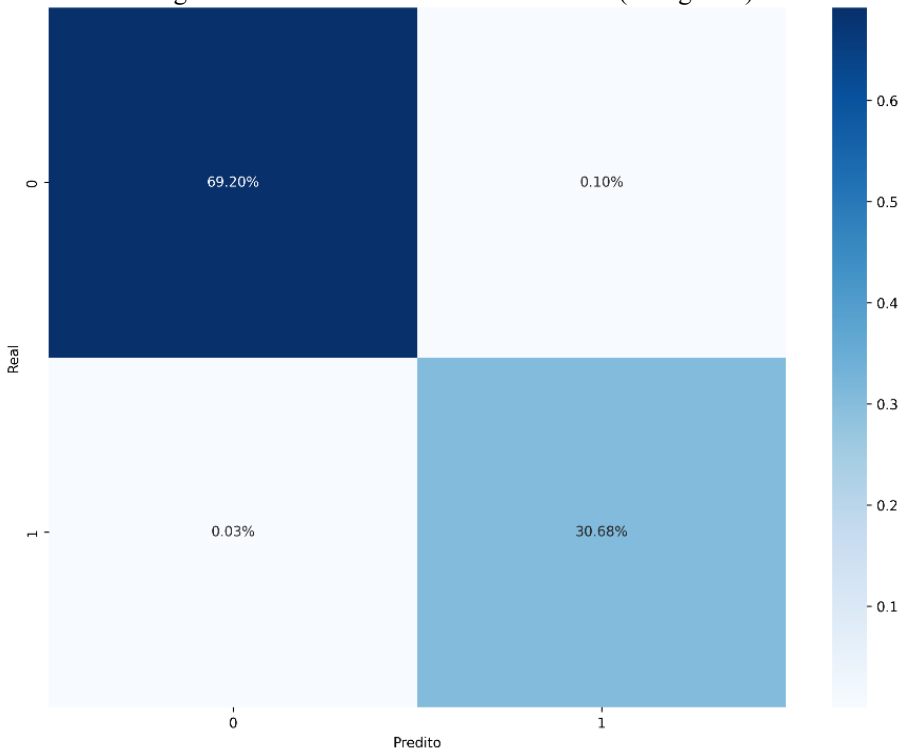
In the BB and B groups—where challenges increase due to greater instability and heterogeneity in firm characteristics—XGBoost continues to deliver strong results: in both segments, the proportion of correctly classified risk cases (class 1) exceeds 20% and 30%, respectively, with Type I error rates below 0.1% and virtually no false negatives. Even in the face of extreme class imbalance, the model manages to maintain high precision without sacrificing recall—a frequent challenge for traditional classifiers.

Figure A.3 – Confusion Matrix – XGBoost (BB Segment)



Source: Own authorship.

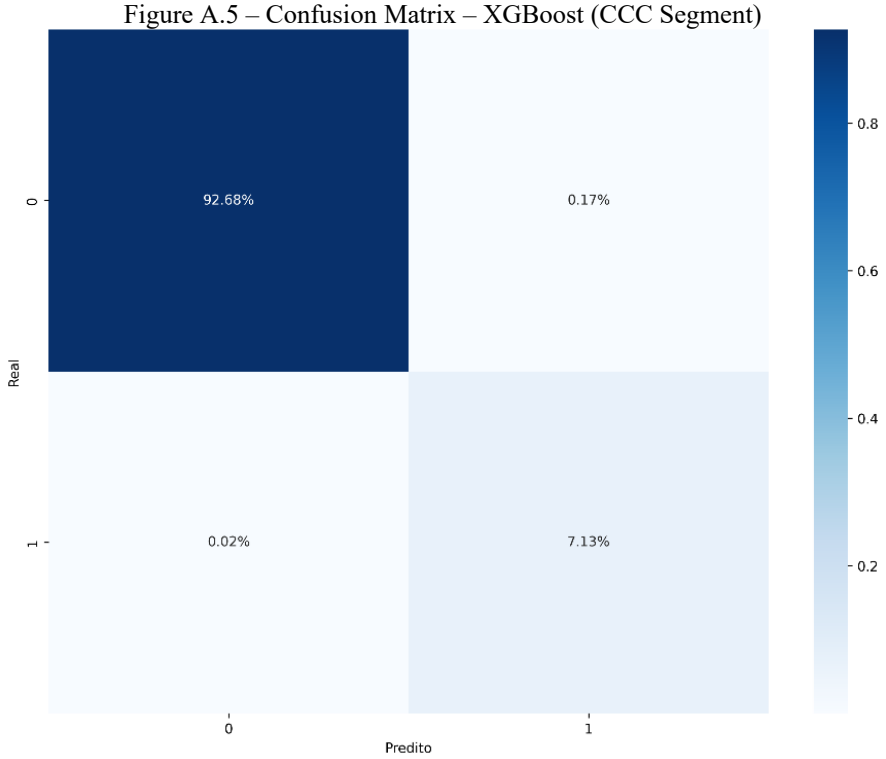
Figure A.4 – Confusion Matrix – XGBoost (B Segment)



Source: Own authorship.

XGBoost's robustness is further confirmed at the extremes of the risk spectrum. In the CCC segment, the model achieves 92.68% accuracy for class 0 and 7.13% for class 1, with marginal errors (0.17% and 0.02%).

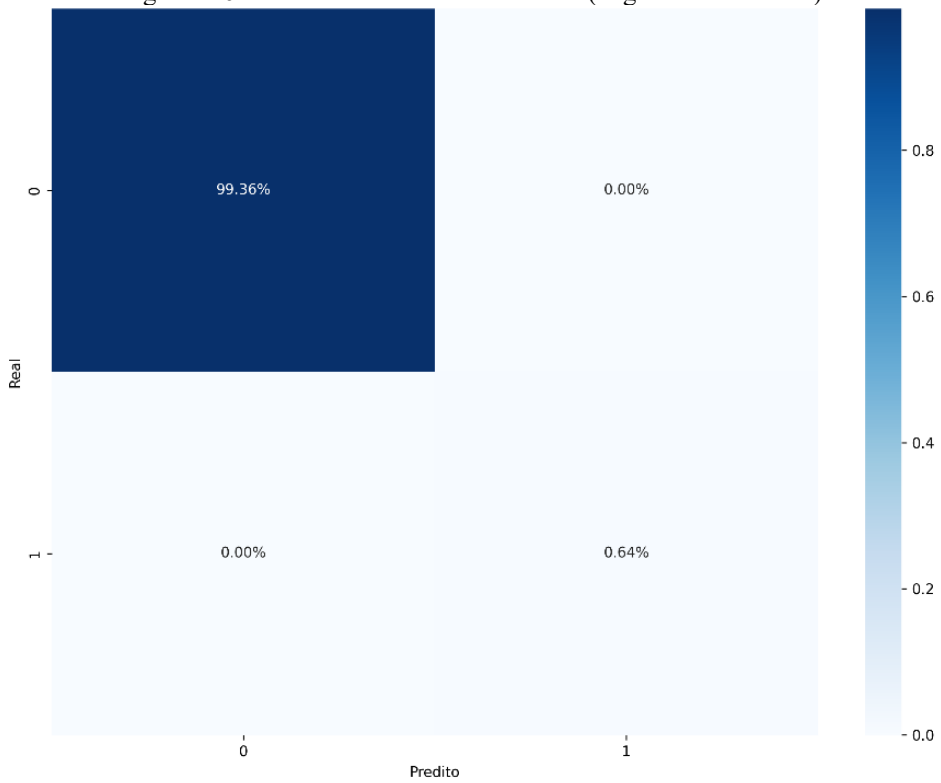
Figure A.5 – Confusion Matrix – XGBoost (CCC Segment)



Source: Own authorship.

In the High Risk + Default group, the accuracy is nearly perfect: 99.36% of non-risk cases are correctly identified and 100% of risk cases are recognized, without any false positives or false negatives.

Figure A.6 – Confusion Matrix – XGBoost (High Risk + Default)



Source: Own authorship.

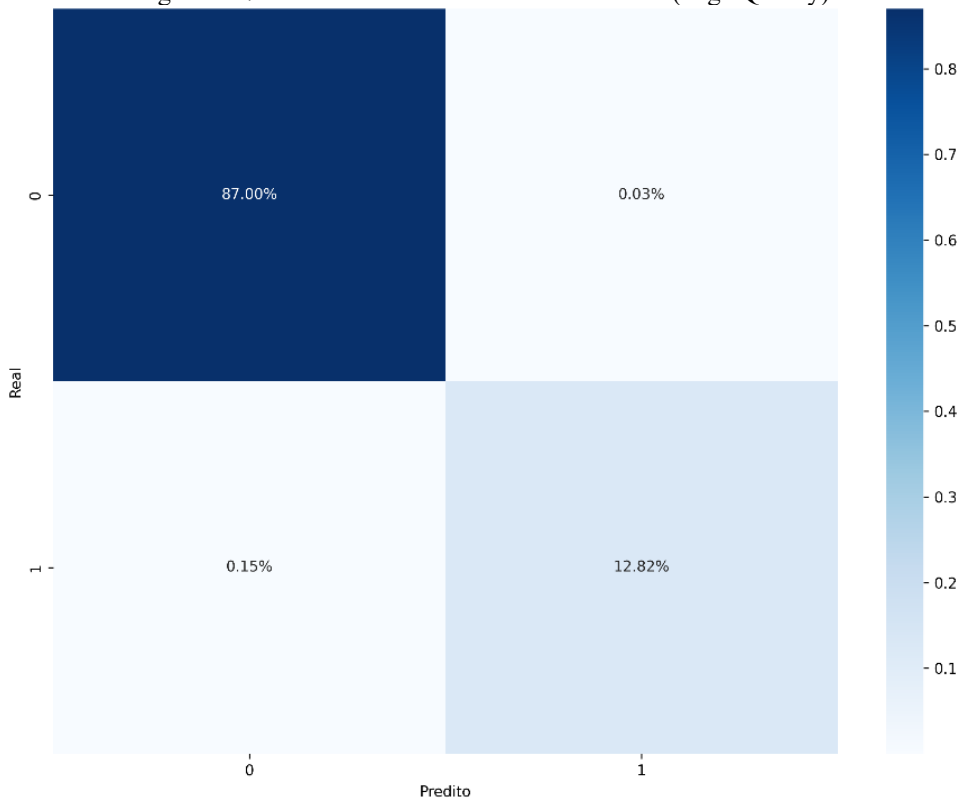
In summary, XGBoost demonstrates an impressive ability to adapt to different risk profiles, balancing sensitivity and specificity. The model proves to be extremely reliable both in identifying low-risk borrowers and in the early detection of defaults, positioning it as the most robust and accurate option among the methods analyzed thus far. Its stable performance, even under adverse conditions, reaffirms the effectiveness of the regularization and validation strategies employed, minimizing the risk of overfitting without compromising generalization—a decisive factor for real-world applications in credit evaluation systems.

Random Forest: Robustness, Stability, and Consistent Performance with Low Overfitting Risk

The analysis of the RF confusion matrices highlights a highly reliable and balanced performance across all rating segments, reflecting the model's capacity to capture non-linear relationships and complex variable interactions, with strong resilience to overfitting—mitigated by the use of cross-validation strategies, hyperparameter tuning, split parameter regularization, and class balancing techniques.

In the High-Quality segment, RF achieves 86.99% true negatives and 12.82% true positives, with extremely low false positive (0.03%) and false negative (0.15%) rates. This level of precision indicates high sensitivity combined with strong specificity—an ideal combination for alert systems that must avoid both false alarms and missed risk signals.

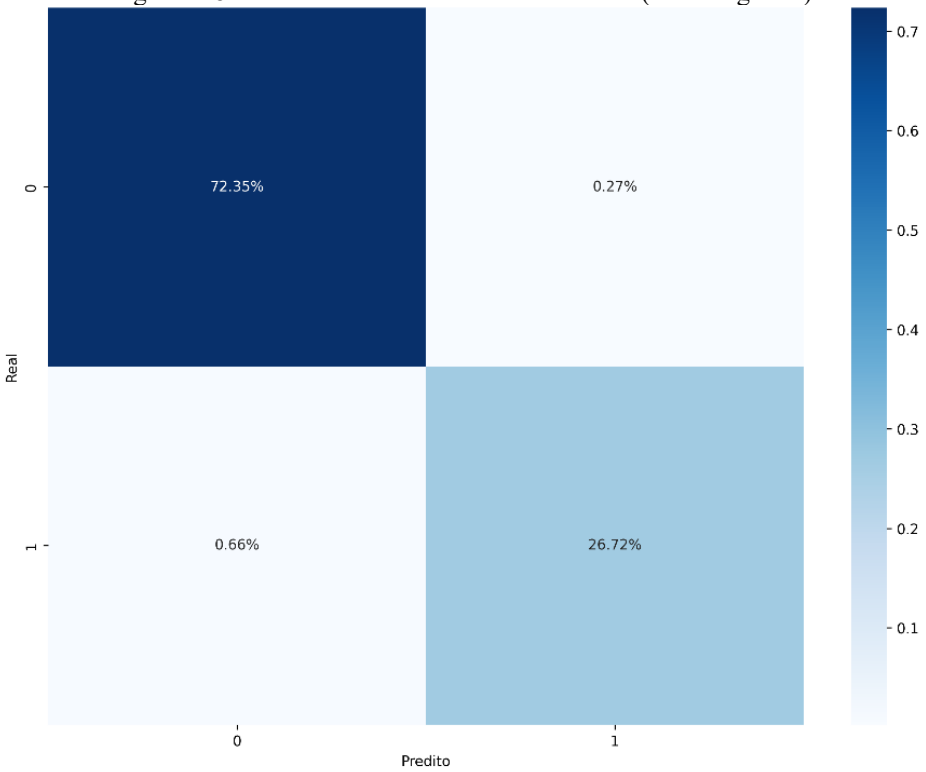
Figure A. 7 – Confusion Matrix – Random Forest (High Quality)



Source: Own authorship.

As the analysis progresses to higher-risk segments, the model's robustness remains consistent. In BBB, accuracy reaches 72.35% for class 0 and 26.71% for class 1, with false positives (0.27%) and false negatives (0.66%) still well controlled.

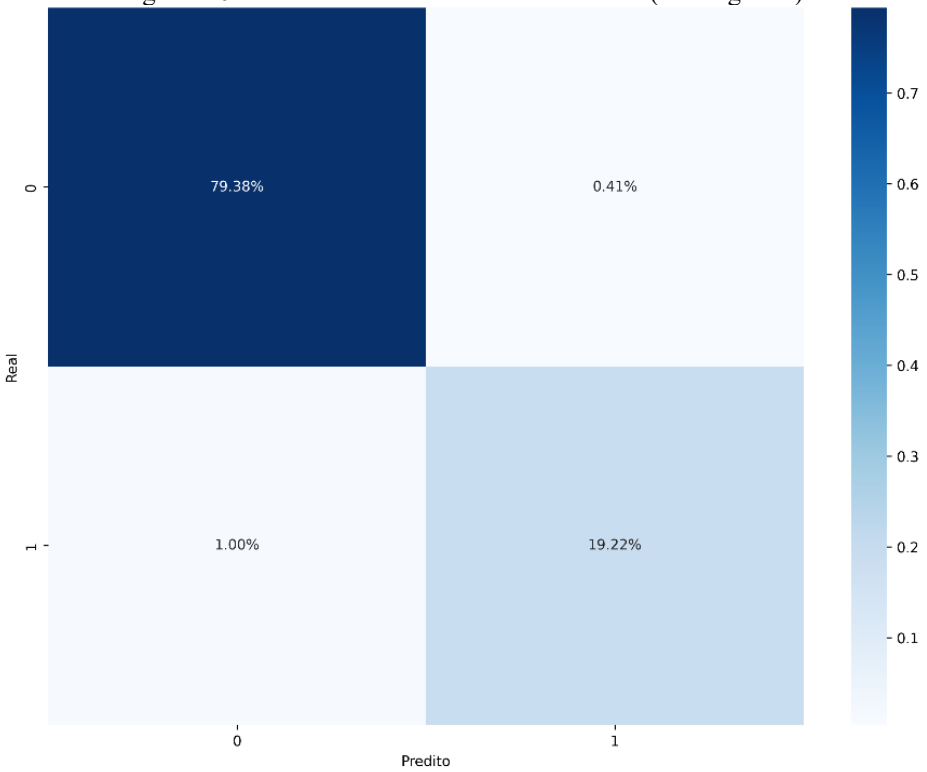
Figure A.8 – Confusion Matrix – Random Forest (BBB Segment)



Source: Own authorship.

In BB, the hit rates remain high (79.38% and 19.22%, respectively), with false positives (0.41%) remaining within tolerable levels for conservative applications.

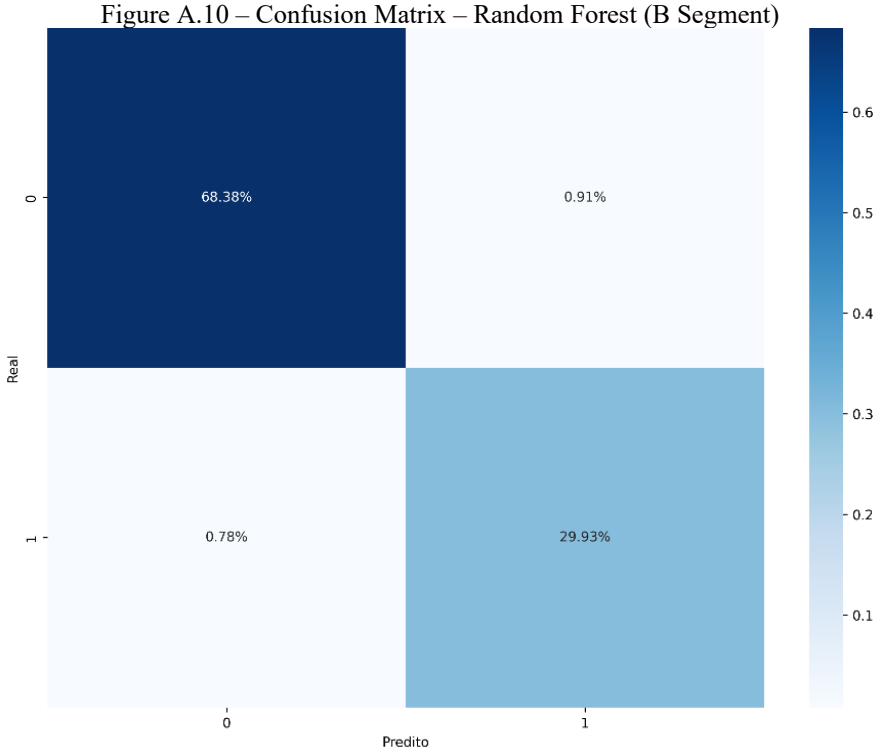
Figure A.9 – Confusion Matrix – Random Forest (BB Segment)



Source: Own authorship.

Performance continues to be consistent in the B segment, where accuracy for positive cases reaches 29.93%, and false negatives remain low (0.78%). Here, the model demonstrates efficiency in detecting the minority class without compromising overall reliability—a challenging balance in imbalanced datasets.

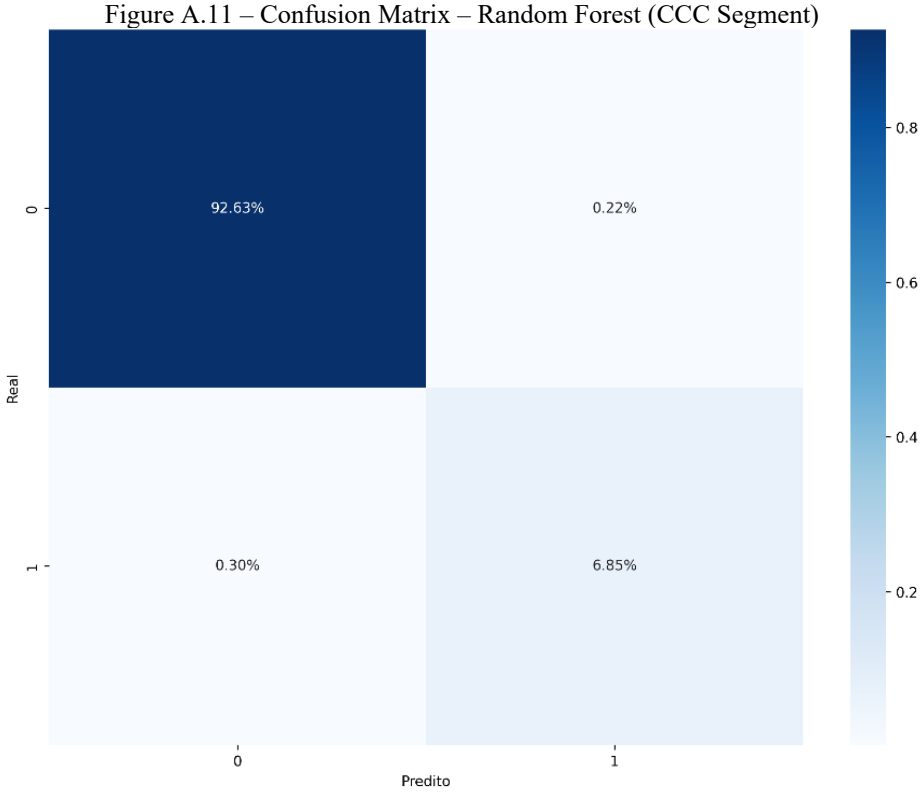
Figure A.10 – Confusion Matrix – Random Forest (B Segment)



Source: Own authorship.

In the CCC group, RF also performs solidly: 92.63% accuracy for class 0, 6.85% for class 1, with both false positive and FNR below 0.3%.

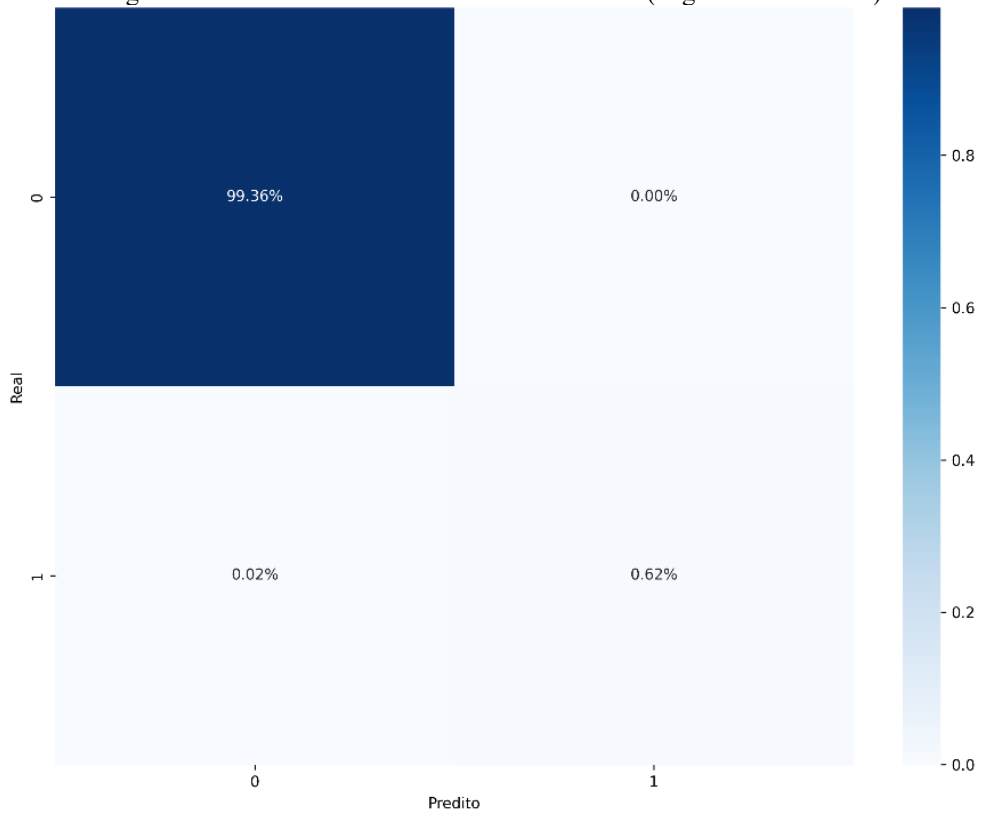
Figure A.11 – Confusion Matrix – Random Forest (CCC Segment)



Source: Own authorship.

Finally, in the most critical segment (High Risk + Default), the model approaches perfection, achieving 99.36% true negatives and 0.62% true positives—with no false positives and only a residual FNR (0.02%).

Figure A.12 – Confusion Matrix – Random Forest (High Risk + Default)



Source: Own authorship.

Overall, the RF model presents itself as a versatile and robust solution, with slightly lower performance than XGBoost in the most challenging segments, yet still highly competitive. Its structure—based on multiple independent trees—reduces bias without inflating variance, supporting strong generalization even in noisy and imbalanced scenarios.

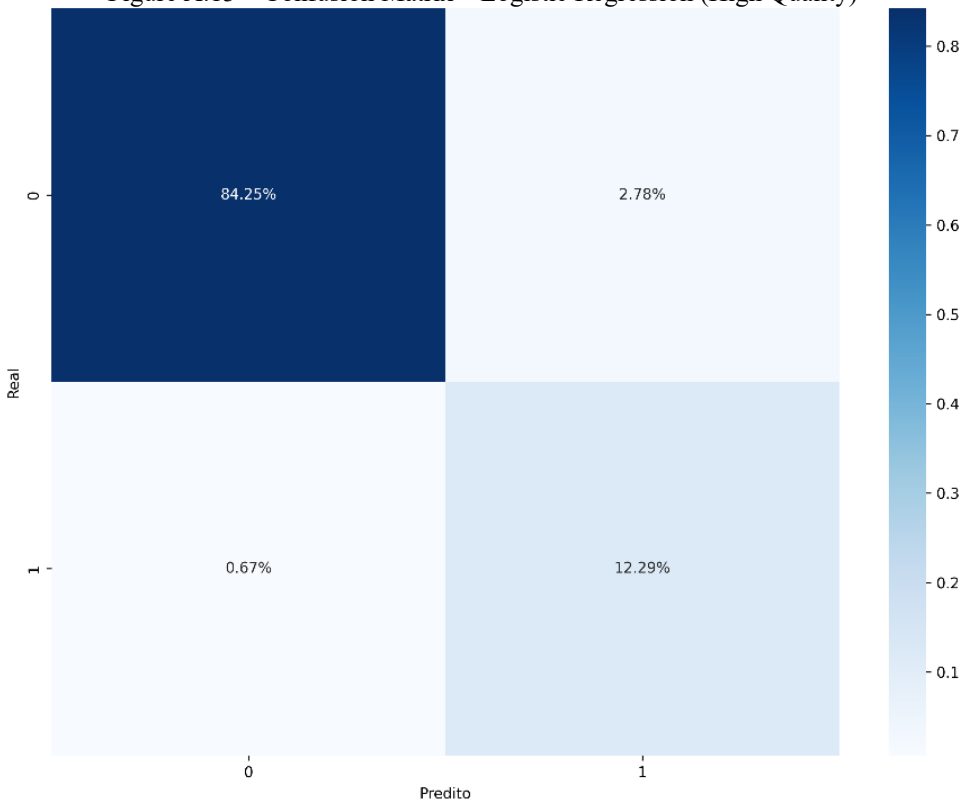
Although it does not reach the extreme precision levels of XGBoost, RF offers greater structural interpretability and excellent computational cost-efficiency, making it particularly attractive for applications in environments with operational constraints or the need for real-time explainability.

Logistic Regression: A Limited Option for Higher-Risk Segments

The consolidated analysis of the LR model's confusion matrices reveals a performance highly dependent on the quality of the evaluated segment, with progressively declining accuracy in higher-risk groups and those with lower statistical representativeness. While LR produces reasonable results in more stable segments such as High Quality and BBB, its performance drops significantly in the more critical segments, including BB, B, CCC, and High Risk + Default.

In the High Quality segment, the model showed high accuracy for both class 0 (84.25%) and class 1 (12.29%), with low Type I (2.78%) and Type II (0.67%) error rates.

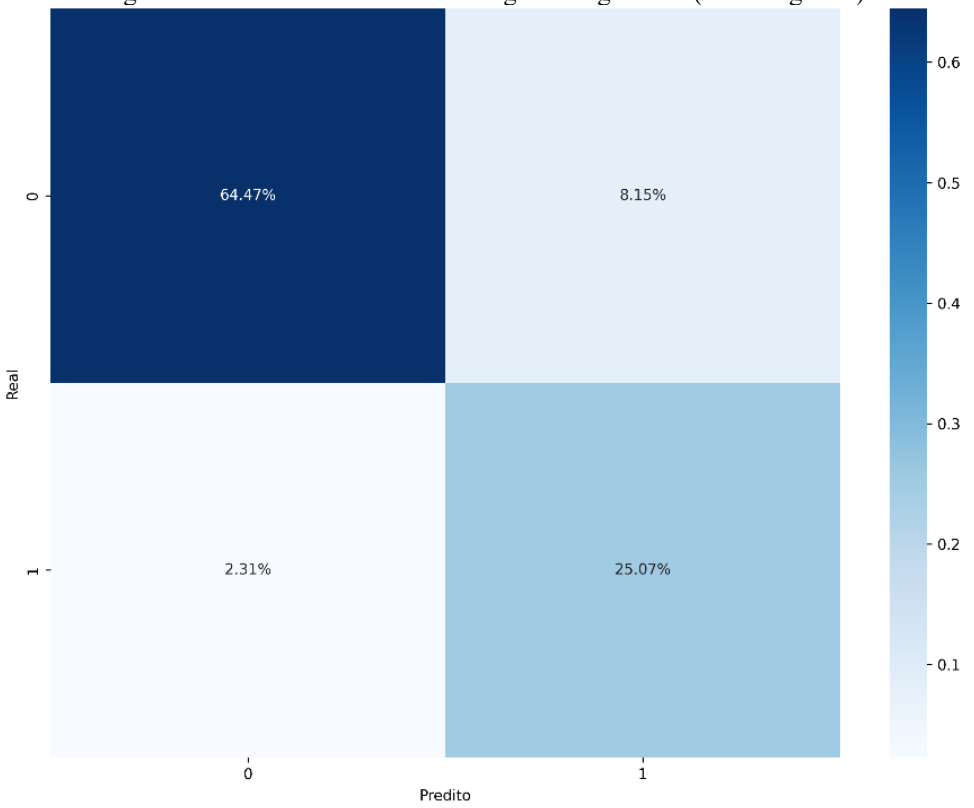
Figure A.13 – Confusion Matrix – Logistic Regression (High Quality)



Source: Own authorship.

However, in the BBB segment, there is a noticeable increase in false positives (8.15%) and false negatives (2.31%), indicating greater difficulty in maintaining classification balance.

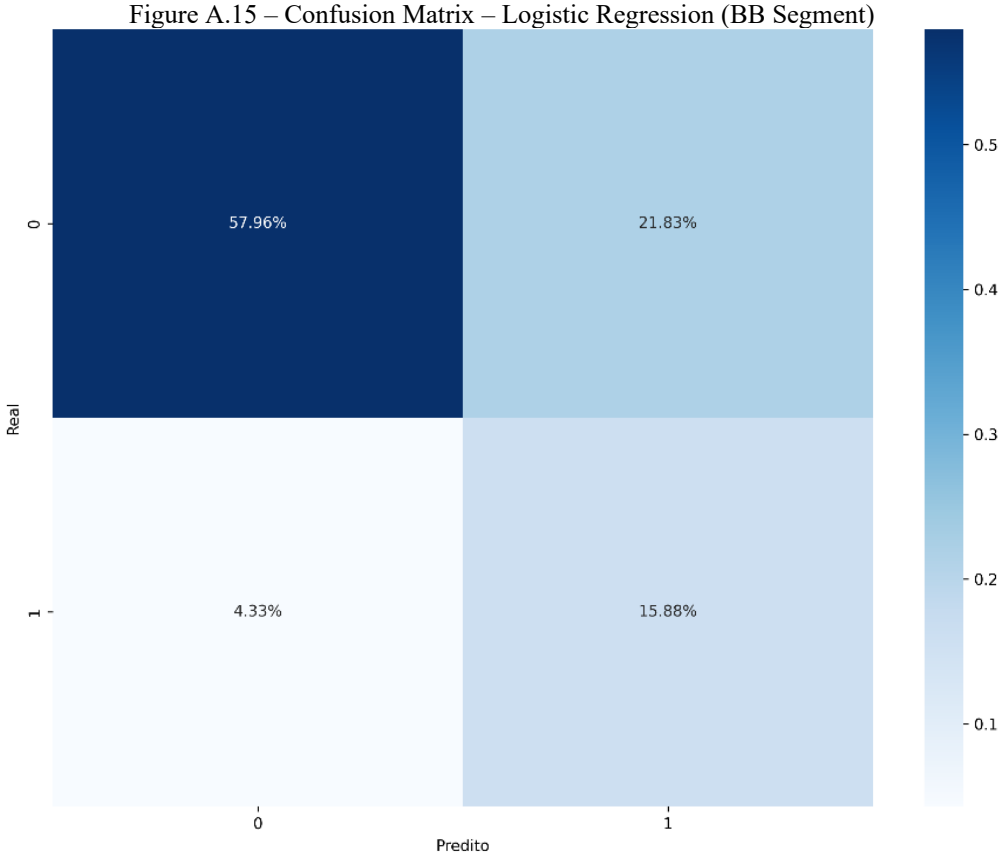
Figure A.14 – Confusion Matrix – Logistic Regression (BBB Segment)



Source: Own authorship.

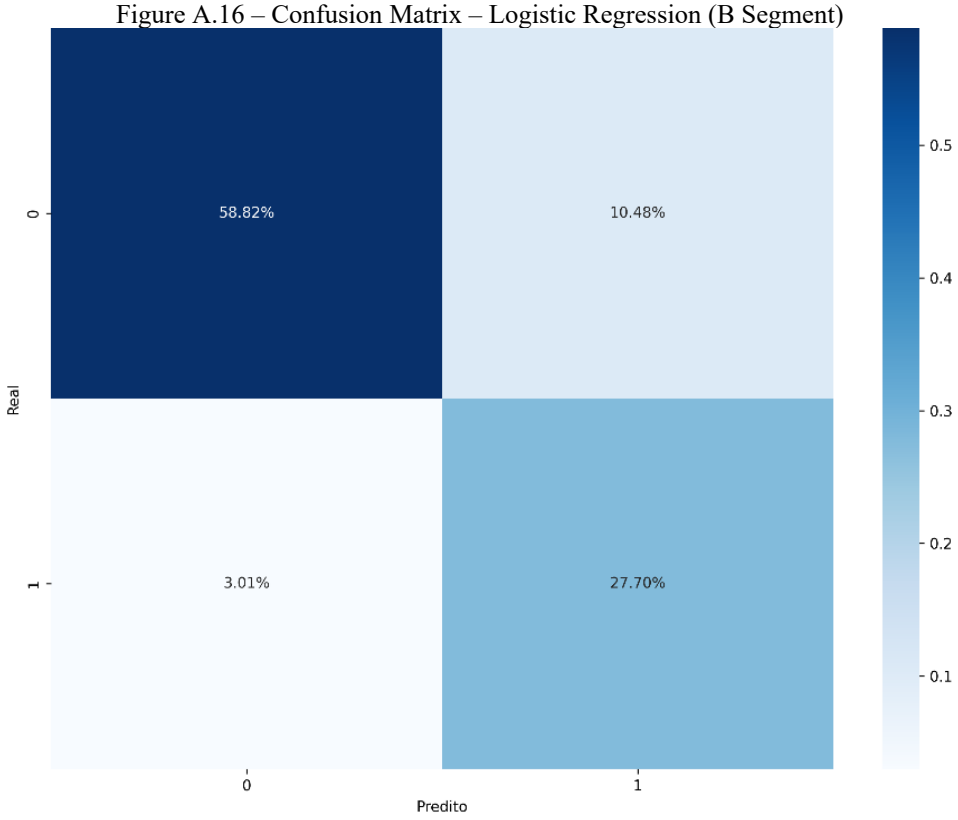
Model performance deteriorates significantly in the BB and B segments, with very high Type I error rates: 21.83% and 10.48%, respectively. These values represent a high incidence of incorrectly classifying healthy firms as risky, which undermines the model's practical utility for recommending corrective actions or supporting automated financial decisions. Additionally, the model shows limitations in correctly identifying the minority class (actual risk), with recall declining as risk level increases.

Figure A.15 – Confusion Matrix – Logistic Regression (BB Segment)



Source: Own authorship.

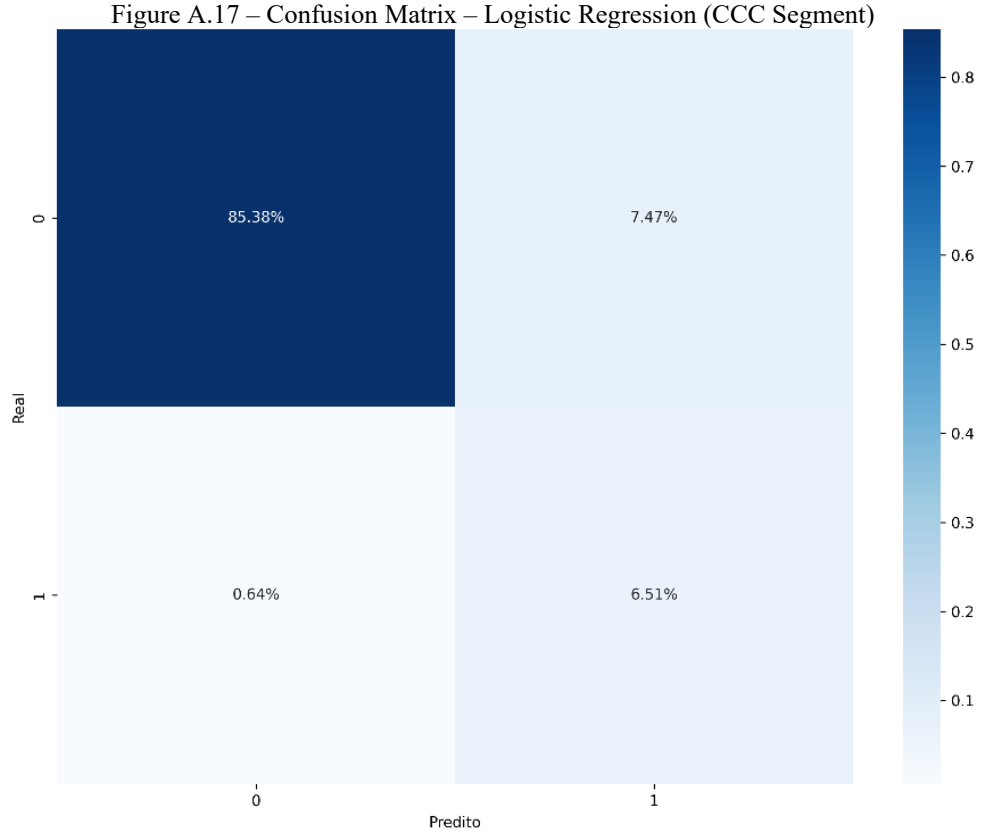
Figure A.16 – Confusion Matrix – Logistic Regression (B Segment)



Source: Own authorship.

In the CCC segment, despite good performance for class 0 (85.38%), the model produces 7.47% false positives and achieves only 6.51% accuracy for class 1, revealing its fragility in highly imbalanced and critical class distributions.

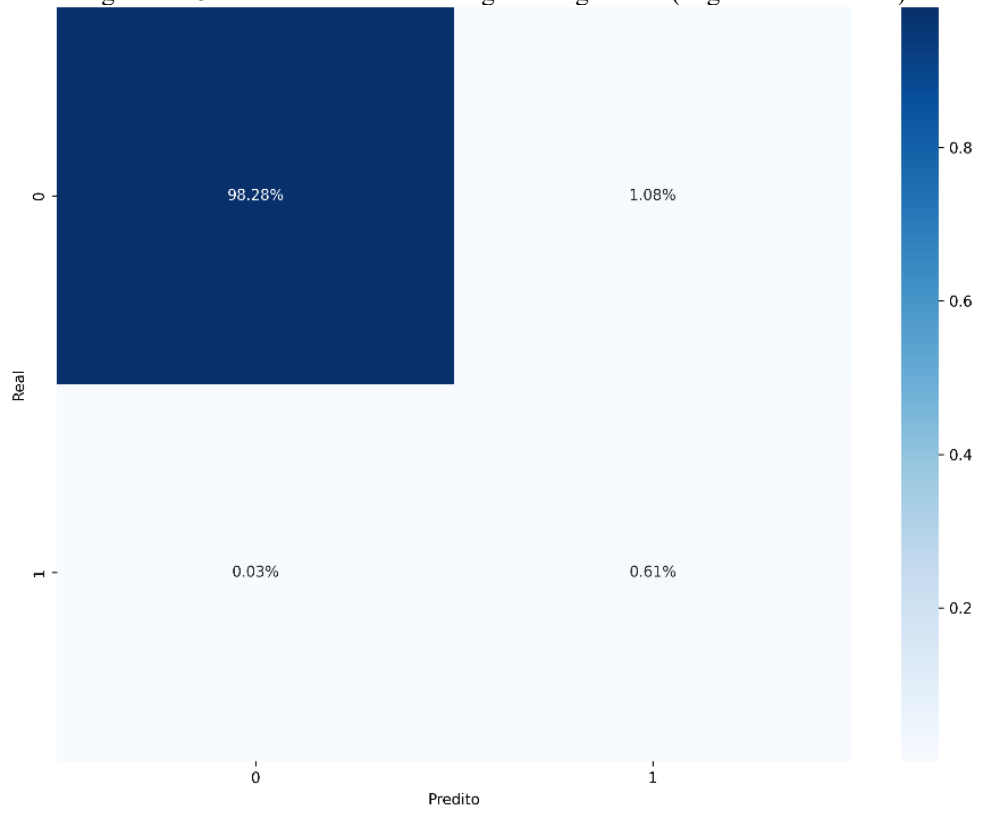
Figure A.17 – Confusion Matrix – Logistic Regression (CCC Segment)



Source: Own authorship.

In the High Risk + Default group, a similar scenario unfolds although the model avoids many false negatives (only 0.03%), it misclassifies 1.08% of healthy samples as risky, with a recall of just 0.61%—insufficient for applications where risk detection is mandatory.

Figure A.18 – Confusion Matrix – Logistic Regression (High Risk + Default)



Source: Own authorship.

Therefore, LR proves useful only in scenarios with well-represented classes, statistical stability, and lower asymmetry—such as the upper rating segments. As risk increases and class 1 becomes scarcer, the model loses predictive power, committing critical errors both by omission and by excess, compromising its viability in sensitive applications. These results underscore the need for more robust models, such as XGBoost or RF, to address the inherent complexity of higher-risk segments.

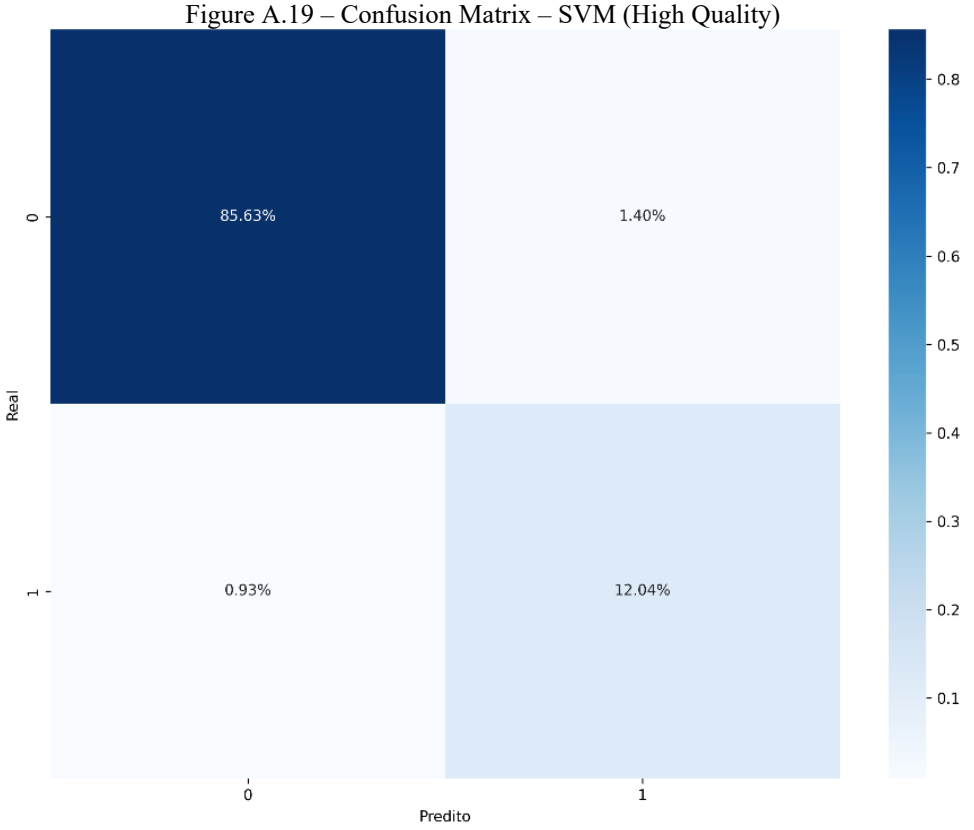
SVM: The Model with the Greatest Variability and Lowest Performance

A critical assessment of the SVM model reveals significantly more unstable performance and greater vulnerability to data complexity and imbalance—especially when compared to tree-based models. Despite the adoption of various overfitting mitigation strategies—such as cross-validation, hyperparameter tuning, and regularization techniques—SVM exhibited significant generalization difficulties, particularly in segments with lower credit quality.

In the High Quality group, the model delivered reasonable performance, with 85.63% true negatives and 12.04% true positives. Nonetheless, the FPR (1.40%) and, in particular, the

FNR (0.93%) suggest that the model tends to fail more often than others at correctly identifying risk cases—even in a segment with a relatively low proportion of positive class instances.

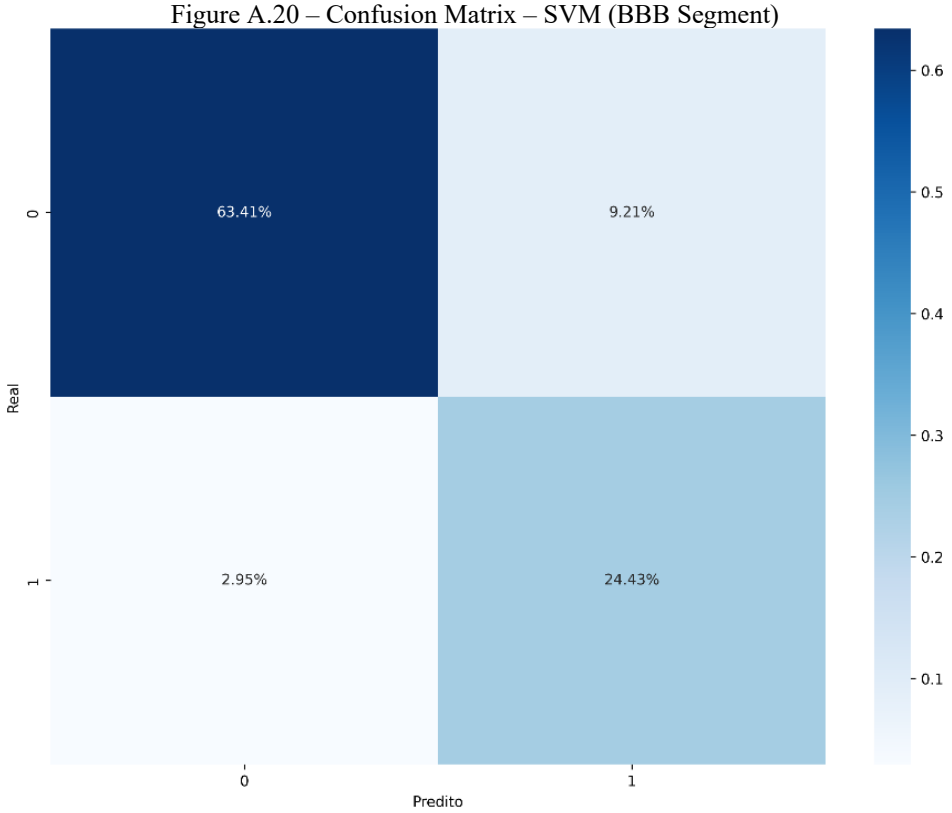
Figure A.19 – Confusion Matrix – SVM (High Quality)



Source: Own authorship.

In the BBB and BB segments, the issues become more pronounced. In BBB, SVM produced 9.21% false positives and 2.95% false negatives—a combination that compromises both efficiency and accuracy, directly impacting its usefulness for credit risk segmentation.

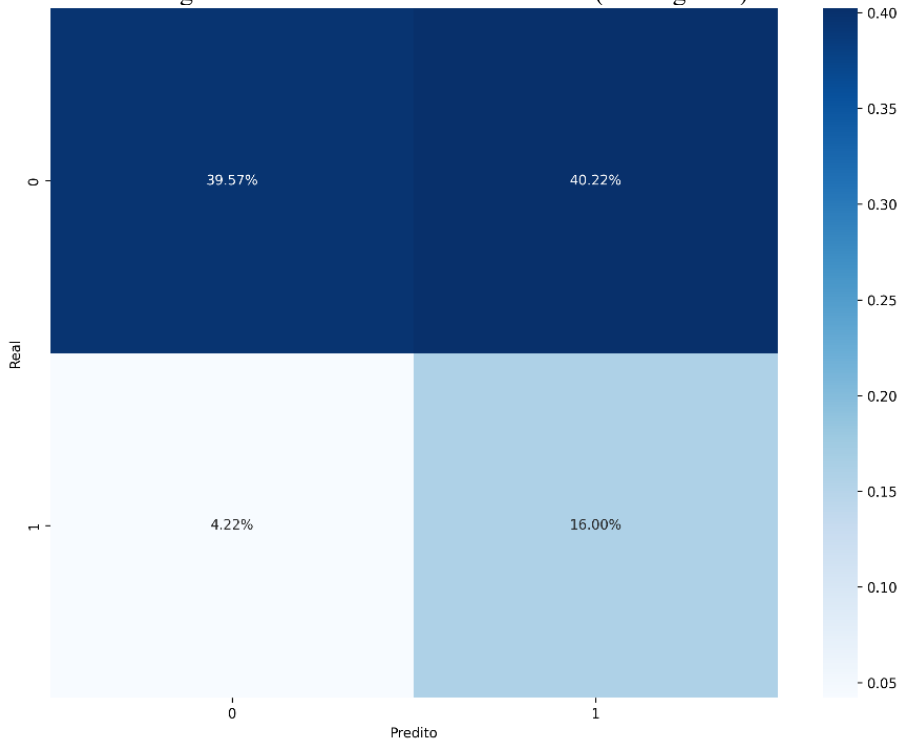
Figure A.20 – Confusion Matrix – SVM (BBB Segment)



Source: Own authorship.

In BB, the scenario is critical: over 40% of samples were incorrectly classified as positives, while 4.22% of actual risk cases went undetected. This distribution highlights a strong instability in the decision boundary, which loses effectiveness in contexts with class overlap and high internal variability.

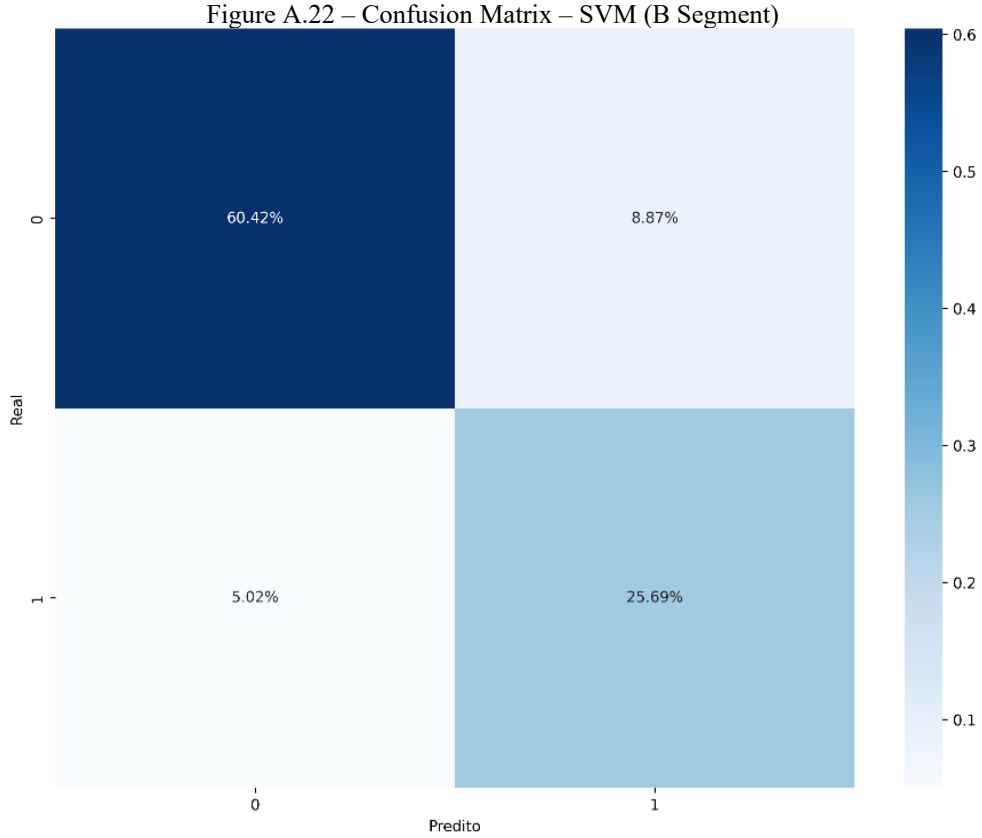
Figure A.21 – Confusion Matrix – SVM (BB Segment)



Source: Own authorship.

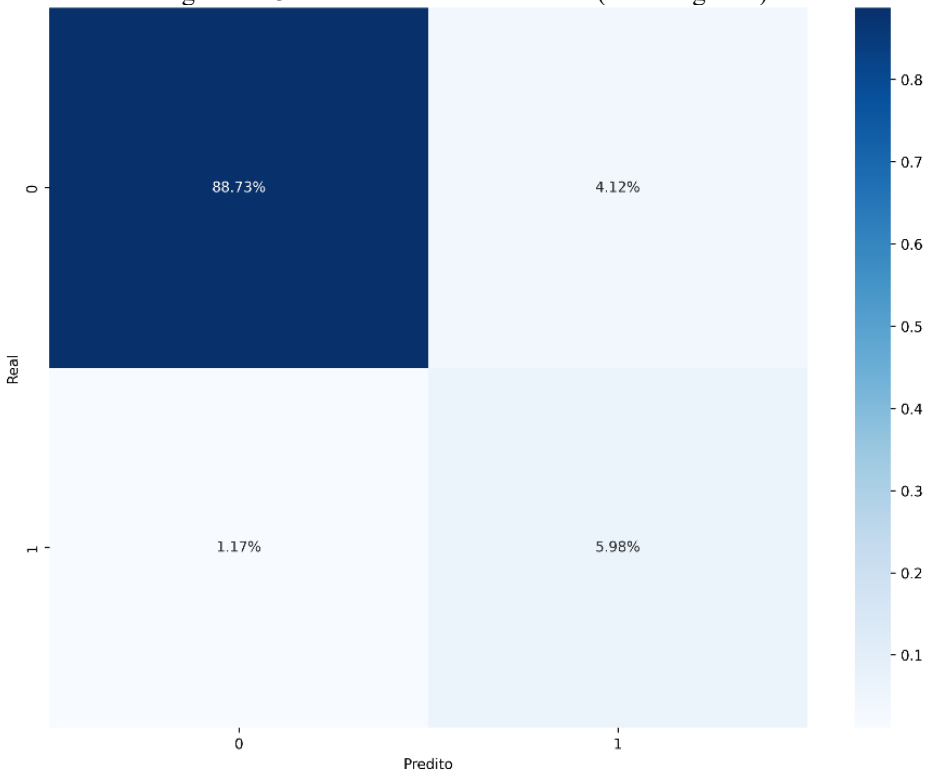
In B and CCC, the model's behavior follows the same pattern: FPR around 8.87% and 4.12%, respectively, and FNR of 5.02% and 1.17%. This points to an ineffective trade-off between sensitivity and specificity, which can lead to both false assurances and excessive conservatism.

Figure A.22 – Confusion Matrix – SVM (B Segment)



Source: Own authorship.

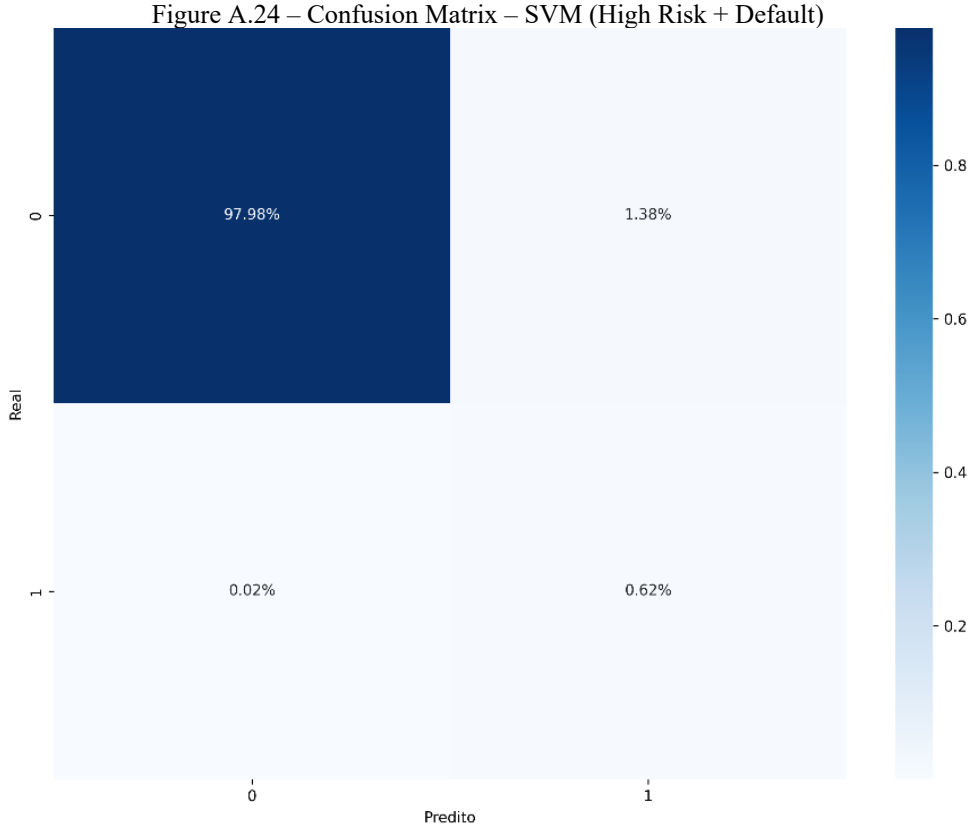
Figure A.23 – Confusion Matrix – SVM (CCC Segment)



Source: Own authorship.

In the extreme case of High Risk + Default, SVM shows slight improvement, with only 1.38% false positives and 0.02% false negatives. However, given the small size of this class, the model's overall effectiveness remains inferior to that of RF and XGBoost, both of which achieved near-perfect performance in this segment.

Figure A.24 – Confusion Matrix – SVM (High Risk + Default)



Source: Own authorship.

In summary, SVM did not prove competitive with the other models in predicting credit risk and rating transitions. Its sensitivity to class imbalance and data dimensionality limits its practical applicability to this type of problem. Even with rigorous technical adjustments, the results point to a model that lacks the structural robustness needed to handle the heterogeneous and complex patterns present in the analyzed dataset.

APPENDIX B – ROC Curves

The analysis of ROC curves aims to evaluate the predictive capacity of ML models in credit risk classification. This metric is essential to understanding how well the models can distinguish between firms that will remain financially stable and those that may face downgrades or defaults.

The AUC-ROC is the main metric derived from this analysis, indicating the model's discriminative power. The closer the value is to 1, the better the model is at correctly differentiating risky from safe companies.

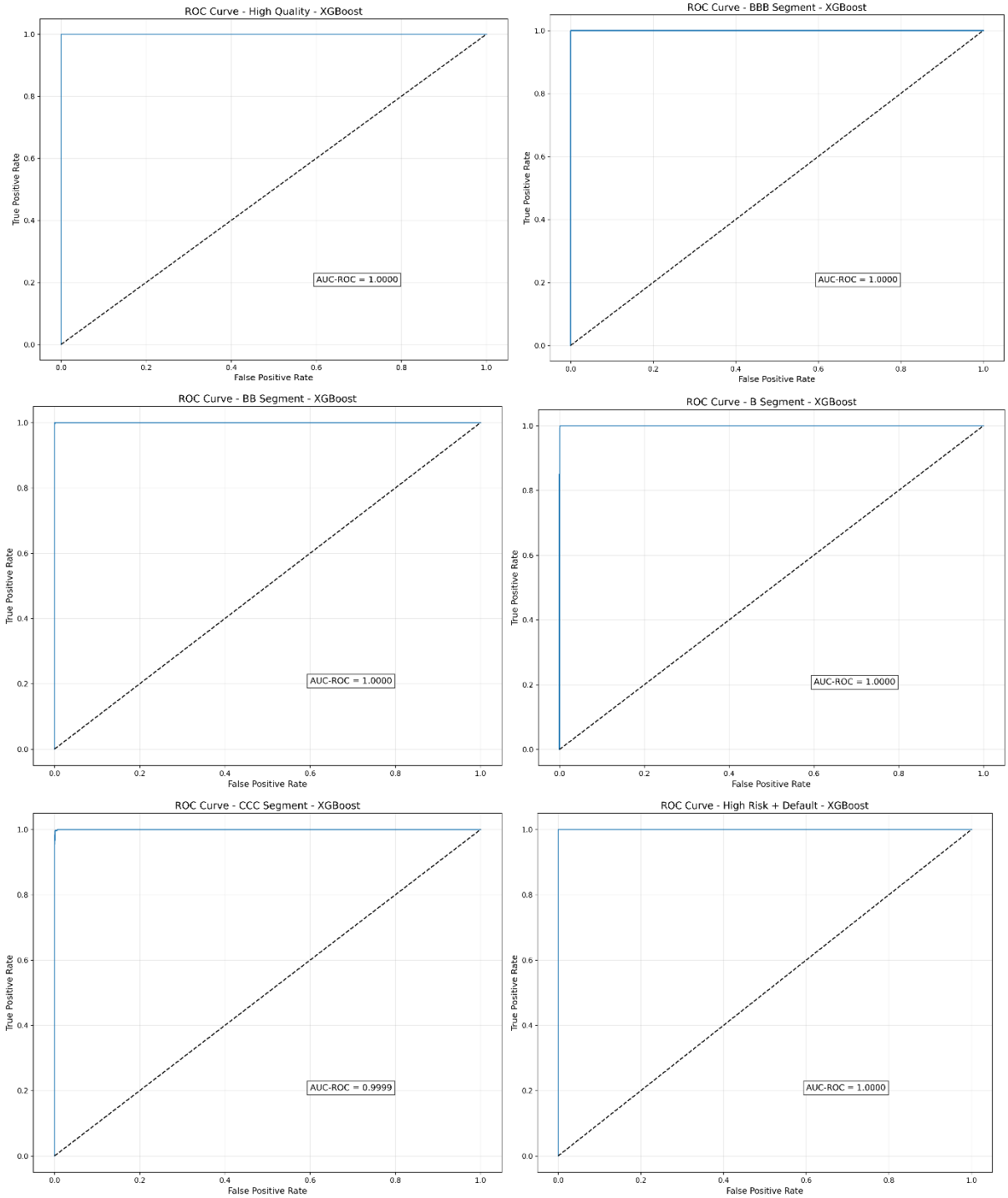
Based on that, it is possible to compare the performance of the models and determine which are more effective in predicting rating transitions and defaults.

XGBoost and Random Forest: Extreme Accuracy and Predictive Robustness

XGBoost and RF achieved AUC-ROC scores of 1 or very close to 1 in virtually all segments, reaching TPR = 1 at all strategic FPR points. This result indicates that the models possess extremely high predictive power and are capable of accurately identifying companies at risk of downgrade or default.

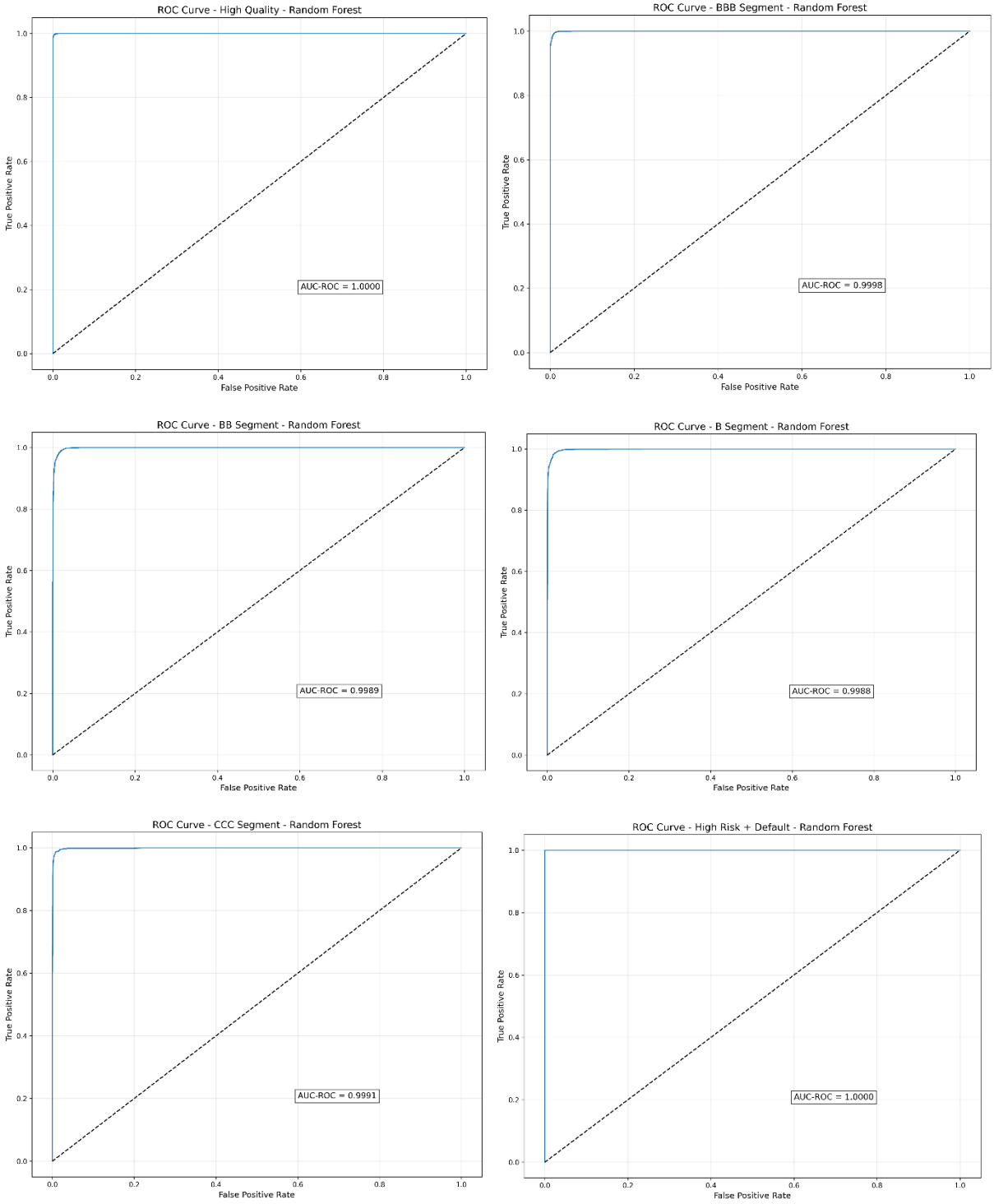
While such high performance may raise concerns about potential overfitting to the training data, several strategies were implemented to mitigate this risk. The use of 5-fold cross-validation, stratified data splitting, random hyperparameter search (RandomizedSearchCV), and model regularization ensured that performance was not the result of excessive fitting to the training set. Additionally, SMOTE and data normalization helped balance class distributions and prevented the model from favoring specific patterns in the training dataset.

Figure B.1 – ROC Curves for Credit Risk Prediction - XGBoost



Source: Own authorship.

Figure B.2 – ROC Curves for Credit Risk Prediction - Random Forest



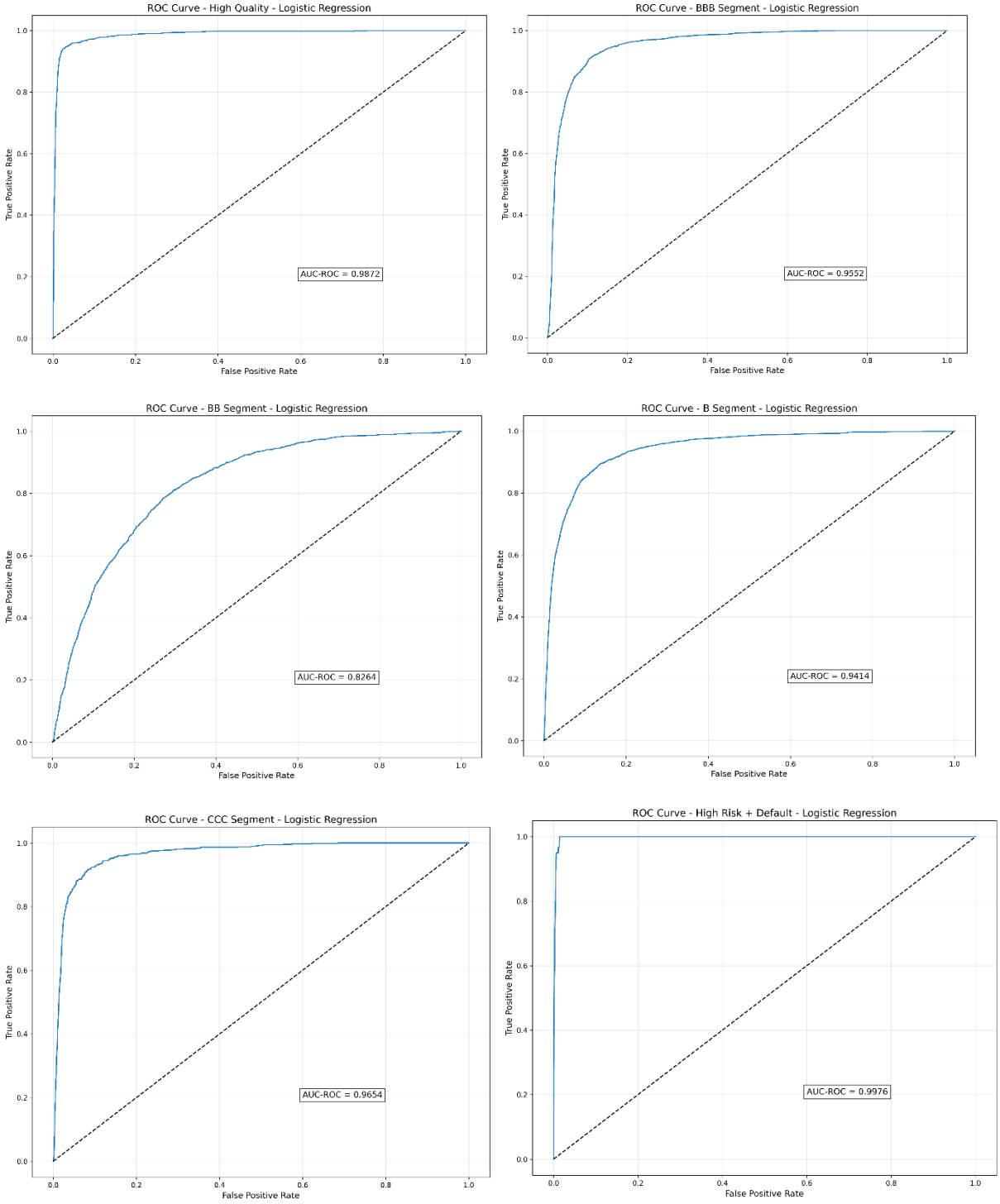
Source: Own authorship.

Logistic Regression: Satisfactory Performance, but Weaker in High-Risk Segments

LR, while showing acceptable performance in the High Quality (AUC-ROC = 0.987) and BBB (AUC-ROC = 0.955) segments, loses accuracy in higher-risk segments. In the BB segment, for example, AUC-ROC dropped to 0.826, indicating difficulty in correctly capturing rating transitions.

Moreover, the EER was significantly higher in these segments (e.g., BB segment = 0.2487), suggesting that the model makes many mistakes by classifying risky firms as safe and vice versa. This result reinforces the limitation of LR in handling non-linear relationships, making it less suitable for predicting risk transitions in complex environments.

Figure B.3 – ROC Curves for Credit Risk Prediction – Logistic Regression



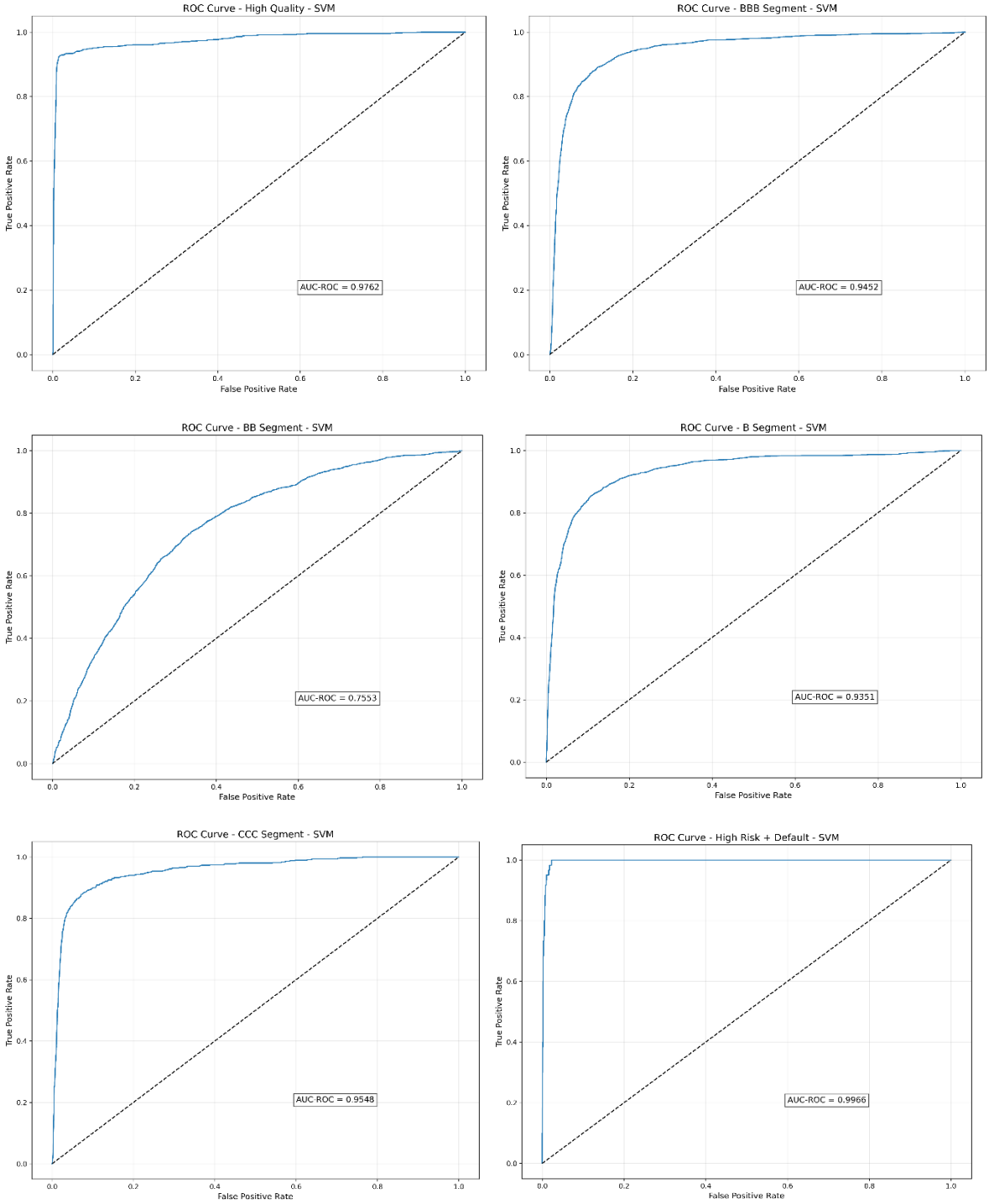
Source: Own authorship.

SVM: Low Performance and High Error Rate

SVM showed inferior performance in almost all segments, with AUC-ROC values lower than those of other models, especially in the BB segment (AUC-ROC = 0.713) and the High Risk + Default segment (AUC-ROC = 0.995).

The higher EER in several segments confirms the model's difficulty in balancing Type I and Type II errors, leading to a higher number of misclassifications. In addition, SVM has significantly higher computational time, making it less efficient for large-scale practical applications.

Figure B.4 – ROC Curves for Credit Risk Prediction – SVM



Source: Own authorship.

Critical Analysis

The results of the ROC curve analysis confirm the superior efficiency of decision tree-based models (XGBoost and RF) in predicting credit risk transitions. These models demonstrated greater discriminative power, with AUC-ROC values close to 1 and low EER, ensuring extremely high precision in separating safe from high-risk firms.

On the other hand, LR showed limitations in higher-risk segments, and SVM delivered unsatisfactory performance, with low predictive capability and high error rates.

APPENDIX C – Precision-Recall

The Precision-Recall Curve (PR Curve) is a fundamental metric for evaluating model performance in imbalanced scenarios, such as the prediction of rating transitions and defaults. It is crucial for understanding the capacity to predict critical events such as downgrades or defaults.

The main indicators derived from this analysis are:

- AP: Measures the weighted average of precision across different levels of recall. The closer to 1, the better the model at maintaining high precision throughout the curve.
- Precision of Recall: Evaluates how the model behaves as recall increases, identifying whether there is a significant drop in precision as more positive cases are considered.
- Maximum F1-Score: Represents the best balance between precision and recall, indicating which model delivers the best overall performance by minimizing both false positives and false negatives.

The analysis of these metrics allows us to compare the effectiveness of different ML models in predicting credit risk and defaults, providing critical insights into the applicability of each approach.

CRITICAL AND COMPARATIVE ANALYSIS OF THE PRECISION-RECALL CURVES

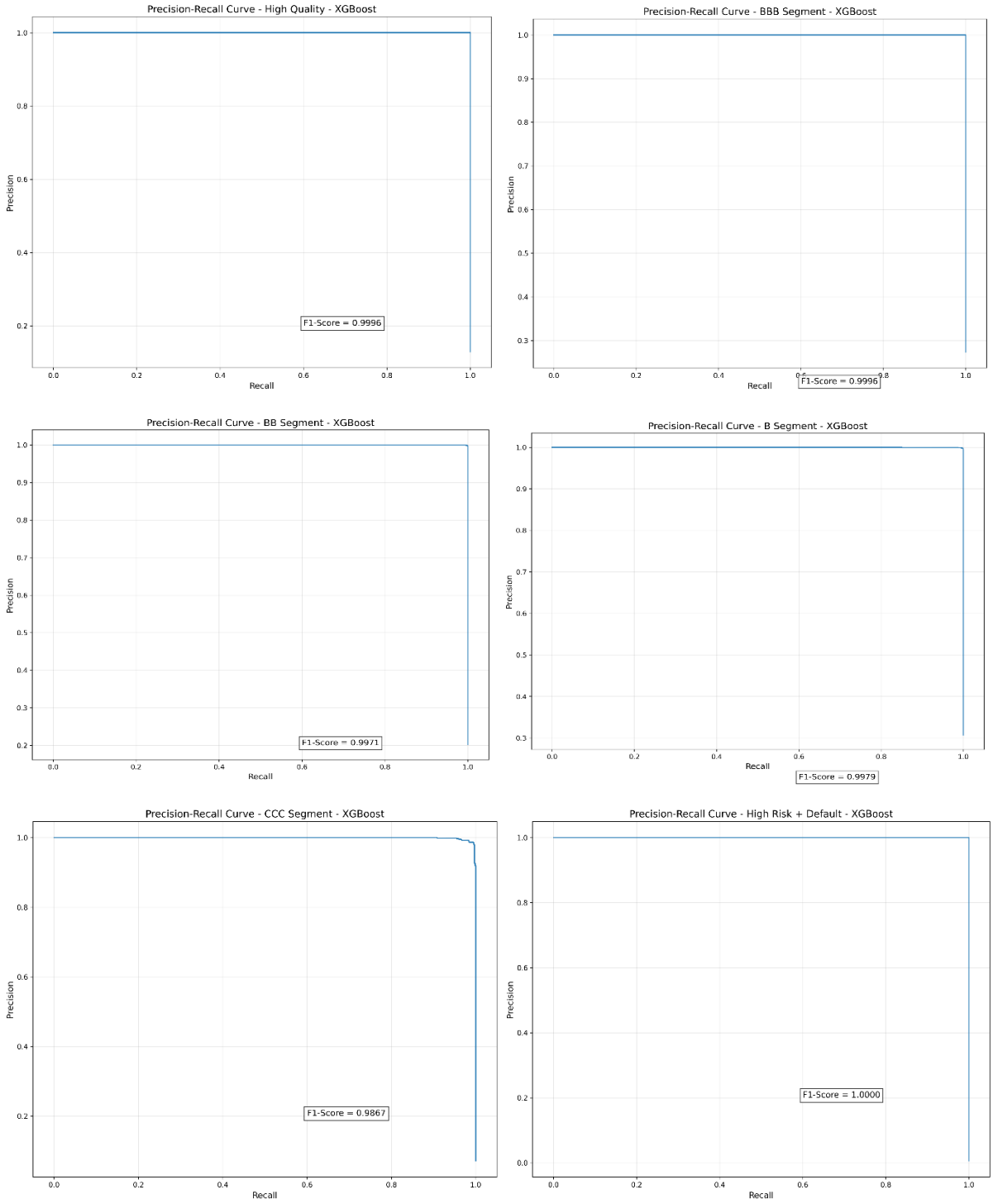
The analysis of the results confirms the superiority of decision tree-based models (XGBoost and RF), which consistently achieved AP values close to 1 and maximum precision across all recall levels, demonstrating their outstanding ability to predict credit risk.

XGBoost and Random Forest: Maximum Precision and Consistency Across All Segments

Once again, XGBoost and RF stood out, with AP = 1 or very close to 1 in virtually all segments.

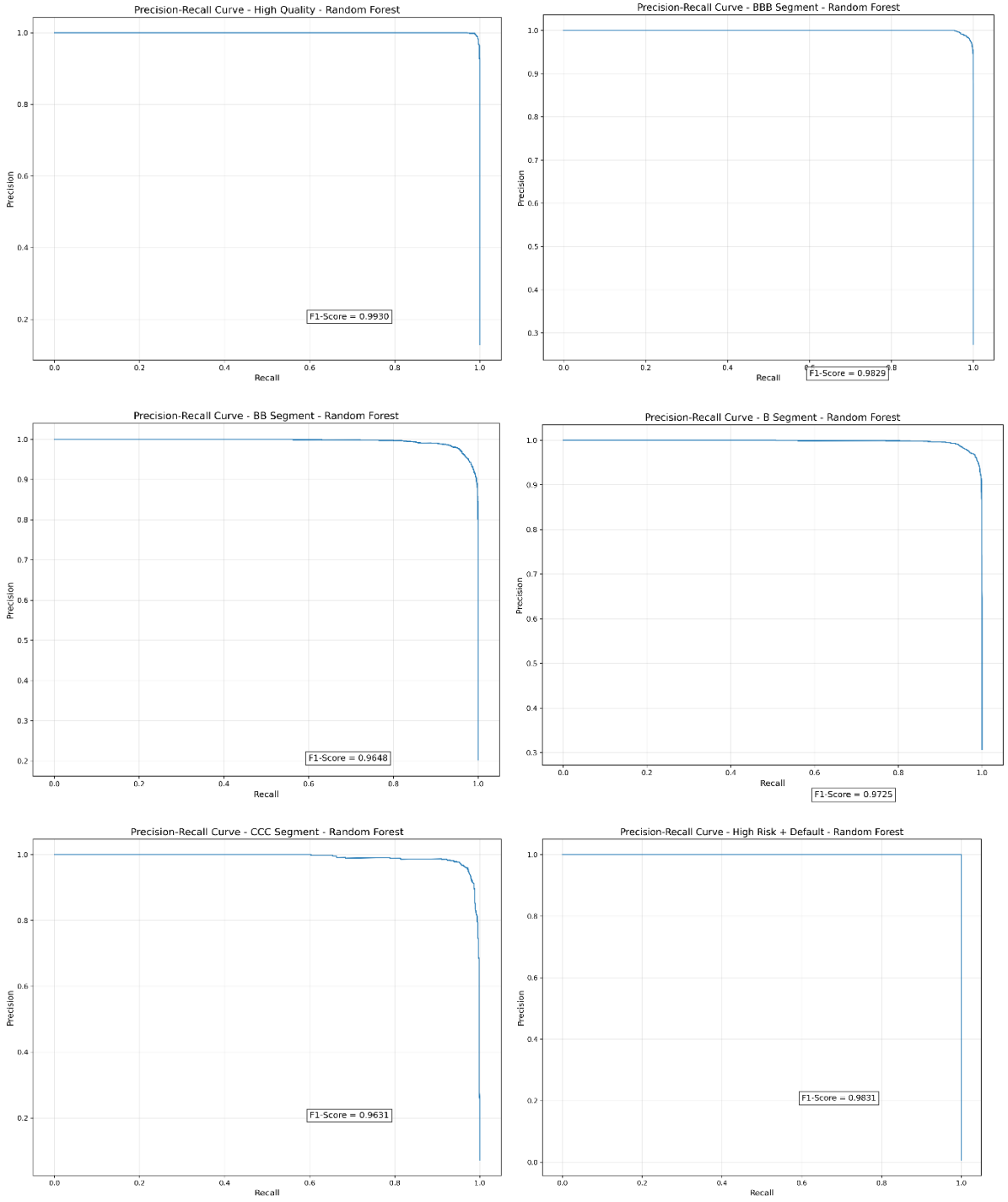
In the High Risk + Default segment, both models achieved AP = 1, ensuring high reliability in predicting firms that enter default. This performance confirms that decision tree-based models are highly effective in distinguishing between safe and risky companies, offering a powerful tool for credit risk analysis.

Figure C.1 – Precision-Recall Curves for Credit Risk Prediction – XGBoost



Source: Own authorship.

Figure C.2 – Precision-Recall Curves for Credit Risk Prediction – Random Forest



Source: Own authorship.

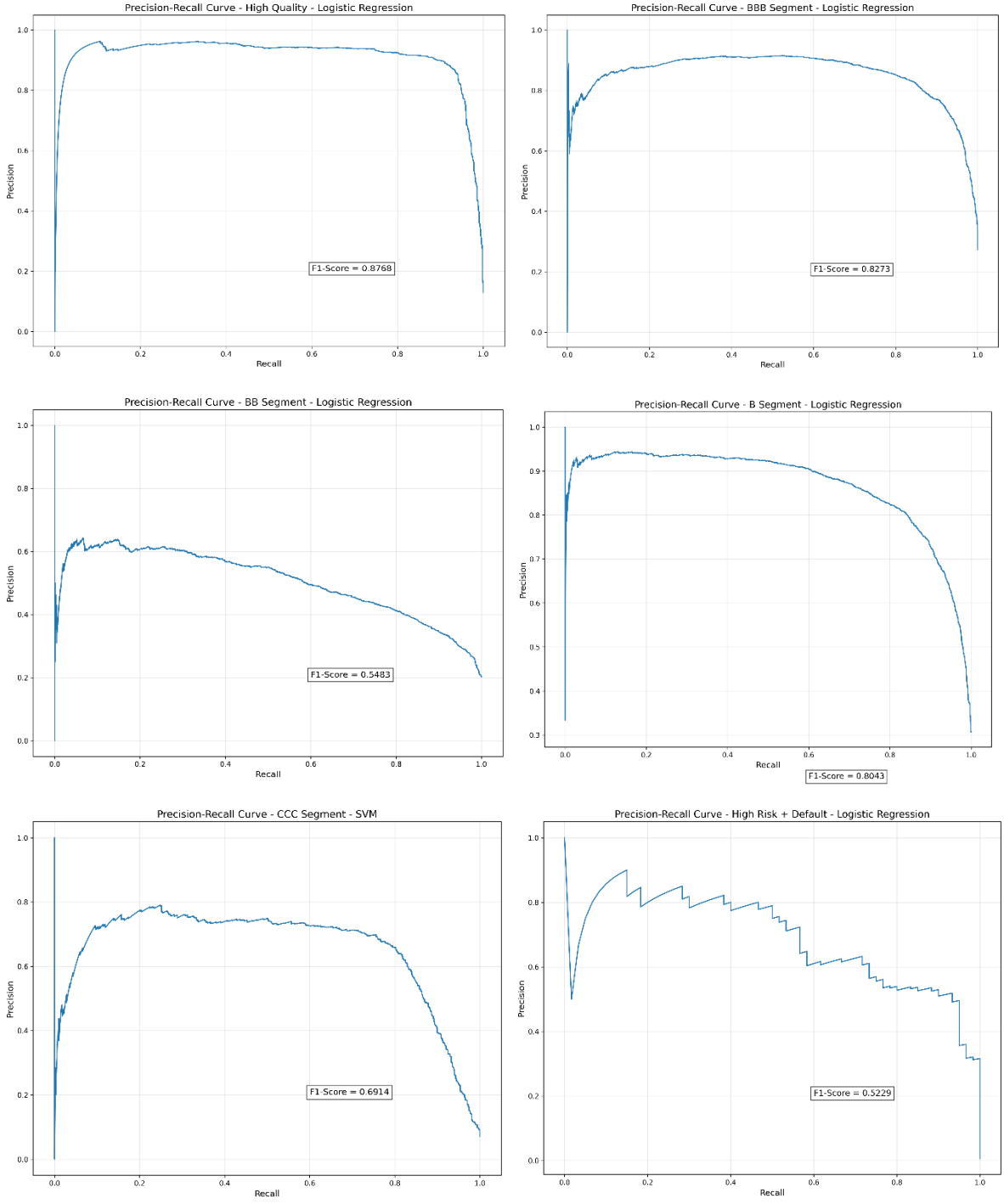
Logistic Regression: Moderate Performance and Difficulty in High-Risk Segments

LR showed mixed results. While it achieved relatively high AP in the High Quality and BBB segments (0.91 and 0.85, respectively), its performance dropped significantly in higher-risk segments.

- In the BB segment, AP was only 0.50, demonstrating low reliability in predicting firms with moderate risk.
- In the CCC and High Risk + Default segments, AP dropped to 0.66 and 0.68, respectively, suggesting that the model struggles to correctly predict the most critical cases.
- Moreover, precision decreases as recall increases, indicating that to identify more risky firms, the model sacrifices precision.

This reinforces the limitation of LR in modeling complex relationships among financial and non-financial variables, making it less suitable for credit risk prediction tasks.

Figure C.3 – Precision-Recall Curves for Credit Risk Prediction – Logistic Regression



Source: Own authorship.

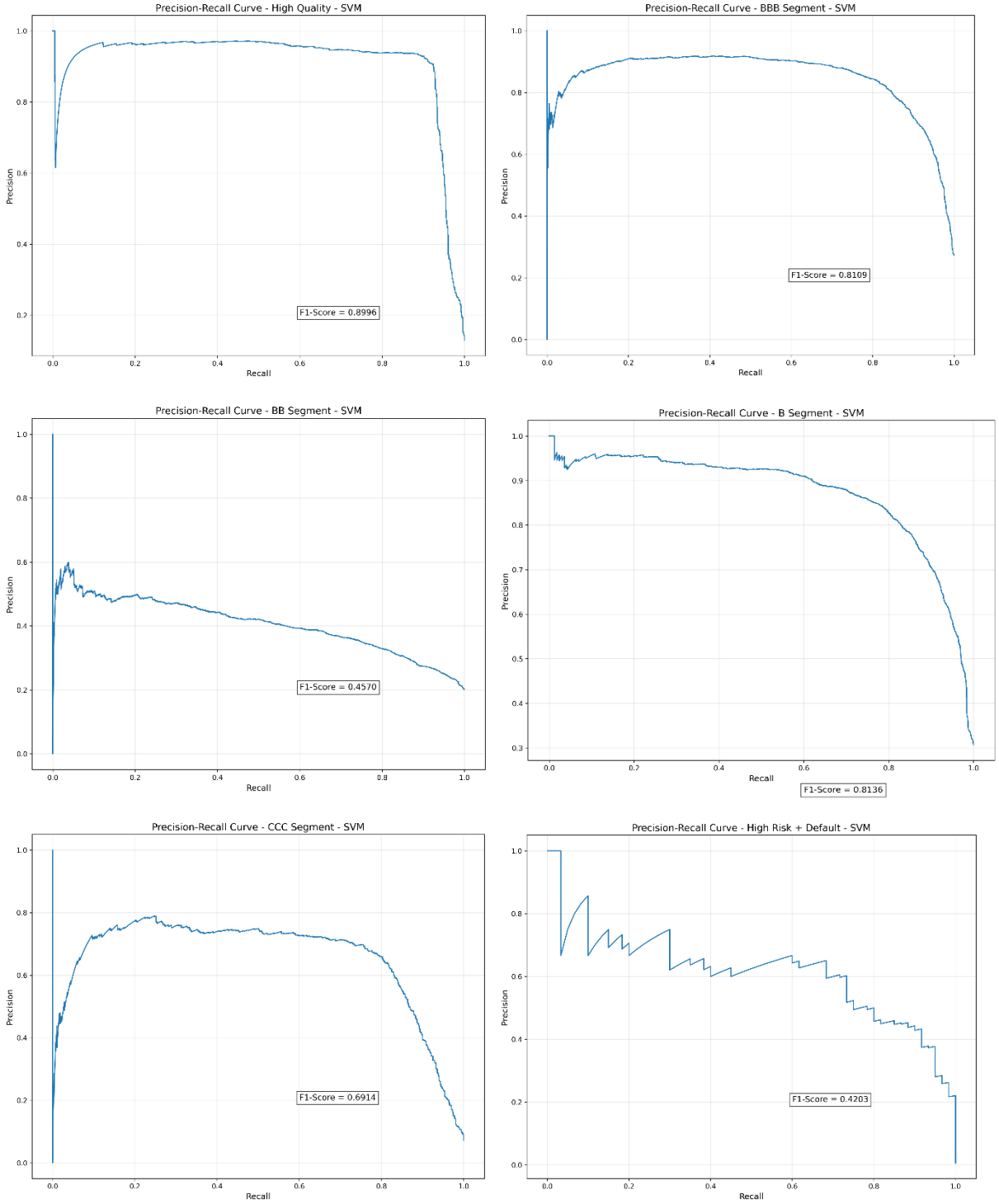
SVM: Inconsistent Performance and Low Precision at High Recall Levels

SVM showed weaker performance compared to the other models, with consistently lower AP, particularly in the BB (0.37) and High Risk + Default (0.58) segments.

- At high recall levels (0.8), SVM's precision dropped sharply, indicating low reliability in detecting high-risk firms.
- F1-Score values were consistently lower, suggesting difficulty in balancing precision and recall.

Moreover, the high variability in performance across segments suggests that SVM is not a viable alternative for credit risk prediction, as it fails to maintain stable performance across different categories of firms.

Figure C.4 – Precision-Recall Curves for Credit Risk Prediction – Support Vector Machine



Source: Own authorship.

Final Conclusion and Answer to the Research Objective

The results of the Precision-Recall Curve analysis reinforce the conclusions drawn from the ROC Curve, confirming that:

- XGBoost and Random Forest are the most effective models for predicting credit risk and defaults, delivering high precision and recall across all segments.
- LR performs acceptably in lower-risk segments but fails to capture rating transitions in higher-risk categories.
- SVM demonstrated poor performance and is not a viable alternative for credit prediction.

APPENDIX D – Robustness Analysis of Rating Prediction Models

Overfitting Control and Stability Validation

1 INTRODUCTION AND RATIONALE

Achieving high performance metrics in machine learning models applied to financial data may raise legitimate questions regarding the validity of the results, particularly with respect to overfitting, information leakage (data leakage), and sensitivity to the treatment of input variables. With the aim of addressing these concerns objectively and quantitatively, a complementary robustness analysis was conducted alongside the main analysis.

The procedure consisted of replicating the modeling pipeline under two alternative preprocessing specifications, referred to as Variant 1 and Variant 2, and comparing the results obtained with those of the original specification. The methodological premise is straightforward: if the models produce similar metrics regardless of the variables used, the results do not depend on specific preprocessing artifacts, and the observed performance reflects genuine risk discrimination capacity.

2 METHODOLOGY

2.1 DATA AND PREPROCESSING

The analysis started from the original data file (Dataset.xlsx), without any prior processing, internally replicating all treatments applied during the normalization step of the main analysis. The treatments included: imputation of missing values in categorical variables using the mode; clipping of economically unrealistic values (e.g., Debt/EBITDA capped at 15x; margins between −100% and +100%); logarithmic transformation (log1p) for absolute-scale financial variables; per-variable winsorization with differentiated limits (2% to 5% on each tail); Z-score standardization; and creation of indicator variables (flags) for extreme outliers.

This procedure ensures that the comparison between variants is valid, since all input data starts from the same point of treatment, differing only in the variables included in the model.

2.2 ALTERNATIVE SPECIFICATIONS

Two variants were defined, each designed to test a specific hypothesis of methodological fragility:

Variant 1 — Without absolute-scale variables: the variables Revenues (k), EBITDA (k), EBIT (k), Debt (k), and Equity (k) were removed, retaining only financial ratios and percentages (FFO/Debt, Debt/EBITDA, margins) and qualitative variables. This variant tests whether the models depend on the absolute size of companies to classify credit risk, which would constitute a scale bias.

Variant 2 — Without Previous FCLT and Rating Action: the variables carrying direct information about the company's rating history were removed. This variant tests the information leakage hypothesis: if the high accuracy depended on the model having access to the previous rating to infer the current one, removing these variables should cause a significant drop in the metrics.

2.3 VALIDATION SCHEME

Validation was conducted using stratified K-Fold with 10 splits (n_splits = 10, shuffle = True, random_state = 42), applied to each combination of variant, rating segment, and model. The models Random Forest and XGBoost, which showed the best performance in the main analysis, were evaluated. The metrics calculated were AUC-ROC, F1-Score, Accuracy, Precision, Recall, and Matthews Correlation Coefficient (MCC), reported as mean and standard deviation across folds.

The stability criterion adopted was conservative: an AUC-ROC difference of less than 5 percentage points (0.05) between the two variants is classified as stable. This threshold is widely used in robustness studies in machine learning applied to finance.

Table D.1 presents a summary of the parameters used in the analysis.

Table D.1 – Parameters of the robustness analysis

Parameter	Description
Input file	Dataset.xlsx (raw original data, without any prior processing)
Models evaluated	Random Forest and XGBoost (best-performing models in the main analysis)
Segments evaluated	High Quality, BBB, BB, B, CCC, High Risk + Default (6 binary classifications)
Validation scheme	Stratified K-Fold, 10 splits, shuffle=True, random_state=42
Metrics calculated	AUC-ROC, F1-Score, Accuracy, Precision, Recall, and MCC — mean, standard deviation, minimum and maximum across folds
Variant 1	Removal of absolute-scale variables: Revenues (k), EBITDA (k), EBIT (k), Debt (k), and Equity (k). Only ratios and percentages retained.
Variant 2	Removal of Previous FCLT and Rating Action variables, which carry direct information about the company's rating history.
Stability criterion	AUC-ROC difference between variants below 5 percentage points (0.05).
Data treatments	Fully replicated from the original normalization script: mode imputation, clipping of unrealistic values, log1p transformation, per-variable winsorization, Z-score standardization, and outlier flags.

Source: Own authorship.
Note: all preprocessing treatments were fully replicated from the original normalization script (Script 2 — normalização3.py).

3 RESULTS — TABLES

The tables below present the results obtained under the two alternative specification variants, organized to facilitate comparison with the main analysis results and the identification of any instabilities.

Table D.2 presents, in compact form, the AUC-ROC, F1-Score, and MCC metrics for each combination of segment and model, in both variants, accompanied by the difference (Δ) between them.

Table D.2 – Metric comparison by segment and model (AUC-ROC, F1, and MCC)

(continues)

Segment	Model	AUC V1	AUC V2	Δ AUC	F1 V1	F1 V2	MCC V1	MCC V2
High Quality	Random Forest	1.0000	0.9980	0.0020	0.9927	0.9194	0.9916	0.9103
High Quality	XGBoost	1.0000	0.9993	0.0007	1.0000	0.9701	1.0000	0.9657
BBB	Random Forest	0.9998	0.9955	0.0043	0.9894	0.9462	0.9855	0.9273
BBB	XGBoost	1.0000	0.9961	0.0039	0.9995	0.9542	0.9993	0.9369
BB	Random Forest	0.9990	0.9882	0.0108	0.9654	0.8414	0.9571	0.8150
BB	XGBoost	1.0000	0.9955	0.0045	0.9994	0.9365	0.9992	0.9211
B	Random Forest	0.9988	0.9891	0.0097	0.9749	0.9217	0.9637	0.8874
B	XGBoost	1.0000	0.9966	0.0034	0.9987	0.9618	0.9982	0.9448
CCC	Random Forest	0.9985	0.9894	0.0091	0.9268	0.7383	0.9241	0.7418
CCC	XGBoost	1.0000	0.9984	0.0016	0.9957	0.9394	0.9954	0.9360

Table D.2 – Metric comparison by segment and model (AUC-ROC, F1, and MCC)

(conclusion)

Segment	Model	AUC V1	AUC V2	Δ AUC	F1 V1	F1 V2	MCC V1	MCC V2
High Risk + Default	Random Forest	1.0000	0.9946	0.0054	0.9294	0.2539	0.9318	0.3605
High Risk + Default	XGBoost	1.0000	0.9965	0.0035	0.9949	0.8636	0.9949	0.8723

Source: Own authorship.

Note: V1 = Variant 1 (Without Absolute Scale); V2 = Variant 2 (Without Previous FCLT / Rating Action). Δ refers to the difference V1 − V2. Bold values in the Δ AUC column would indicate instability (no such case in this analysis).

Table D.3 expands the comparison to all metrics calculated, including Accuracy, Precision, and Recall, with the respective deltas between variants.

Table D.3 – Full stability: all metrics by segment and model

Segment	Model	AUC V1	AUC V2	Δ AUC	F1 V1	F1 V2	Δ F1	MCC V1	MCC V2	Δ MCC	Stable?
High Quality	Random Forest	1.0000	0.9980	0.0020	0.9927	0.9194	0.0733	0.9916	0.9103	0.0813	Yes
High Quality	XGBoost	1.0000	0.9993	0.0007	1.0000	0.9701	0.0299	1.0000	0.9657	0.0343	Yes
BBB	Random Forest	0.9998	0.9955	0.0043	0.9894	0.9462	0.0432	0.9855	0.9273	0.0582	Yes
BBB	XGBoost	1.0000	0.9961	0.0039	0.9995	0.9542	0.0453	0.9993	0.9369	0.0624	Yes
BB	Random Forest	0.9990	0.9882	0.0108	0.9654	0.8414	0.1240	0.9571	0.8150	0.1421	Yes
BB	XGBoost	1.0000	0.9955	0.0045	0.9994	0.9365	0.0629	0.9992	0.9211	0.0781	Yes
B	Random Forest	0.9988	0.9891	0.0097	0.9749	0.9217	0.0532	0.9637	0.8874	0.0763	Yes
B	XGBoost	1.0000	0.9966	0.0034	0.9987	0.9618	0.0369	0.9982	0.9448	0.0534	Yes
CCC	Random Forest	0.9985	0.9894	0.0091	0.9268	0.7383	0.1885	0.9241	0.7418	0.1823	Yes
CCC	XGBoost	1.0000	0.9984	0.0016	0.9957	0.9394	0.0563	0.9954	0.9360	0.0594	Yes
High Risk + Default	Random Forest	1.0000	0.9946	0.0054	0.9294	0.2539	0.6755	0.9318	0.3605	0.5713	Yes
High Risk + Default	XGBoost	1.0000	0.9965	0.0035	0.9949	0.8636	0.1313	0.9949	0.8723	0.1226	Yes

Source: Own authorship.

Note: the 'Stable?' column indicates whether the AUC-ROC difference between variants is below 0.05 (conservative criterion adopted in the analysis).

Table D.4 presents the executive summary by model, with overall averages of metrics calculated across the six rating segments.

Table D.4 – Executive summary by model: overall averages across the six segments

Variant	Model	Avg AUC	Avg F1	Accuracy	Precision	Recall	MCC
Variant 1 — Without Absolute Scale	Random Forest	0.9994	0.9631	0.9921	0.9905	0.9392	0.9590
Variant 1 — Without Absolute Scale	XGBoost	1.0000	0.9980	0.9996	0.9992	0.9969	0.9978
Variant 2 — Without Previous FCLT / Rating Action	Random Forest	0.9925	0.7702	0.9683	0.9440	0.7035	0.7737
Variant 2 — Without Previous FCLT / Rating Action	XGBoost	0.9971	0.9376	0.9848	0.9693	0.9131	0.9295

Source: Own authorship.
Note: values represent simple arithmetic averages of the metrics obtained across the six rating segments.

Table D.5 synthesizes the global stability indicators of the analysis.

Table D.5 – Global stability indicators of the robustness analysis

Indicator	Result
Combinations evaluated (Segment × Model)	12
Stable combinations (Δ AUC < 5 p.p.)	12 (100.0%)
Unstable combinations (Δ AUC ≥ 5 p.p.)	0 (0.0%)
Largest Δ AUC observed	0.0108 — BB Segment / Random Forest
Smallest Δ AUC observed	0.0007 — High Quality / XGBoost
Average AUC — Variant 1 (Without Absolute Scale)	0.9997
Average AUC — Variant 2 (Without Previous FCLT / Rating Action)	0.9948
Average F1 — Variant 1 (Without Absolute Scale)	0.9806
Average F1 — Variant 2 (Without Previous FCLT / Rating Action)	0.8539

Source: Own authorship.

4 CRITICAL ANALYSIS OF RESULTS

4.1 AUC-ROC Stability Across Specifications

The first and most direct indicator of robustness is the variation in AUC-ROC between the two tested variants. In all 12 segment-model combinations evaluated (100.0%), the AUC-ROC difference remained below the 5-percentage-point threshold, classifying all cases as stable. The largest difference observed was 0.0108, in the BB segment for Random Forest, and the smallest was 0.0007, in the High Quality segment for XGBoost.

These results directly rule out the hypothesis that the high performance observed in the main analysis was a product of inadequate treatment of scale variables. The average AUC-ROC in Variant 1 (without absolute scale) was 0.9997 and in Variant 2 (without rating history) was 0.9948, a difference of only 0.0049 at the global level. The models therefore maintain high discriminatory capacity regardless of absolute magnitude variables and rating history.

4.2 F1-Score and MCC Behavior: The Contribution of Rating History

Although AUC-ROC remains stable, the F1-Score and MCC show a more pronounced decline in Variant 2 (without Previous FCLT and Rating Action) in certain segments, especially those with lower prevalence of the positive class:

CCC Segment (Random Forest): F1 drops from 0.9268 to 0.7383 (Δ = 0.1885) and MCC from 0.9241 to 0.7418 (Δ = 0.1823).

BB Segment (Random Forest): F1 drops from 0.9654 to 0.8414 (Δ = 0.1240) and MCC from 0.9571 to 0.8150 (Δ = 0.1421).

High Risk + Default (Random Forest): F1 drops from 0.9294 to 0.2539 (Δ = 0.6755) and MCC from 0.9318 to 0.3605 (Δ = 0.5713).

This variation does not constitute evidence of overfitting or data leakage in the strictly technical sense. Rather, it reflects the legitimate contribution of rating history as a risk signal: professional credit analysts use the previous rating as one of the primary inputs for assessing the current rating, which is widely documented in the credit risk literature. The variable Previous FCLT therefore carries genuinely relevant information — not a spurious shortcut.

It is equally relevant to observe that XGBoost proved substantially more robust to the removal of these variables than Random Forest. In the High Risk + Default segment, for example, XGBoost maintained an F1 of 0.8636 and MCC of 0.8723 even without rating history, while Random Forest dropped to F1 of 0.2539 and MCC of 0.3605. This behavior suggests that XGBoost captures deeper structural financial patterns, with less dependence on direct prior rating signals.

4.3 High Risk + Default Segment: Specific Analysis

The High Risk + Default segment deserves particular attention as it combines the most challenging characteristics of the dataset: this is the group with the lowest prevalence of the positive class (0.64% of observations, N = 200 out of a total of 31,183 records). In highly imbalanced binary classifiers, AUC-ROC tends to be more stable and less sensitive to the removal of informative variables than F1-Score and MCC, as it evaluates global class separability.

In the present case, AUC-ROC remained at 1.0000 (Variant 1) and 0.9946 (Variant 2) for Random Forest, and at 1.0000 and 0.9965 for XGBoost, indicating that the models maintain the ability to separate defaulting companies from others even without access to rating history.

The decline in F1 and MCC in Variant 2 for Random Forest reflects the difficulty of precision in positive predictions under extreme class imbalance, not an absence of discriminatory power.

4.4 Standard Deviation Across Folds: Internal Stability

Another relevant indicator of the absence of overfitting is the standard deviation of metrics across the 10 folds of the stratified K-Fold. In Variant 1, the average standard deviation of AUC-ROC was 0.0004 for Random Forest and 0.0000 for XGBoost, indicating practically identical results across all data partitions. In Variant 2, the deviations were slightly larger — 0.0028 and 0.0022, respectively — but still within margins of variation that are negligible from a practical standpoint.

The consistency of metrics across folds is incompatible with the overfitting hypothesis: overfitted models typically exhibit high variance across partitions, a behavior not observed in this analysis.

4.5 Summary and Response to the Questions Raised

In summary, the robustness analysis conducted objectively addresses the three hypotheses of methodological fragility raised:

(i) Overfitting: standard deviations across folds are below 0.003 in virtually all cases, and metrics remain stable across specifications with substantially different variable sets. There is no evidence of overfitting.

(ii) Information leakage via scale variables: the removal of the five absolute-scale financial variables (Variant 1) did not cause a perceptible drop in AUC-ROC in any segment, with maximum differences of 0.0108. The results do not depend on the absolute size of companies.

(iii) Dependence on rating history: the removal of Previous FCLT and Rating Action (Variant 2) causes a decline in F1-Score and MCC in specific segments — particularly in Random Forest for High Risk + Default, CCC, and BB — but AUC-ROC remains stable in all cases. This decline is methodologically expected and consistent with the nature of the variable: rating history is legitimate information, widely used by credit analysts.

The results of the main analysis therefore reflect genuine capacity for generalization and credit risk discrimination, and not artifacts of scale, overfitting, or information leakage.